MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, HIGH SPEED, CMOS, BUS TRANSCEIVERS WITH THREE-STATE OUTPUTS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, high speed, CMOS, logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device types shall be as follows:

Device type	Circuit
01	Quadruple inverting bus transceivers with three-state outputs
02	Quadruple non-inverting bus transceivers with three-state outputs
03	Octal non-inverting bus transceivers with three-state outputs
04, 05	To be included later
06	Octal inverting bus transceivers with three-state outputs
07	Octal non-inverting or inverting bus transceivers with three-state outputs
08	Octal non-inverting bus transceivers with three-state outputs
09	Octal inverting bus transceivers with three-state outputs
53	Octal non-inverting bus transceivers with three-state outputs and TTL-input voltage compatibility

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
L	D-9 (24-lead, 1/4" x 1 1/4"), dual-in-line package
R	D-8 (20-lead, 1/4" x 1 1/16"), dual-in-line package
2	C-2 (20-terminal, .350" x .350") square chip carrier package
3	C-4 (28-terminal, .450" x .450"), square chip carrier package

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

1.3 Absolute maximum ratings.

Supply voltage (V_{CC})	-0.5 V dc +7.0 V dc
DC input voltage (V_{IN})	-0.5 V dc to V_{CC} +0.5 V dc
DC output voltage (V_{OUT})	-0.5 V dc to V_{CC} +0.5 V dc
Clamp diode current (I_{IK}, I_{OK})	±20 mA
DC output current per pin (I_{OUT})	±35 mA
DC V_{CC} or GND current per pin (I_{OC})	±70 mA
Storage temperature range (T_{STG})	-65°C to +150°C
Maximum power dissipation (P_D)	300 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases C, L, and R	(See MIL-M-38510, appendix C)
Case 2	60°C/W
Case 3	TBD
Junction temperature (T_J)	+175°C

1.4 Recommended operating conditions.

Device types 01, 02, 03, 06, 07, 08, 09:

Input low (V_{IL}) maximum voltage	0.3 V at V_{CC} = 2 V 0.9 V at V_{CC} = 4.5 V 1.2 V at V_{CC} = 6 V
Input high (V_{IH}) minimum voltage	1.5 V at V_{CC} = 3 V 3.15 V at V_{CC} = 4.5 V 4.2 V at V_{CC} = 6 V
Operating ambient temperature range T_A	-55°C to +125°C
Width of clock pulse (t_{CLK}):	
Device types 08 and 09	27 ns minimum
Setup time before clock (t_{setup}):	
Device types 08 and 09	30 ns minimum
Hold time (t_{hold}):	
Device types 08 and 09	8 ns minimum
Supply voltage (V_{CC})	2 V dc to 6 V dc
Output voltage	0 V dc to V_{CC}

Input rise and fall times (t_r, t_f) maximum:

V_{CC} = 2 V	1000 ns
V_{CC} = 4.5 V	500 ns
V_{CC} = 6 V	400 ns

Device type 53:

Input low (V_{IL}) maximum voltage	0.8 V at V_{CC} = 4.5 V - 5.5 V
Input high (V_{IH}) minimum voltage	2.0 V at V_{CC} = 4.5 V - 5.5 V
Supply voltage (V_{CC})	4.5 V dc to 5.5 V dc
Output voltage	0 V dc to V_{CC}
Operating ambient temperature T_A	-55°C to +125°C

Input rise and fall times (t_r, t_f):

V_{CC} = 4.5 V	500 ns maximum

2. APPLICABLE DOCUMENTS

2.1 Government specification and standard. Unless otherwise specified, the following specification and standard, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended operating ambient temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.6.1. Total dose radiation hardness identifier. Total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and 4.5.4 herein.

3.6.2 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.3 Correctness of indexing and marking. All devices shall be subjected to the final electrical tests specified in table II after part number marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 37 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in 3.1.10 through 3.1.14 of method 5004 and substitute lines 1 through 7 of table II herein.

b. Burn-in (method 1015 of MIL-STD-883).

(1) Static tests (test condition A) using circuit shown on figure 3, or equivalent. Ambient temperature (T_A) shall be +125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

(2) Dynamic test (test condition D) using circuit shown on figure 3, or equivalent. Ambient temperature shall be +125°C minimum. Test duration shall be in accordance with table I of method 1015.

c. Interim and final electrical parameters shall be as specified in table II herein.

d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failures shall be cumulative for determining the PDA.

c. The PDA for class B devices shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics, after burn-in, exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5, 6, 7, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_I measurement) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

d. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady-state life test circuits shall be submitted to the qualifying activity for approval. When the alternate steady-state life test is used, the circuit on figure 3, or equivalent, shall be used.

b. Electrostatic Discharge Sensitivity (ESDS) testing shall be performed in accordance with MIL-STD-883, method 3015. The option to categorize devices as ESD sensitive without performing the test is not allowed. Device types categorized as ESD sensitive shall be further tested using method 3015 modified as follows:

(1) Table I pin combinations 4 and 5 shall be deleted.

(2) The test sequence specified in paragrapn 3.b shall be repeated an additional four times rather than the two specified.

(3) Only those device types that pass ESDS testing at 1,000 volts or greater shall be considered as conforming to the requirements of this specification.

c. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 5 of group B inspections and shall consist of tests specified in table IV herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D and as specified in 4.5.2 herein and as shown on figure 3 (note 3), or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration, 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

Text continues on page 77.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified	Device type 2/	V_{CC}	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH1} 3/	V_{IH} = 1.5 V V_{IL} = 0.3 V I_{OH} = -20 µA	01-09	2.0 V	1.95	---	V
	V_{OH2} 3/	V_{IH} = 3.15 V V_{IL} = 0.9 V I_{OH} = -20 µA	01-09	4.5 V	4.44	---	V
	V_{OH3}	V_{IH} = 4.2 V V_{IL} = 1.2 V I_{OH} = -20 µA	01-09	6.0 V	5.95	---	V
	V_{OH4} 3/	V_{IH} = 3.15 V V_{IL} = 0.9 V I_{OH} = -6.0 mA	01-09	4.5 V	3.7	---	V
	V_{OH5}	V_{IH} = 4.2 V V_{IL} = 1.2 V I_{OH} = -7.8 mA	01-09	6.0 V	5.2	---	V
	V_{OH6}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OH} = -20 µA	53	4.5 V	4.4	---	V
	V_{OH7}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OH} = -6.0 mA	53	5.5 V	3.7	---	V
Low-level output voltage	V_{OL1} 3/	V_{IL} = 0.8 V V_{IH} = 1.5 V I_{OL} = 20 µA	01-09	2.0 V	---	0.05	V
	V_{OL2} 3/	V_{IL} = 0.9 V V_{IH} = 3.15 V I_{OL} = 20 µA	01-09	4.5 V	---	0.05	V
	V_{OL3}	V_{IL} = 1.2 V V_{IH} = 4.2 V I_{OL} = 20 µA	01-09	6.0 V	---	0.05	V
	V_{OL4} 3/	V_{IL} = 0.9 V V_{IH} = 3.15 V I_{OL} = +6.0 mA	01-09	4.5 V	---	0.4	V
	V_{OL5}	V_{IL} = 1.2 V V_{IH} = 4.2 V I_{OL} = 7.8 mA	01-09	6.0 V	---	0.4	V

See footnotes at end of table.

1.2.1.3 Case characteristic. The die characteristic applies

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified	Device type 2/	V_{CC}	Limits Min	Limits Max	Unit
Low-level output voltage	V_{OL6}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OH} = +20 µA	53	4.5 V	---	0.05	V
	V_{OL7}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OH} = +6.0 mA	53	5.5 V	---	0.4	V
Positive input clamp voltage	V_{IC+}	V_{CC} = GND I_{IN} = +1 mA T_C = +25°C	All	---	8/	+1.5	V
Negative input clamp voltage	V_{IC-}	V_{CC} = Open I_{IN} = -1 mA T_C = +25°C	All	---	---	-1.5	V
Control input current low	I_{IL}	V_{IN} = GND	01-09	6.0 V	---	-0.1	µA
			53	5.5 V		-0.1	µA
Control input current high	I_{IH}	V_{IN} = V_{CC}	01-09	6.0 V	---	0.1	µA
			53	5.5 V		0.1	µA
Input/output current, low-level, outputs disabled	I_{IL}/ I_{OZL} 4/	V_{IN} control = 4.2 V or 1.2 V V_{OUT} = GND	01-09	6.0 V	---	-2.0	µA
		V_{IN} control = 2.0 V or 0.8 V V_{OUT} = GND	53	5.5 V	---	-2.0	µA
Input/output current, high-level, outputs disabled	I_{IH}/ I_{OZH} 4/	V_{IN} control = 4.2 V or 1.2 V V_{OUT} = 6 V	01-09	6.0 V	---	2.0	µA
		V_{IN} control = 2.0 V or 0.8 V V_{OUT} = 5.5 V	53	5.5 V	---	2.0	µA
Short circuit output current	I_{OS1} 3/	T_C = -55°C to +125°C V_O = GND V_I = GND or V_{CC}	01-09	2.0 V	-2	-60	mA
	I_{OS2} 3/		All	4.5 V	-15	-165	
	I_{OS3} 3/		01-09	6.0 V	-25	-210	
	I_{OS4}		All	4.0 V	-10	-135	
	I_{OS5} 3/		53	5.5 V	-25	-210	

See footnotes at end of table.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified	Device type 2/	V_{CC}	Limits Min	Limits Max	Unit
Supply current quiescent	I_{CC}	V_I = 6.0 V	01,02, 03,06,07, 08,09	6.0 V	---	20	µA
			53	5.5 V	---	20	µA
	I_{CCZ}	V_I = 6.0 V	01,02 03,06,07, 08,09	6.0 V	---	10	µA
			53	5.5 V	---	10	µA
Additional supply current quiescent per input pin (one unit load)	$I_{CC\Delta}$ 9/	V_{IL} = 0.8 V V_{IH} = 2.4 V Test pin at V_{IN} = 2.4 V Other pins at 0.8 V or 5.5 V I_O = 0 V	53	5.5 V		1.5	mA
Maximum clock frequency	f_{MAX}	V_I = 0-V_{CC}, 50% DC C_L = 50 pF ±10% 5/	08,09	4.5 V	23	---	MHz
Control capacitance	C_C	T_C = +25°C	All		---	10	pF
Input/output capacitance	$C_{I/O}$	T_C = +25°C	All		---	20	pF
Power dissipation capacitance	C_{PD} 2/ 3/	T_C = +25°C	01,02, 03,53, 04,05, 06,07, 08,09		--- --- --- --- ---	70 40 120 55	pF
Transition delay times 6/, 7/	t_{THL}, t_{TLH}	V_{CC} = 4.5 V C_L = 50 pF ±10%	All	4.5 V	2	16	ns
Propagation delay time, clock to bus	t_{PLH1}, t_{PHL1}		08	4.5 V	8	51	ns
			09	4.5 V	8	56	ns
Propagation delay time, bus to bus	t_{PLH2}, t_{PHL2}		01,	4.5 V	3	23	ns
			02,	4.5 V	4	23	ns
			03,53	4.5 V	4	33	ns
			06,07,	4.5 V	4	26	ns
			08,09	4.5 V	5	40	ns
Propagation delay time, select (with bus input high) to bus	t_{PLH3}, t_{PHL3}		08,09	4.5 V	10	51	ns

See footnotes at end of table.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified	Device type 2/	V_{CC}	Limits Min	Limits Max	Unit
Propagation delay time, select (with bus input low) to bus	t_{PLH4}, t_{PHL4}	V_{CC} = 4.5 V C_L = 50 pF ±10%	08,09	4.5 V	10	51	ns
Propagation delay time, disabled-to-high level output	t_{PZH1}	V_{CC} = 4.5 V C_L = 50 pF ±10% R_L = 1 kΩ ±5% 6/, 7/	01	4.5 V	4	35	ns
			02	4.5 V	4	36	ns
			03	4.5 V	9	59	ns
			53	4.5 V	6	54	ns
Propagation delay time, disabled-to-high level output, enable to bus	t_{PZH2}		06,07,	4.5 V	9	59	ns
			08,09	4.5 V	6	57	ns
Propagation delay time, disabled-to-high level output direction to bus	t_{PZH3}		08,09	4.5 V	6	57	ns
Propagation delay time, disabled-to-low level output	t_{PZL1}		01	4.5 V	4	35	ns
			02	4.5 V	4	36	ns
			03	4.5 V	9	59	ns
			53	4.5 V	6	54	ns
Propagation delay time, disabled-to-low level, output enable to bus	t_{PZL2}		06,07,	4.5 V	9	59	ns
			08,09	4.5 V	6	57	ns
Propagation delay time, disabled-to-low level, direction to bus	t_{PZL3}		08,09	4.5 V	6	57	ns
Propagation delay time, high-level to disabled output	t_{PHZ1}		01	4.5 V	4	40	ns
			02	4.5 V	4	42	ns
			03	4.5 V	8	54	ns
			53	4.5 V	6	54	ns
Propagation delay time, high-level-to-disabled output enable to bus	t_{PHZ2}		06,07,	4.5 V	8	53	ns
			08,09	4.5 V	6	57	ns
Propagation delay time, high-level-to-disabled, output direction to bus	t_{PHZ3}		08,09	4.5 V	6	57	ns
Propagation delay time, low-level-to-disabled output	t_{PLZ1}		01	4.5 V	4	40	ns
			02	4.5 V	4	42	ns
			03	4.5 V	8	54	ns
			53	4.5 V	6	54	ns

See footnotes at end of table.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified	Device type 2/	V_{CC}	Limits Min	Limits Max	Unit
Propagation delay time, low-level-to-disabled output, enable to bus	t_{PLZ2}	V_{CC} = 4.5 V C_L = 50 pF ±10% R_L = 1 kΩ ±5% 6/, 7/	06,07,	4.5 V	8	53	ns
			08,09	4.5 V	6	57	ns
Propagation delay time, low-level-to-disabled output, direction to bus	t_{PLZ3}		08,09	4.5 V	6	57	ns

1/ Complete terminal conditions shall be as specified in table III.

2/ Power dissipation capacitance (C_{PD}) typically per transceiver.

3/ Guaranteed but not tested.

4/ I_{OZL} = output set to high state. Three state output conditions are required.
I_{OZH} = output set to low state. Three state output conditions are required.

5/ Tested at V_{CC} = 4.5 V at +125°C for sample testing and V_{CC} = 4.5 V at +25°C for screening. Guaranteed at other V_{CC} voltages and temperatures, see table IB and exception in 4.4.1d.

6/ Tested at V_{CC} = 4.5 V at +125°C for sample testing and V_{CC} = 4.5 V at +25°C for screening. Guaranteed at other V_{CC} voltages and temperatures, see table IA and exception in 4.4.1d.

7/ For propagation and transition delay times at V_{CC} = 2.0 V, increase limit by a factor of 5. For propagation and transition delay times at V_{CC} = 6.0 V, decrease limit by a factor of .85.

8/ A minimum limit of 0.4 V applies only when test equipment limitations do not allow the GND pin to be open during testing.

9/ Total supply current = I_{CC} + $I_{CC\Delta}$.

TABLE IA. Calculated dynamic figures at -55/+25 ambient temperature (°C).

V_{CC}	T_A = (°C)	
	+125	-55/+25
2.0 V	5X	5Y
4.5 V	X = 1	0.75X = Y
6.0 V	0.85X	0.85Y

Normalized numbers
(+125°C equals 1)

NOTE: The 2.0 V and 6.0 V numbers are derived from their 4.5 V integer value. Rounding off according 5/4.

TABLE IB. Calculated formula for f_{MAX} at -55/+25 ambient temperature.

V_{CC}	T_A = (°C)	
	+125	-55/+25
2.0 V	0.2X	0.2Y
4.5 V	X = 1	1.33X = Y
6.0 V	1.18X	1.18Y

Normalized numbers
(+125°C equals 1)

NOTE: The 2.0 V and 6.0 V numbers are derived from their 4.5 V integer value. Rounding off according 5/4.

TABLE II. Burn-in and electrical test requirements.

Line no.	Applicable tests and MIL-STD-883 test method	Class S device 1/				Class B device 1/			
		Ref. paragraph	Table III subgroups 2/	Table IV delta limits 3/	Test circuit figure	Ref. paragraph	Table III subgroups 2/	Table IV delta limits 3/	Test circuit figure
1	Interim electrical parameters (method 5004)		1				1		
2	Static burn-in I (method 1015)	4.2b 4.5.2	Req'd		3		Not req'd		
3	Same as line 1		1	Δ					
4	Static burn-in II (method 1015)	4.2b 4.5.2	Req'd		3	4.2b 4.5.2	4/ Req'd		3
5	Same as line 1	4.2d	1*	Δ		4.2d	1* 4/	Δ	
6	Dynamic burn-in (method 1015)	4.2b 4.5.2	Req'd		1/		Not req'd		
7	Same line 1	4.2d	1	Δ					
8	Final electrical parameters (method 5004)		1*,2,3, 7,8,9				1*,2,7,9 4/		
9	Group A test requirements (method 5005)	4.4.1	1,2,3,4, 7,8,9, 10,11			4.4.1	1,2,3,4, 7,8,9, 10,11		
10	Group B end-point electrical parameters (method 5005)	4.4.2	1,2,3,9, 10,11 5/	Δ			1 5/		
11	Group C end-point electrical parameters (method 5005)					4.4.3	1,2	Δ	
12	Group D end-point electrical parameters (method 5005)	4.4.4	1,2,3			4.4.4	1,2		

1/ Blank spaces indicate tests are not applicable.

2/ * indicates PDA applies to subgroup 1 (see 4.2.1).

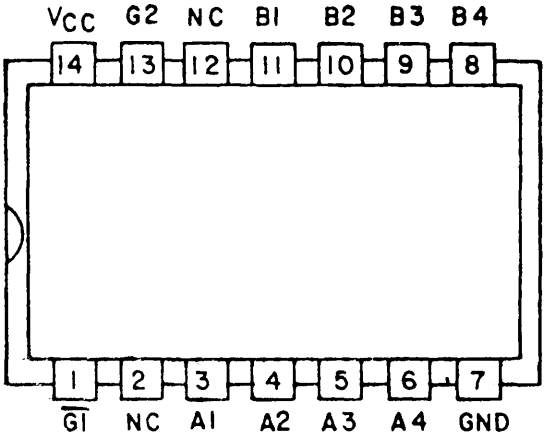
3/ Δ indicates delta limit shall be required only on table III subgroup 1, where specified, and the delta values shall be computed with reference to the previous interim electrical parameters (line 1).

4/ The device manufacturer may at his option either complete subgroup 1 electrical parameter measurements, including delta measurements, within 96 hours after burn-in completion (removal of bias); or may complete subgroup 1 electrical measurements without delta measurements within 24 hours after burn-in completion (removal of bias).

5/ indicates also applies to electrostatic discharge sensitivity tests.

Device types 01 and 02

Case C



Device types 01 and 02

Case 2

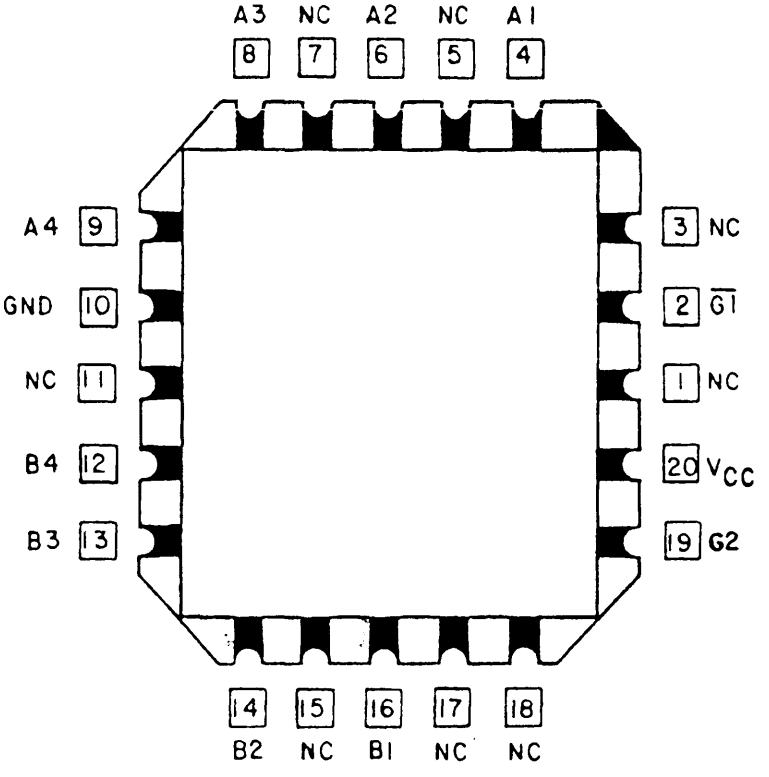
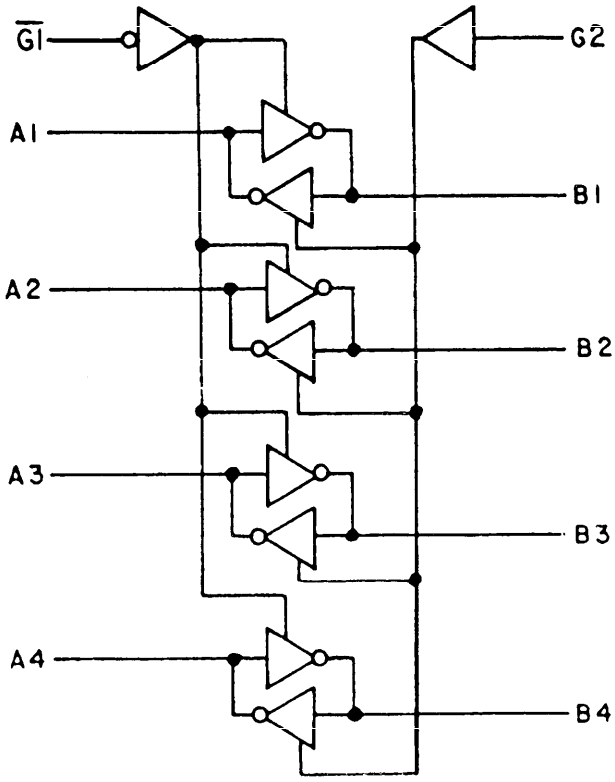


FIGURE 1. Logic diagrams and terminal connections (top view).

Device type 01



Device type 02

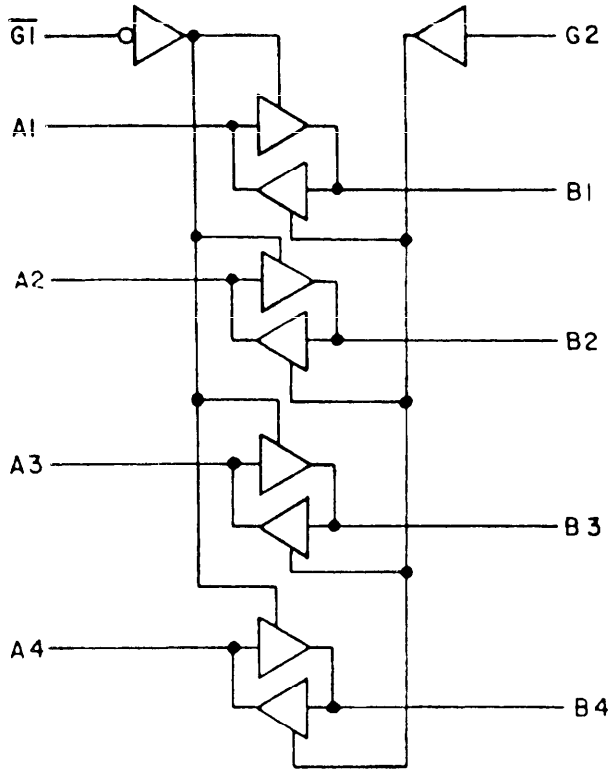
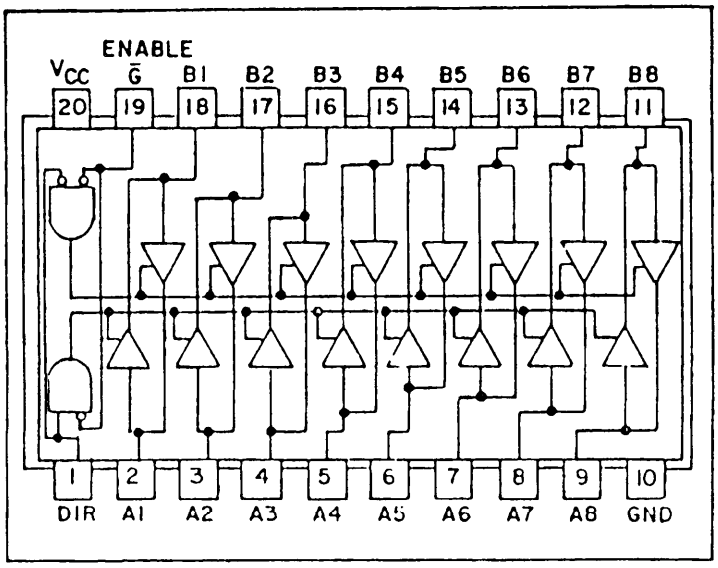


FIGURE 4. Switching time test circuit and waveforms - Continued.

Device types 03 and 53

Cases R



Device types 03 and 53

Case 2

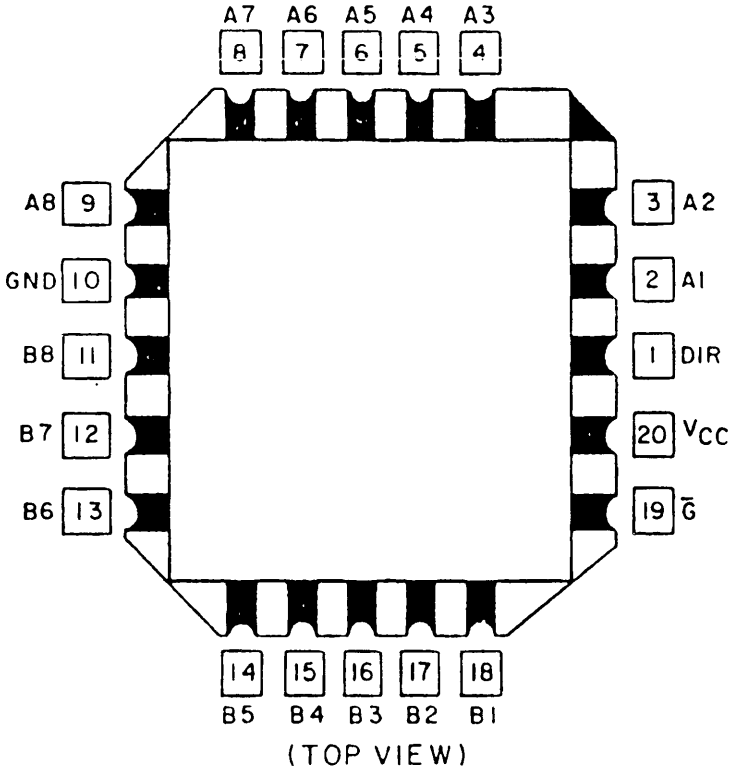
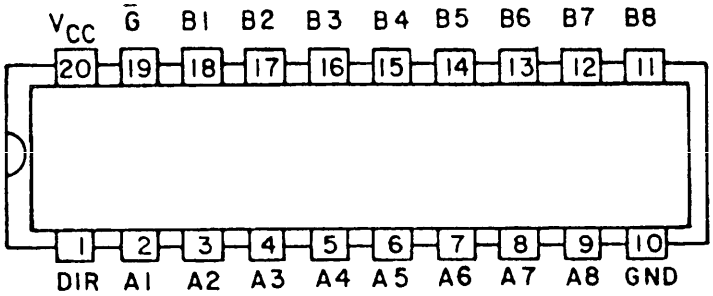


FIGURE 1. Logic diagrams and terminal connections (top view) - Continued.

Device types 06, 07
Case R



Device types 06, 07
Case 2

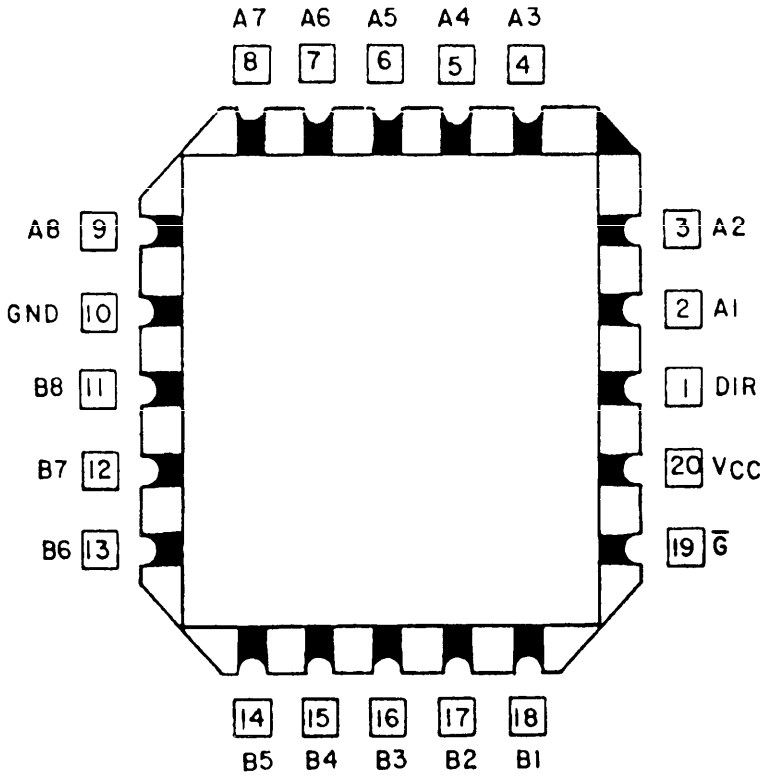
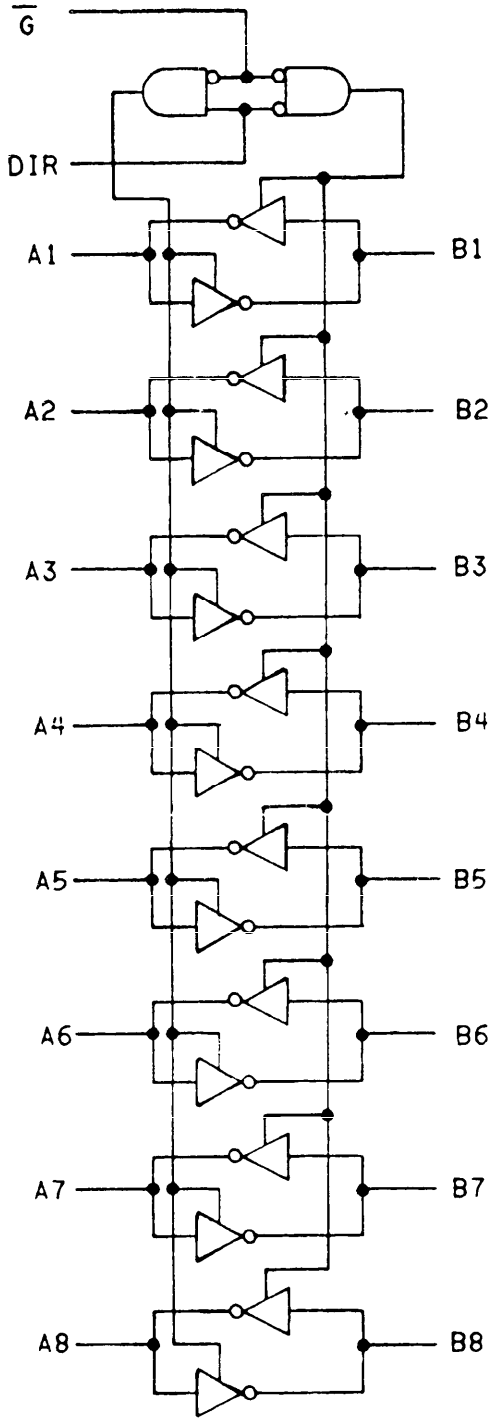


FIGURE 1. Logic diagrams and terminal connections (top view) - Continued.

Device type 06



Device type 07

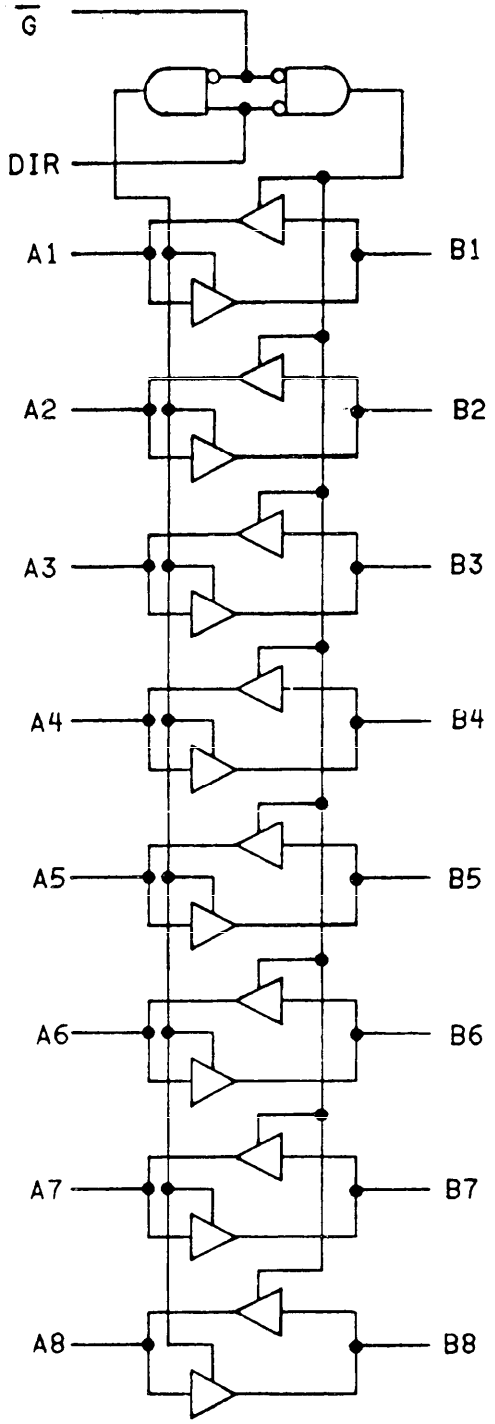
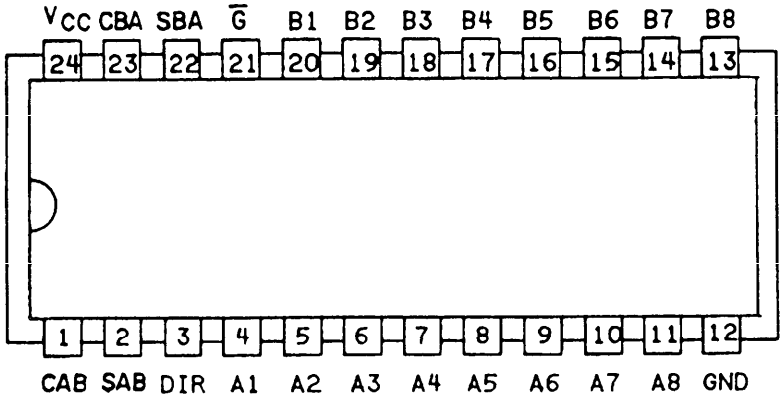


FIGURE 1. Logic diagrams and terminal connections (top view) - Continued.

Device types 08 and 09

(Terminal connections)

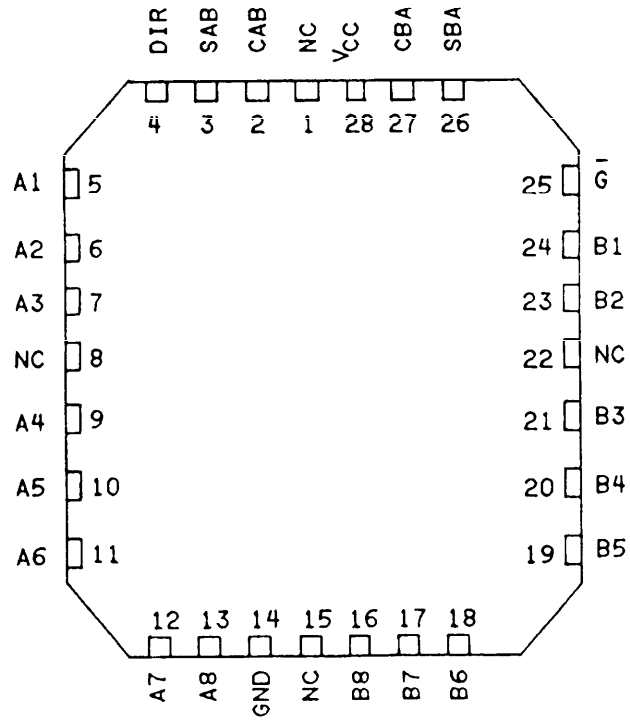
Case L



Device types 08 and 09

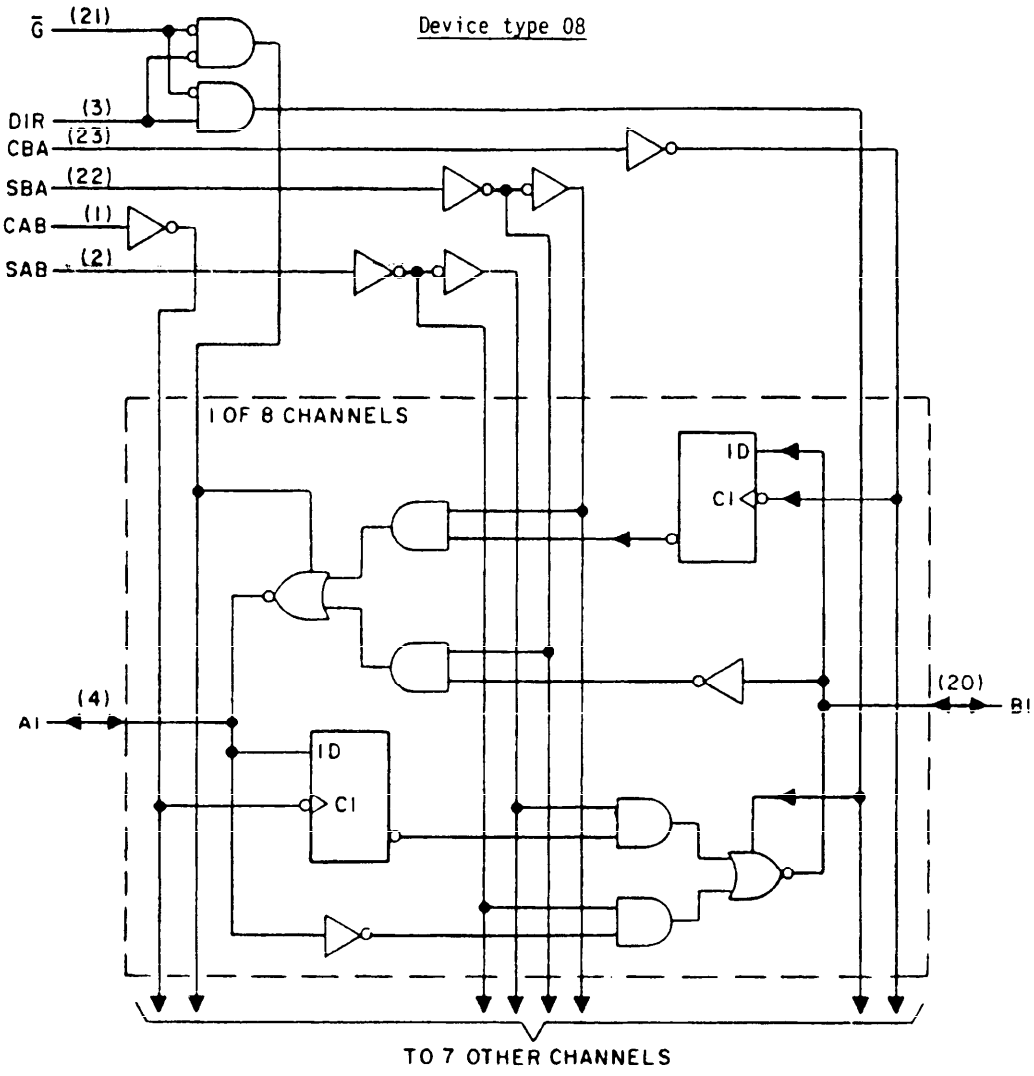
(Top view)

Case 3



NC-NO internal connection

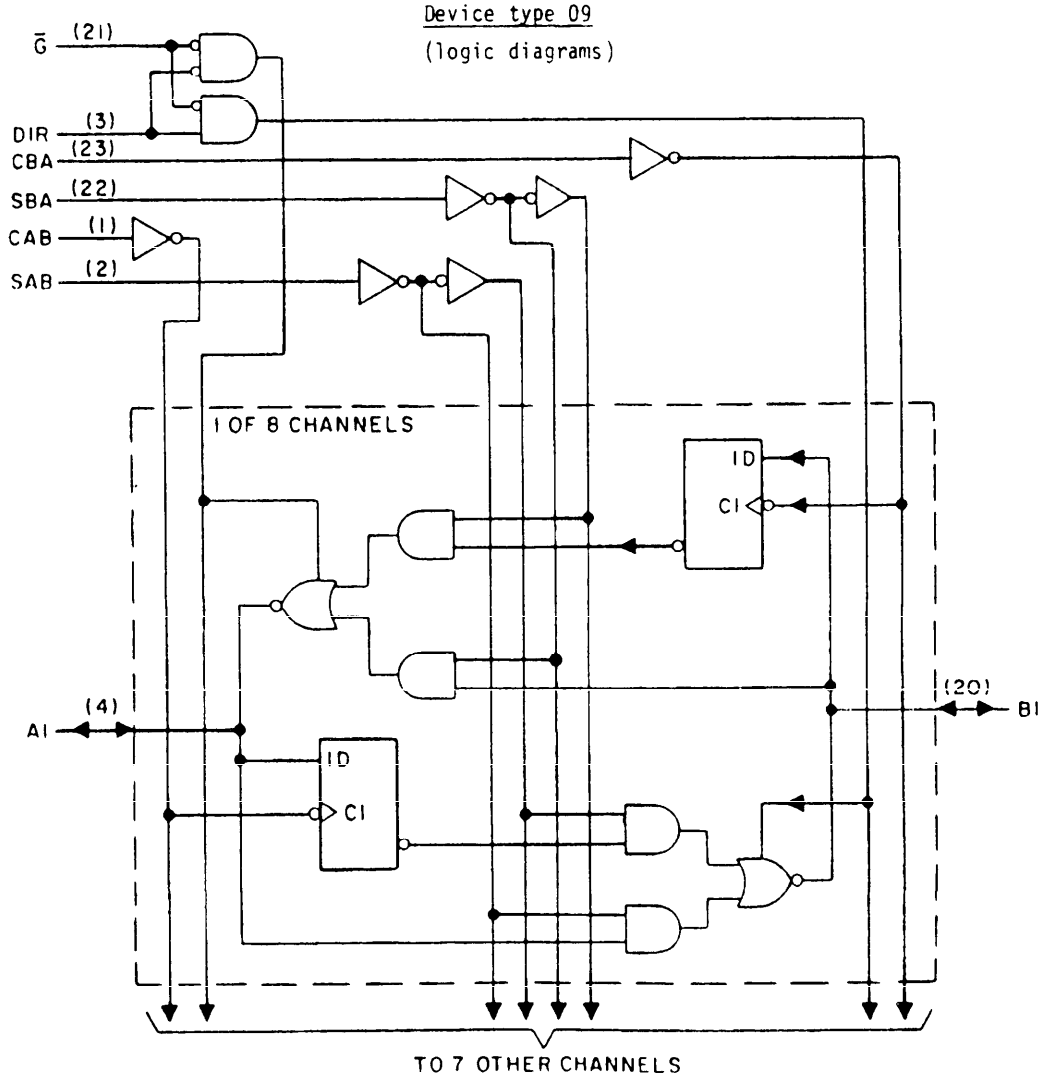
FIGURE 1. Logic diagrams and terminal connections (top view) - Continued.



NOTE: Terminal connections shown in parentheses are for case L.

FIGURE 1. Logic diagrams and terminal connections (top view) - Continued.

Device type 09
(logic diagrams)



NOTE: Terminal connections shown in parentheses are for case L.

FIGURE 1. Logic diagrams and terminal connections (top view) - Continued.

inspections and tests is arbitrary and may be varied to suit the

Device type 01

Inputs		Outputs
$\overline{G}_1$	G_2	
L	L	$\overline{A}$ to B
H	H	$\overline{B}$ to A
H	L	Isolation
L	H	Isolation

H = high level, L = low level

Device type 02

Inputs		Outputs
$\overline{G}AB$	GBA	
L	L	A to B
H	H	B to A
H	L	Isolation
L	H	Isolation

H = high level, L = low level

Device types 03 and 53

Enable $\overline{G}$	Direction control DIR	Operation
L	L	B data to A bus
L	H	A data to B bus
H	X	Isolation

H = high level, L = low level, X = irrelevant

FIGURE 2. Truth tables.

Certified by a VA Inspector. Technician and C

Device type 06

Enable $\overline{G}$	Direction control DIR	Operation
L	L	$\overline{B}$ data to A bus
L	H	$\overline{A}$ data to B bus
H	X	Isolation

H = High level, L = Low level, X = Irrelevant

Device type 07

Enable $\overline{G}$	Direction control DIR	Operation
L	L	B data to A bus
L	H	$\overline{A}$ data to B bus
H	X	Isolation

H = High level, L = Low level, X = Irrelevant

FIGURE 2. Truth tables - Continued.

Device type 08

Inputs						Data I/O*		Operation or function
G	DIR	CAB	CAB	SAB	SBA	A1 thru A8	B1 thru B8	
H	X	H or L	H or L	X	X	Input	Input	Isolation
H	X			X	X			store A and B data
L	L	X	X	X	L	Output	Input	Real time B data to A bus
L	L	X	H or L	X	H			store B and A data
L	H	X	X	L	X	Input	Output	Real time B data to A bus
L	H	H or L	X	H	X			store A and B data
X	X	↑	X	X	X	Input	Not specified	Store A, B undefined
X	X	X	↑	X	X	not specified	input	Store B, A undefined

H = High Level L = Low Level X = Irrelevant ↑ = Low-to-high level transition
*The data output functions may be enabled or disabled by various signals at the G and DIR inputs. Data input functions are always enabled, i.e., data at the bus pins will be stored on every low-to-high transition on the clock inputs.

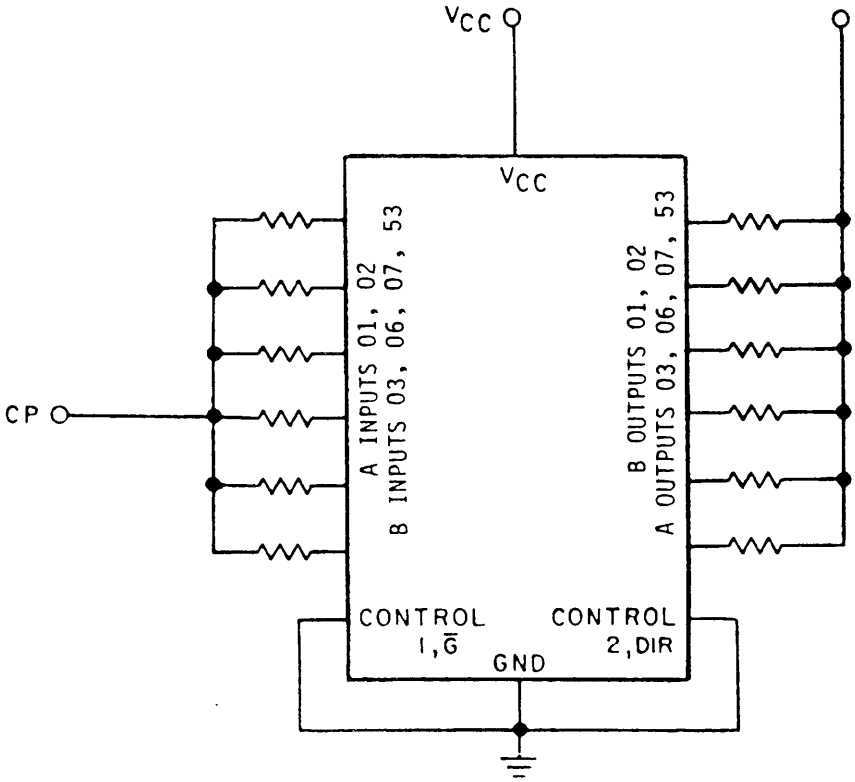
Device type 09

Inputs						Data I/O*		Operation or function
G	DIR	CAB	CAB	SAB	SBA	A1 thru A8	B1 thru B8	
H	X	H or L	H or L	X	X	Input	Input	Isolation
H	X			X	X			store A and B data
L	L	X	X	X	L	Output	Input	Real time B data to A bus
L	L	X	H or L	X	H			store B and A data
L	H	X	X	L	X	Input	Output	Real time B data to A bus
L	H	H or L	X	H	X			store A and B data
X	X	↑	X	X	X	Input	Not specified	Store A, B undefined
X	X	X	↑	X	X	not specified	input	Store B, A undefined

H = High Level L = Low Level X = Irrelevant ↑ = Low-to-high level transition
*The data output functions may be enabled or disabled by various signals at the G and DIR inputs. Data input functions are always enabled, i.e., data at the bus pins will be stored on every low-to-high transition on the clock inputs.

FIGURE 2. Truth tables - Continued.

Device types 01, 02, 03, 06, 07 and 53

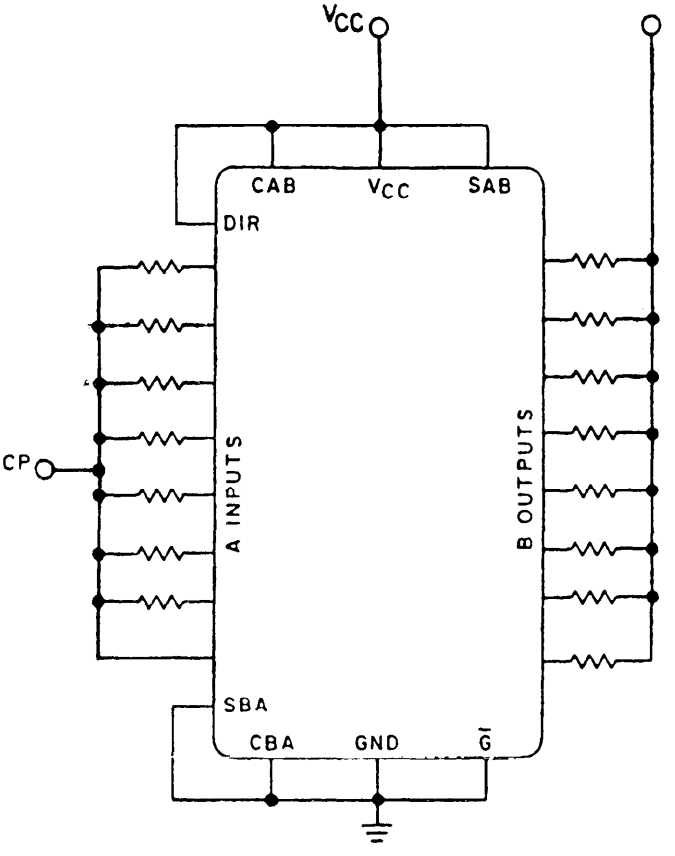


NOTES:

1. For static burn-in I, all inputs shall be connected to GND. Outputs shall be open or connected to $V_{CC}/2$. Resistors are optional on outputs if open. Resistors are required on inputs, and required on outputs connected to $V_{CC}/2$. R = 680Ω to 47 kΩ.
2. For static burn-in II, all inputs shall be connected through the resistors to V_{CC}. Outputs shall be open or connected to $V_{CC}/2$. Resistors are optional on outputs if open. Resistors are required on inputs, and required on outputs connected to $V_{CC}/2$. R = 680Ω to 47 kΩ.
3. For dynamic burn-in, all inputs shall be connected through the resistors in parallel to a common CP. Outputs shall be connected to $V_{CC}/2$ ±.5 V through the resistors. R = 680Ω ±5% for outputs, 680Ω to 47 kΩ for inputs.
4. CP = 25 kHz to 1 MHz square wave; duty cycle = 50 ±15%; V_{IH} = 4.5 V to V_{CC}; V_{IL} = 0 ±.5 V, transition time ≤ 0.5 µs.
5. V_{CC} = 6.0 V ±0.5 V.

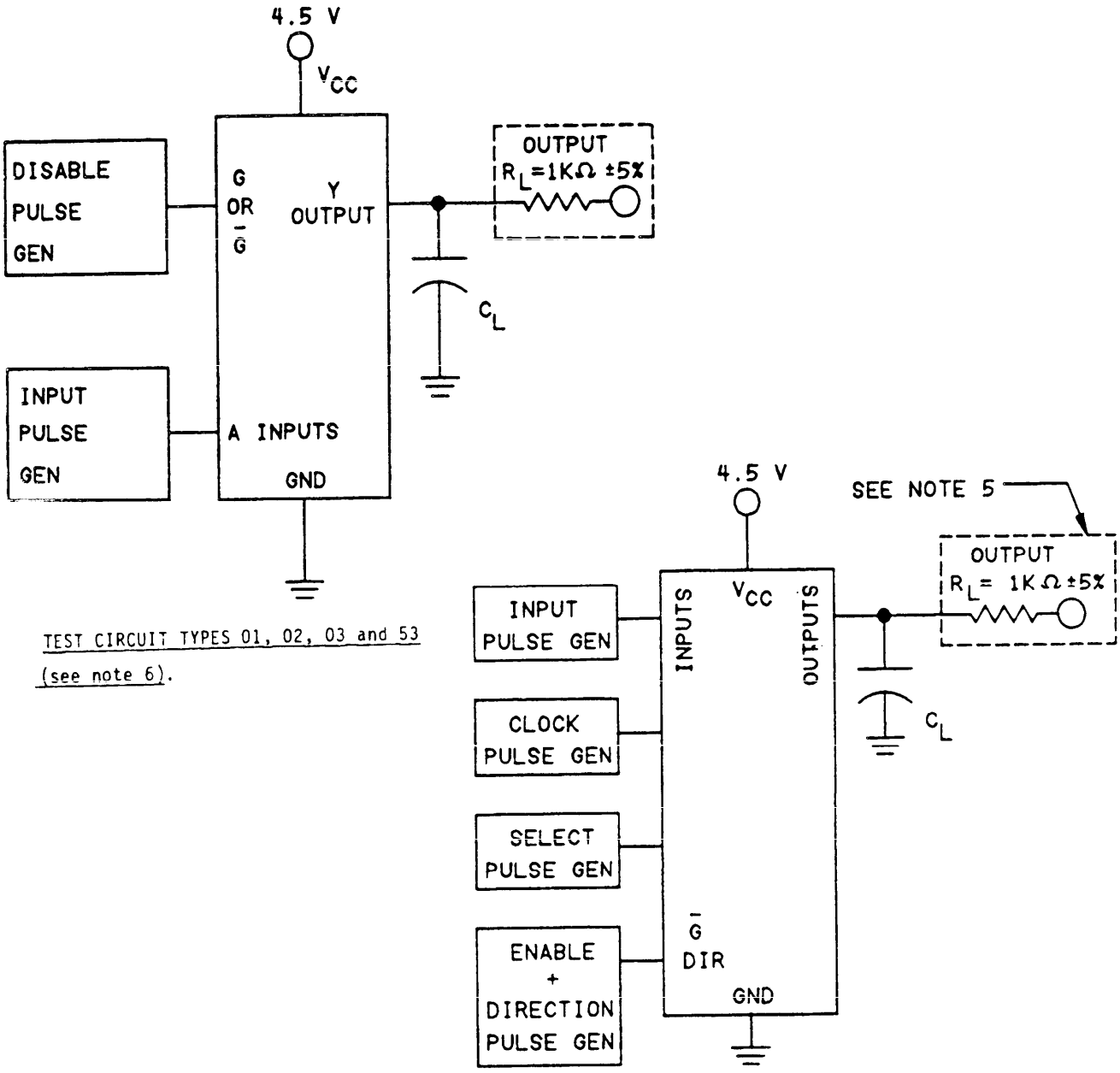
FIGURE 3. Burn-in and life test circuits.

Device types 08 and 09



NOTES:
1. For static burn-in I, all inputs shall be connected to GND. Outputs shall be open or connected to $V_{CC}/2$. Resistors are optional on outputs if open. Resistors are required on inputs, and required on outputs connected to $V_{CC}/2$. R = 680Ω to 47 kΩ.
2. For static burn-in II, all inputs shall be connected through the resistors to V_{CC}. Outputs shall be open or connected to $V_{CC}/2$. Resistors are optional on outputs if open. Resistors are required on inputs, and required on outputs connected to $V_{CC}/2$. R = 680Ω to 47 kΩ.
3. For dynamic burn-in, all inputs shall be connected through the resistors in parallel to a common CP. Outputs shall be connected to $V_{CC}/2$ ±.5 V through the resistors. R = 680Ω ±5% for outputs, 680Ω to 47 kΩ for inputs.
4. CP = 25 kHz to 1 MHz square wave; duty cycle = 50 ±15%; V_{IH} = 4.5 V to V_{CC}; V_{IL} = 0 ±.5 V, transition time ≤ 0.5 µs.
5. V_{CC} = 6.0 V ±0.5 V.

FIGURE 3. Burn-in and life test circuits - Continued.



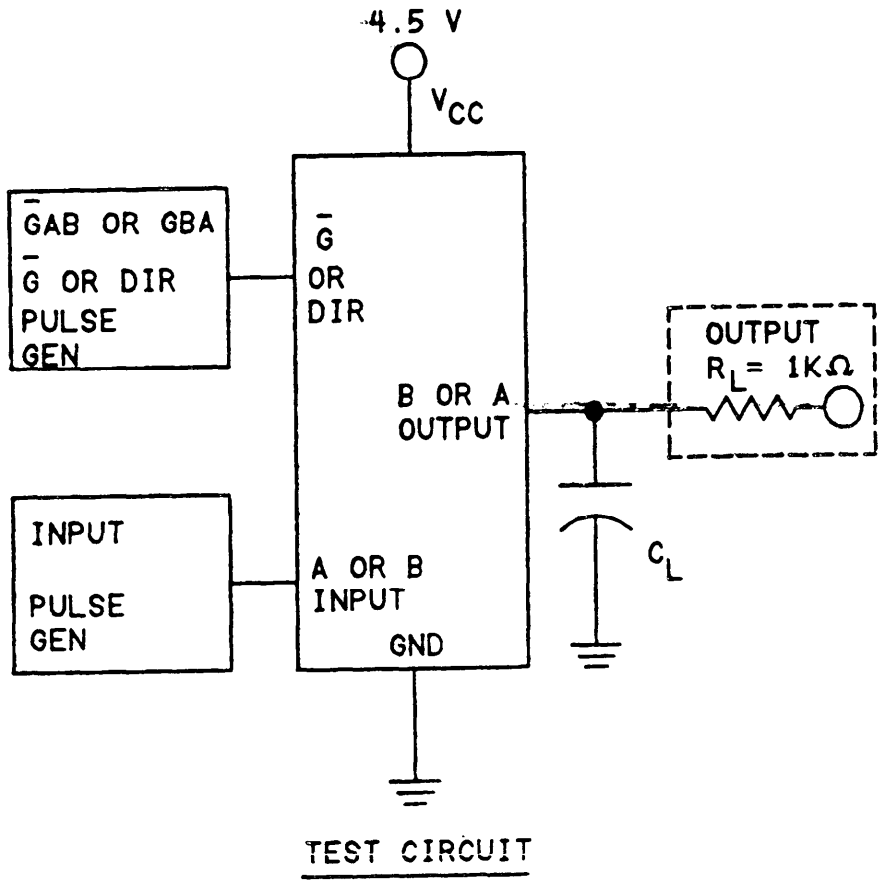
TEST CIRCUIT TYPES 01, 02, 03 and 53 (see note 6).

TEST CIRCUIT TYPES 08 and 09

NOTES:
1. R_L = 1kΩ ±5%.
2. C_L = 50 pF ±10% including probe and jig capacitance.
3. The pulse generators have the following characteristics: V_{gen} = 4.5 V, PRR ≤ 1 MHz, $t_{TLH} \leq 6$ ns, $t_{THL} \leq 6$ ns, Z_{OUT} = 50Ω.
4. Clock pulse characteristics: t_p (CLK) = 18 ns, t_{setup} = 20 ns, t_{hold} = 5 ns.
5. Output connected to V_{CC} for t_{PLZ} and t_{PZL}; and to GND for t_{PHZ} and t_{PZH}, see figure 4.
6. The input signal for the 53 device type (HCT) will be 0-3 volts; however, the 50% V_{CC} measure point is 1.3 volts for input and output signals. The 10% V_{CC} and 90% V_{CC} points are, 3 V and 2.7 V, respectively.

FIGURE 4. Switching time test circuit and waveforms.

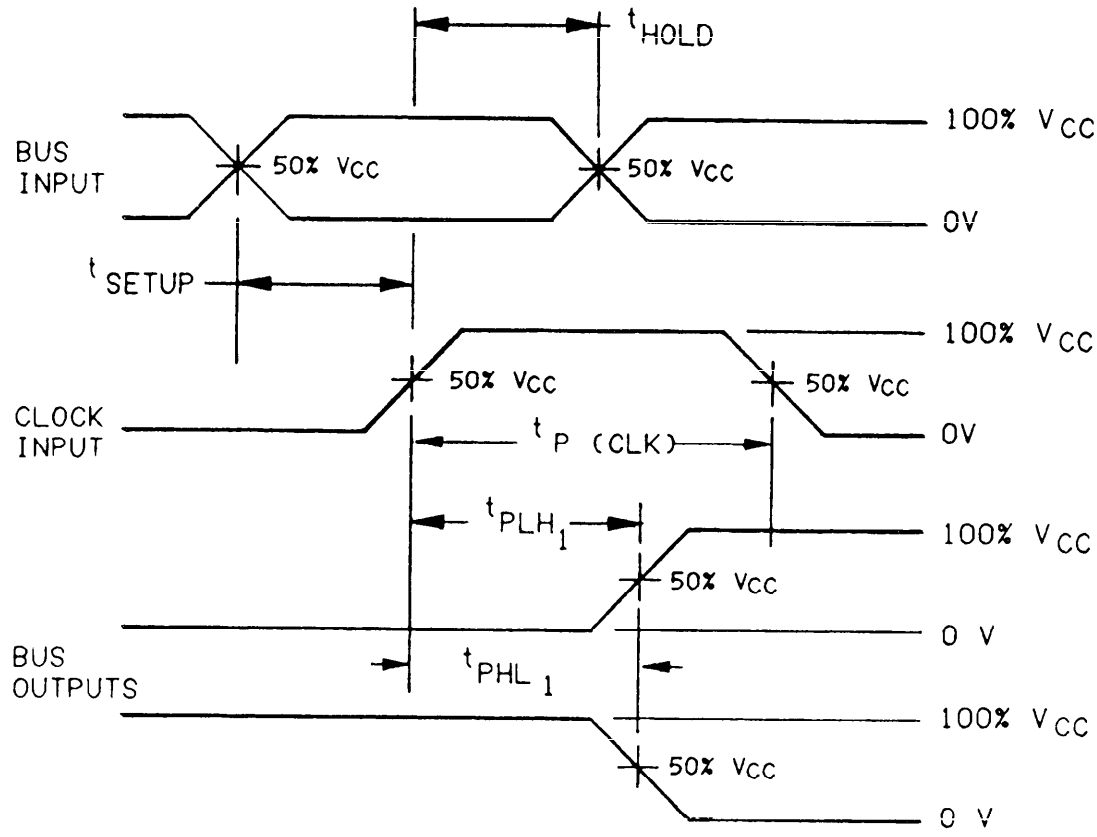
Device types 06 and 07



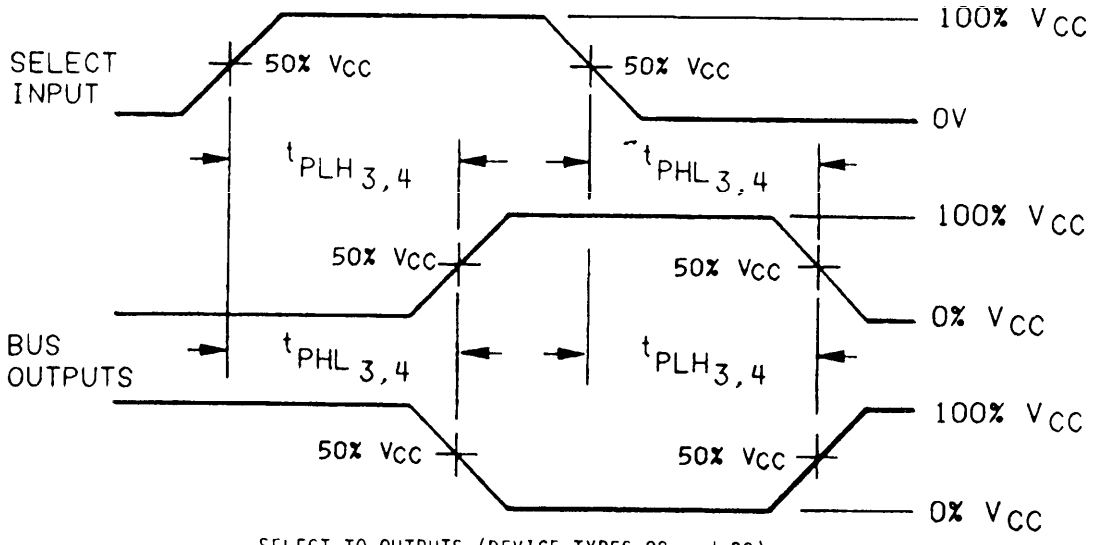
TEST CIRCUIT

NOTES:
1. The input pulse shall have the following characteristics:
   $t_f = t_r \leq 6\mu s$, Vgen = 4.5 V
2. R_L = 1 kΩ ±5%.
3. C_L = 50 pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
4. Voltage measurements are to be made with respect to network ground terminal.
5. Output connected to V_{CC} for t_{PLZ} and t_{PZL}; and to GND for t_{PHZ} and t_{PZH}, see figure 4A.

FIGURE 4. Switching time test circuit and waveforms - Continued.



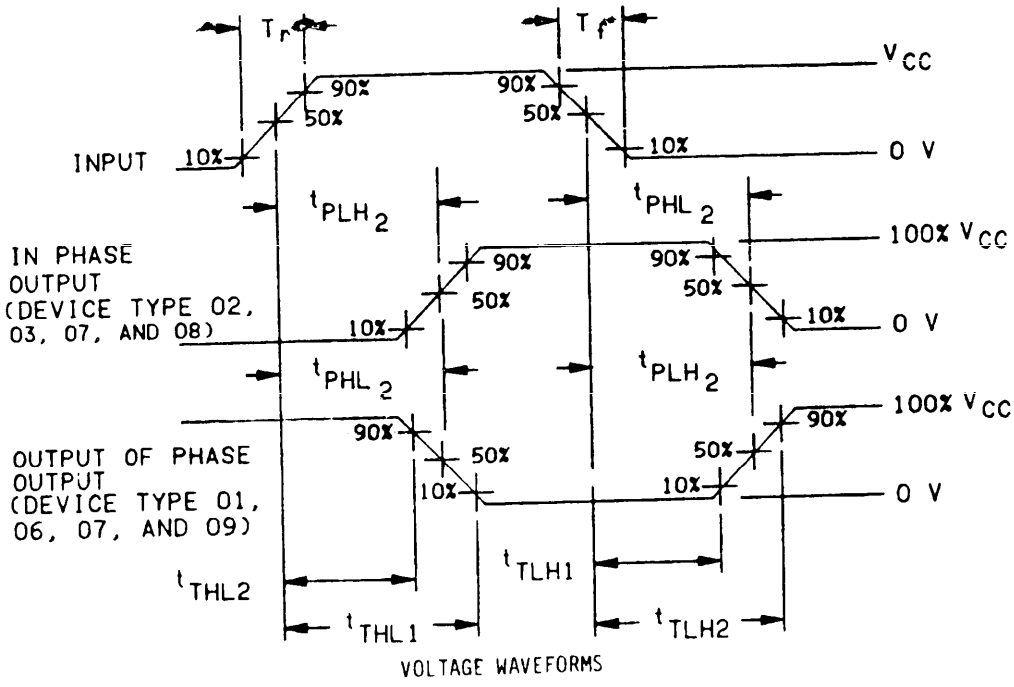
CLOCK TO OUTPUT (DEVICE TYPES 08 and 09)

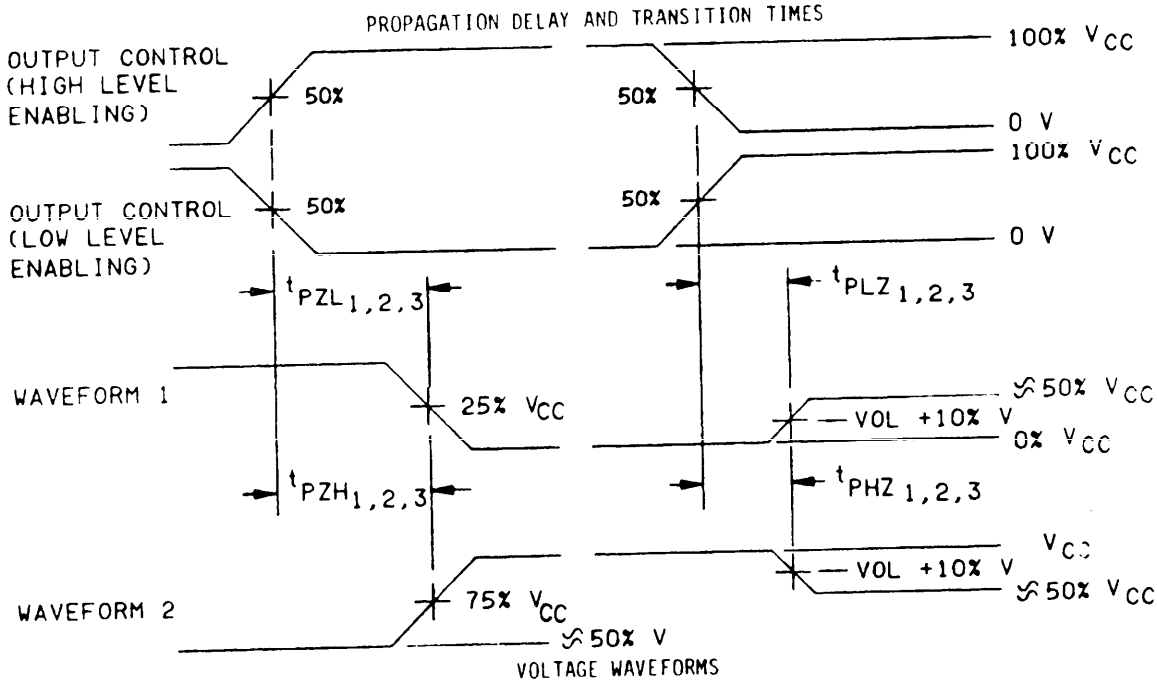


SELECT TO OUTPUTS (DEVICE TYPES 08 and 09)

NOTE: $t_{HOLD} \leq 5$ ns, $t_{setup} \leq 30$ ns, are tp (clock) ≤ 30 ns.

FIGURE 4. Switching time test circuit and waveforms - Continued.





ENABLE AND DISABLE TIMES, THREE-STATE OUTPUTS, ALL DEVICE TYPES

NOTES:
1. Input pulse shall have following characteristics: $t_r = t_f \leq 6$ ns; PRR $\leq$ 1 MHz; duty cycle = 50%.
2. All unused inputs are tied to V_{CC}.
3. $t_{THL1} - t_{THL2} = t_{THL}$; $t_{TLH2} - t_{TLH1} = t_{TLH}$

FIGURE 4. Switching time test circuit and waveforms - Continued.

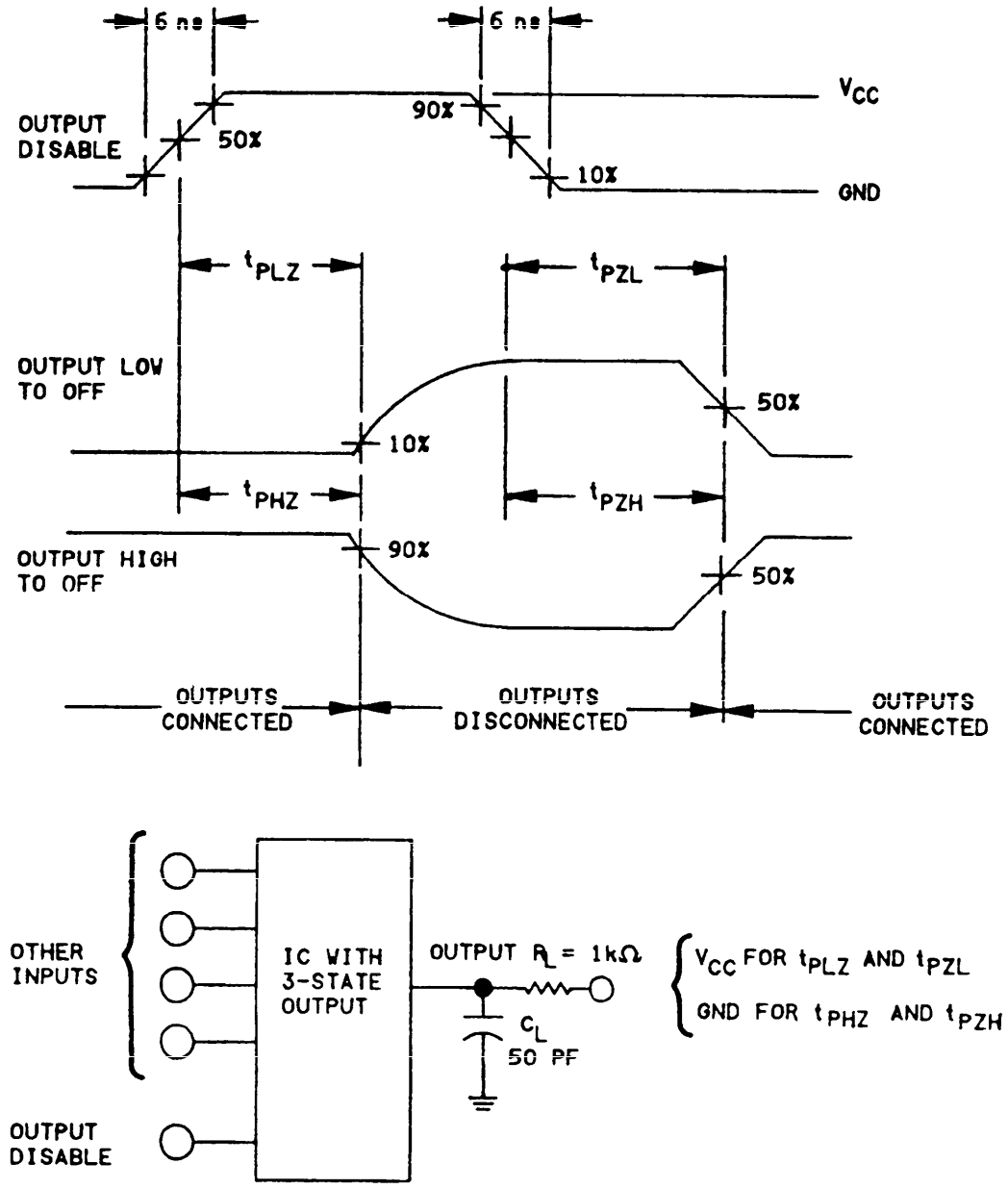


FIGURE 4A. Three-state propagation delay wave shapes and test circuit.

TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Case 2 / Case C	Terminal conditions 1/ 2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Test limits: Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		Test no.	1 GI	2 NC	3 A1	4 A2	5 A3	6 A4	7 GND	8 B4	9 B3	10 B2	11 B1	12 NC	13 G2	14 V_{CC}								
V_{IC} (pos)		1	1 mA						1/							GND	GI	0.4	1.5					V
		2							I/						1 mA	GND	G2	0.4	1.5					
V_{IC} (neg)		3	-1 mA						GND							1/	GI		-1.5					
		4													-1 mA	I/	G2		-1.5					
I_{CC}	3005	5	GND		6 V	GND	6 V	GND							GND	6 V	V_{CC}		0.2		20			μA
		6	GND		6 V	6 V	6 V	6 V							GND									
		7	6 V							GND	6 V	GND	6 V		6 V									
		8	6 V							6 V	6 V	6 V	6 V		6 V									
I_{CCZ}		9	6 V												GND						10			
V_{OH3}	3006	10	4.2 V		-20 μA							1.2 V	1.2 V		4.2 V		A1	5.95		5.95		5.95		V
		11				-20 mA					1.2 V						A2							
		12					-20 mA										A3							
		13						-20 mA		1.2 V							A4							
		14	1.2 V		1.2 V								-20 mA		1.2 V		B1							
		15				1.2 V						-20 mA					B2							
		16					1.2 V				-20 mA						B3							
		17						1.2 V		-20 mA							B4							
V_{OH5}	3006	18	4.2 V		-7.8 mA							1.2 V	1.2 V		4.2 V		A1	5.48		5.2		5.48		
		19				-7.8 mA					1.2 V						A2							
		20					-7.8 mA										A3							
		21						-7.8 mA		1.2 V							A4							
		22	1.2 V		1.2 V								-7.8 mA		1.2 V		B1							
		23				1.2 V						-7.8 mA					B2							
		24					1.2 V				-7.8 mA						B3							
		25						1.2 V		-7.8 mA							B4							
V_{OL3}	3007	26	4.2 V		20 μA							4.2 V	4.2 V		4.2 V	6 V	A1		.05		.05		.05	
		27				20 μA					4.2 V						A2							
		28					20 μA										A3							
		29						20 μA		4.2 V							A4							
		30	1.2 V		4.2 V							20 μA	20 μA		1.2 V		B1							
		31				4.2 V					20 μA						B2							
		32					4.2 V										B3							
		33						4.2 V		20 μA							B4							
V_{OL5}	3007	34	4.2 V		7.8 mA							4.2 V	4.2 V		4.2 V		A1		0.26		0.4		0.26	
		35				7.8 mA					4.2 V						A2							
		36					7.8 mA										A3							
		37						7.8 mA		4.2 V							A4							
		38	1.2 V		4.2 V							7.8 mA	7.8 mA		1.2 V		B1							
		39				4.2 V					7.8 mA						B2							
		40					4.2 V										B3							
		41						4.2 V		7.8 mA							B4							
I_{OS4}	3011	42	4 V		GND							GND	GND		4 V	4 V	A1	-10	-135	-10	-135	-10	-135	mA
		43				GND					GND						A2							
		44					GND										A3							
		45						GND		GND							A4							
		46	GND		GND							GND	GND		GND		B1							
		47				GND					GND						B2							
		48					GND										B3							
		49						GND		GND							B4							

See footnotes at end of table.

TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Case 2 / Case C	Terminal conditions 1/														Measured terminal	Test Limits						Unit
			2 / 1	3 / 2	4 / 3	6 / 4	8 / 5	9 / 6	10 / 7	12 / 8	13 / 9	14 / 10	16 / 11	18 / 12	19 / 13	20 / 14		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{G1}$	NC	A1	A2	A3	A4	GND	B4	B3	B2	B1	NC	G2	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{IH}/ I_{OZH}		50	2/		0 V				GND				2/		4.2 V	6 V	A1		.1		2.0			µA
		51	"			6 V			"			2/			"	"	A2		"		"			"
		52	"				6 V		"		2/				"	"	A3		"		"			"
		53	"					6 V	"	2/					"	"	A4		"		"			"
		54	1.2 V		2/				"				6 V		2/	"	B1		"		"			"
		55	"			2/			"			6 V			"	"	B2		"		"			"
		56	"				2/		"		6 V				"	"	B3		"		"			"
		57	"					2/	"	6 V					"	"	B4		"		"			"
I_{IL}/ I_{OZL}		58	2/		GND				"				2/		4.2 V	"	A1		-.1		-2.0			"
		59	"			GND			"			2/			"	"	A2		"		"			"
		60	"				GND		"		2/				"	"	A3		"		"			"
		61	"					GND	"	2/					"	"	A4		"		"			"
		62	1.2 V		2/				"				GND		2/	"	B1		"		"			"
		63	"			2/			"			GND			"	"	B2		"		"			"
		64	"				2/		"		GND				"	"	B3		"		"			"
		65	"					2/	"	GND					"	"	B4		"		"			"
I_{IH}	3010	66	6 V						"							"	$\overline{G1}$		.05		.1			"
		67							"						6 V	"	G2		"		"			"
I_{IL}	3009	68	GND						"							"	$\overline{G1}$		-.05		-.1			"
		69							"						GND	"	G2		"		"			"

Symbol	MIL-STD-883 method	Case 2 / Case C	Terminal conditions 1/														Measured terminal	Test Limits		Unit
			2 / 1	3 / 2	4 / 3	6 / 4	8 / 5	9 / 6	10 / 7	12 / 8	13 / 9	14 / 10	16 / 11	18 / 12	19 / 13	20 / 14		Subgroup 4 T_C = +25°C		
		Test no.	$\overline{G1}$	NC	A1	A2	A3	A4	GND	B4	B3	B2	B1	NC	G2	V_{CC}		Min	Max	
$C_{I/O}$ 7/		70	GND		3/				"						6 V	6 V	A1		20	"
		71	"			3/			"						"	"	A2		"	"
		72	"				3/		"						"	"	A3		"	"
		73	"					3/	"						"	"	A4		"	"
		74	6 V						"				3/		GND	"	B1		"	"
		75	"						"			3/			"	"	B2		"	"
		76	"						"		3/				"	"	B3		"	"
		77	"						"	3/					"	"	B4		"	"
C_C		78	3/						"							GND	$\overline{G1}$		10	"
		79							"						3/	GND	G2		10	"

See footnotes at end of table.

TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Case 2 / Case C / Test no.	Terminal conditions 1/ 2 (1) $\overline{GT}$	3 (2) NC	4 (3) A1	6 (4) A2	8 (5) A3	9 (6) A4	10 (7) GND	12 (8) B4	13 (9) B3	14 (10) B2	16 (11) B1	18 (12) NC	19 (13) G2	20 (14) V_{CC}	Measured terminal	Test limits: Subgroup 7 T_C = +25°C 4/ Min	Subgroup 7 Max	Subgroup 8 T_C = +125°C Min	Subgroup 8 Max	Subgroup 8 T_C = -55°C 4/ Min	Subgroup 8 Max	Unit
Truth table tests 5/		80	B		A	A	A	A	GND	L	L	L	L		B	4.5 V	6/							
		81	B		B	B	B	B	"	H	H	H	H		B	"	"							
		82	A		L	L	L	L	"	A	A	A	A		A	"	-							
		83	A		H	H	H	H	"	B	B	B	B		A	"	-							
		84	A		Z	Z	Z	Z	"	Z	Z	Z	Z		B	"	7/							
		85	B		Z	Z	Z	Z	"	Z	Z	Z	Z		A	"	7/							

Symbol	MIL-STD-883 method	Case 2 / Case C / Test no.	Terminal conditions 1/ 2 (1) $\overline{GT}$	3 (2) NC	4 (3) A1	6 (4) A2	8 (5) A3	9 (6) A4	10 (7) GND	12 (8) B4	13 (9) B3	14 (10) B2	16 (11) B1	18 (12) NC	19 (13) G2	20 (14) V_{CC}	Measured terminal	Test limits: Subgroup 9 T_C = +25°C 4/ Min	Subgroup 9 Max	Subgroup 10 T_C = +125°C Min	Subgroup 10 Max	Subgroup 11 T_C = -55°C 4/ Min	Subgroup 11 Max	Unit
t_{THL}	3004 (Fig. 4)	86	4.5 V		OUT				GND			IN	IN		4.5 V	4.5 V	A1	2	12	2	16	2	12	ns
		87	"			OUT			"		IN				"	"	A2	"	"	"	"	"	"	"
		88	"				OUT		"						"	"	A3	"	"	"	"	"	"	"
		89	"					OUT	"	IN					"	"	A4	"	"	"	"	"	"	"
		90	GND		IN				"				OUT		GND	"	B1	"	"	"	"	"	"	"
		91	"			IN			"			OUT			"	"	B2	"	"	"	"	"	"	"
		92	"				IN		"		OUT				"	"	B3	"	"	"	"	"	"	"
		93	"					IN	"	OUT					"	"	B4	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 4)	94 to 101	Same tests, terminal conditions, and limits as specified above for t_{THL}.															2	"	2	"	2	"	"
t_{PHL2}	3003 (Fig. 4)	102	GND		IN				GND			OUT	OUT		GND	4.5 V	A1 to B1	3	17	3	23	3	17	"
		103	"			IN			"		OUT				"	"	A2 to B2	"	"	"	"	"	"	"
		104	"				IN		"						"	"	A3 to B3	"	"	"	"	"	"	"
		105	"					IN	"	OUT					"	"	A4 to B4	"	"	"	"	"	"	"
		106	4.5 V		OUT				"			OUT	IN		4.5 V	"	B1 to A1	"	"	"	"	"	"	"
		107	"			OUT			"						"	"	B2 to A2	"	"	"	"	"	"	"
		108	"				OUT		"		IN				"	"	B3 to A3	"	"	"	"	"	"	"
		109	"					IN	"	OUT					"	"	B4 to A4	"	"	"	"	"	"	"
t_{PLH2}	3003 (Fig. 4)	110 to 117	Same terminal conditions and terminals as specified above for t_{PHL2}.															3	"	3	"	3	"	"
t_{PZH1} 10/	3003 (Fig. 4)	118	IN		GND				GND			OUT	OUT		GND	4.5 V	$\overline{GT}$ to B1	4	26	4	35	4	26	"
		119	"			GND			"		OUT				"	"	$\overline{GT}$ to B2	"	"	"	"	"	"	"
		120	"				GND		"						"	"	$\overline{GT}$ to B3	"	"	"	"	"	"	"
		121	"					GND	"	OUT					"	"	$\overline{GT}$ to B4	"	"	"	"	"	"	"
		122	4.5 V		OUT				"			GND	GND		IN	"	G2 to A1	"	"	"	"	"	"	"
		123	"			OUT			"						"	"	G2 to B2	"	"	"	"	"	"	"
		124	"				OUT		"		GND				"	"	G2 to B3	"	"	"	"	"	"	"
		125	"					GND	"	OUT					"	"	G2 to B4	"	"	"	"	"	"	"
t_{PZL1} 9/	3003 (Fig. 4)	126	IN		4.5 V				"			OUT	OUT		GND	"	$\overline{GT}$ to B1	"	"	"	"	"	"	"
		127	"			4.5 V			"		OUT				"	"	$\overline{GT}$ to B2	"	"	"	"	"	"	"
		128	"				4.5 V		"						"	"	$\overline{GT}$ to B3	"	"	"	"	"	"	"
		129	"					4.5 V	"	OUT					"	"	$\overline{GT}$ to B4	"	"	"	"	"	"	"
		130	4.5 V		OUT				"			4.5 V	4.5 V		IN	"	G2 to A1	"	"	"	"	"	"	"
		131	"			OUT			"						"	"	G2 to B2	"	"	"	"	"	"	"
		132	"				OUT		"		4.5 V				"	"	G2 to B3	"	"	"	"	"	"	"
		133	"					4.5 V	"	4.5 V					"	"	G2 to B4	"	"	"	"	"	"	"

See footnotes at end of table.

TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Case 2 / Case C	Terminal conditions 1/ 2 / 1	3 / 2	4 / 3	6 / 4	8 / 5	9 / 6	10 / 7	12 / 8	13 / 9	14 / 10	16 / 11	18 / 12	19 / 13	20 / 14	Measured terminal	Test limits: Subgroup 9 T_C = +25°C 4/ Min	Subgroup 9 Max	Subgroup 10 T_C = +125°C Min	Subgroup 10 Max	Subgroup 11 T_C = -55°C 4/ Min	Subgroup 11 Max	Unit
		Test no.	$\overline{GI}$	NC	A1	A2	A3	A4	GND	B4	B3	B2	B1	NC	G2	V_{CC}								
t_{PHZ1} 10/	3003 (Fig. 4)	134	IN		GND				GND				OUT		GND	4.5 V	$\overline{GI}$ to B1	4	30	4	40	4	30	ns
		135	"			GND			"			OUT			"	"	$\overline{GI}$ to B2	"	"	"	"	"	"	"
		136	"				GND		"		OUT				"	"	$\overline{GI}$ to B3	"	"	"	"	"	"	"
		137	"					GND	"	OUT					"	"	$\overline{GI}$ to B4	"	"	"	"	"	"	"
		138	4.5 V		OUT				"				GND		IN	"	G2 to A1	"	"	"	"	"	"	"
		139	"			OUT			"			GND			"	"	G2 to B2	"	"	"	"	"	"	"
		140	"				OUT		"		GND				"	"	G2 to B3	"	"	"	"	"	"	"
		141	"					OUT	"	GND					"	"	G2 to B4	"	"	"	"	"	"	"
t_{PLZ1} 9/	3003 (Fig. 4)	142	IN		4.5 V				"				OUT		GND	4.5 V	$\overline{GI}$ to B1	"	"	"	"	"	"	"
		143	"			4.5 V			"			OUT			"	"	$\overline{GI}$ to B2	"	"	"	"	"	"	"
		144	"				4.5 V		"		OUT				"	"	$\overline{GI}$ to B3	"	"	"	"	"	"	"
		145	"					4.5 V	"	OUT					"	"	$\overline{GI}$ to B4	"	"	"	"	"	"	"
		146	4.5 V		OUT				"				4.5 V		IN	"	G2 to A1	"	"	"	"	"	"	"
		147	"			OUT			"			4.5 V			"	"	G2 to B2	"	"	"	"	"	"	"
		148	"				OUT		"		4.5 V				"	"	G2 to B3	"	"	"	"	"	"	"
		149	"					OUT	"	4.5 V					"	"	G2 to B4	"	"	"	"	"	"	"

See footnotes at end of table.

TABLE III. Group A inspection for device type 02.

Symbol	MIL-STD-883 method	Case 2 / Case C / Test no.	Terminal conditions 1/ 2 GT	3 NC	4 A1	6 A2	8 A3	9 A4	10 GND	12 B4	13 B3	14 B2	16 B1	18 NC	19 G2	20 V_{CC}	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Subgroup 1 Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 Max	Unit
V_{IC} (pos)		1	1 mA						1/							GND	GT	0.4	1.5					V
		2							I/						1 mA	GND	G2	0.4	1.5					"
V_{IC} (neg)		3	-1 mA						GND							1/	GT		-1.5					"
		4							"						-1 mA	I/	G2		-1.5					"
I_{CC}	3005	5	GND		6 V	GND	6 V	GND	"						GND	6 V	V_{CC}		.2		20.0			μA
		6	GND		6 V	6 V	6 V	6 V	"						GND	"	"		"		"			"
		7	6 V						"	GND	6 V	GND	6 V		6 V	"	"		"		"			"
		8	6 V						"	6 V	6 V	6 V	6 V		6 V	"	"		"		"			"
I_{CCZ}		9	6 V						"						GND	"	"		"		10.0			"
V_{OH3}	3006	10	4.2 V		-20 μA				"				1.2 V		4.2 V	"	A1	5.95		5.95		5.95		V
		11	"			-20 μA			"			1.2 V			"	"	A2	"		"		"		"
		12	"				-20 μA		"		1.2 V				"	"	A3	"		"		"		"
		13	"					-20 μA	"	1.2 V					"	"	A4	"		"		"		"
		14	1.2 V		1.2 V				"				-20 μA		1.2 V	"	B1	"		"		"		"
		15	"			1.2 V			"			-20 μA			"	"	B2	"		"		"		"
		16	"				1.2 V		"		-20 μA				"	"	B3	"		"		"		"
		17	"					1.2 V	"	-20 μA					"	"	B4	"		"		"		"
V_{OH5}	3006	18	4.2 V		-7.8 mA				"				1.2 V		4.2 V	"	A1	5.48		5.2		5.48		"
		19	"			-7.8 mA			"			1.2 V			"	"	A2	"		"		"		"
		20	"				-7.8 mA		"		1.2 V				"	"	A3	"		"		"		"
		21	"					-7.8 mA	"	1.2 V					"	"	A4	"		"		"		"
		22	1.2 V		1.2 V				"				-7.8 mA		1.2 V	"	B1	"		"		"		"
		23	"			1.2 V			"			-7.8 mA			"	"	B2	"		"		"		"
		24	"				1.2 V		"		-7.8 mA				"	"	B3	"		"		"		"
		25	"					1.2 V	"	-7.8 mA					"	"	B4	"		"		"		"
V_{OL3}	3007	26	4.2 V		20 μA				"				4.2 V		4.2 V	6 V	A1		.05		.05		.05	"
		27	"			20 μA			"			4.2 V			"	"	A2		"		"		"	"
		28	"				20 μA		"		4.2 V				"	"	A3		"		"		"	"
		29	"					20 μA	"	4.2 V					"	"	A4		"		"		"	"
		30	1.2 V		4.2 V				"				20 μA		1.2 V	"	B1		"		"		"	"
		31	"			4.2 V			"			20 μA			"	"	B2		"		"		"	"
		32	"				4.2 V		"		20 μA				"	"	B3		"		"		"	"
		33	"					4.2 V	"	20 μA					"	"	B4		"		"		"	"
V_{OL5}	3007	34	4.2 V		7.8 mA				"				4.2 V		4.2 V	"	A1		0.26		0.4		0.26	"
		35	"			7.8 mA			"			4.2 V			"	"	A2		"		"		"	"
		36	"				7.8 mA		"		4.2 V				"	"	A3		"		"		"	"
		37	"					7.8 mA	"	4.2 V					"	"	A4		"		"		"	"
		38	1.2 V		4.2 V				"				7.8 mA		1.2 V	"	B1		"		"		"	"
		39	"			4.2 V			"			7.8 mA			"	"	B2		"		"		"	"
		40	"				4.2 V		"		7.8 mA				"	"	B3		"		"		"	"
		41	"					4.2 V	"	7.8 mA					"	"	B4		"		"		"	"
I_{OS4}	3011	42	4 V		GND				"				4 V		4 V	4 V	A1	-10	-135	-10	-135	-10	-135	mA
		43	"			GND			"			4 V			"	"	A2	"	"	"	"	"	"	"
		44	"				GND		"		4 V				"	"	A3	"	"	"	"	"	"	"
		45	"					GND	"	4 V					"	"	A4	"	"	"	"	"	"	"
		46	GND		4 V				"				GND		GND	"	B1	"	"	"	"	"	"	"
		47	"			4 V			"			GND			"	"	B2	"	"	"	"	"	"	"
		48	"				4 V		"		GND				"	"	B3	"	"	"	"	"	"	"
		49	"					4 V	"	GND					"	"	B4	"	"	"	"	"	"	"

See footnotes at end of table.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case 2 / Case C	Terminal conditions 1/ 2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Test Limits: Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		Case C: 1	2	3	4	5	6	7	8	9	10	11	12	13	14									
		Test no.	GI	NC	A1	A2	A3	A4	GND	B4	B3	B2	B1	NC	G2	V_{CC}								
I_{IH}/ I_{OZH}		50	2/		6 V				GND				2/		4.2 V	6 V	A1		.1		2.0			uA
		51	"			6 V			"			2/			"	"	A2		"		"			"
		52	"				6 V		"		2/				"	"	A3		"		"			"
		53	"					6 V	"	2/					"	"	A4		"		"			"
		54	1.2 V		2/				"				6 V		2/	"	B1		"		"			"
		55	"			2/			"			6 V			"	"	B2		"		"			"
		56	"				2/		"		6 V				"	"	B3		"		"			"
		57	"					2/	"	0 V					"	"	B4		"		"			"
I_{IL}/ I_{OZL}		58	2/		GND				"	2/					4.2 V	"	A1		-.1		-.2			"
		59	"			GND			"		2/				"	"	A2		"		"			"
		60	"				GND		"			2/			"	"	A3		"		"			"
		61	"					GND	"				2/		"	"	A4		"		"			"
		62	1.2 V		2/				"				GND		2/	"	B1		"		"			"
		63	"			2/			"			GND			"	"	B2		"		"			"
		64	"				2/		"		GND				"	"	B3		"		"			"
		65	"					2/	"	GND					"	"	B4		"		"			"
I_{IH}	3010	66	6 V						"							"	GI		.05		.1			"
		67							"						6 V	"	G2		"		"			"
I_{IL}	3009	68	GND						"							"	GI		-.05		-.1			"
		69							"						GND	"	G2		"		"			"

Symbol	MIL-STD-883 method	Case 2 / Case C	Terminal conditions 1/ 2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Test Limits: Subgroup 4 T_C = +25°C Min	Subgroup 4 T_C = +25°C Max	Unit
		Case C: 1	2	3	4	5	6	7	8	9	10	11	12	13	14					
		Test no.	GI	NC	A1	A2	A3	A4	GND	B4	B3	B2	B1	NC	G2	V_{CC}				
$C_{I/O}$		70	6 V		3/				"						GND	6 V	A1		20	pF
		71	"			3/			"						"	"	A2		"	"
		72	"				3/		"						"	"	A3		"	"
		73	"					3/	"						"	"	A4		"	"
		74	GND						"				3/		6 V	"	B1		"	"
		75	"						"			3/			"	"	B2		"	"
		76	"						"		3/				"	"	B3		"	"
		77	"						"	3/					"	"	B4		"	"
C_C		78	3/						"							GND	GI		10	"
		79							"						3/	GND	G2		10	"

See footnotes at end of table.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case 2 / Case C	Terminal conditions 1/														Measured terminal	Test limits						Unit
			2 / 1	3 / 2	4 / 3	6 / 4	8 / 5	9 / 6	10 / 7	12 / 8	13 / 9	14 / 10	16 / 11	18 / 12	19 / 13	20 / 14		Subgroup 9 T_C = +25°C 4/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 4/		
		Test no.	$\overline{GI}$	NC	A1	A2	A3	A4	GND	B4	B3	B2	B1	NC	G2	V_{CC}		Min	Max	Min	Max	Min	Max	
Truth table tests 5/		80	B		A	A	A	A	GND	H	H	H	H		B	4.5 V	6/							
		81	B		B	B	B	B	"	L	L	L	L		B	"	"							
		82	A		H	H	H	H	"	A	A	A	A		A	"	"							
		83	A		L	L	L	L	"	B	B	B	B		A	"	"							
		84	A		Z	Z	Z	Z	"	Z	Z	Z	Z		B	"	7/							
		85	B		Z	Z	Z	Z	"	Z	Z	Z	Z		A	"	7/							
t_{THL}	3004 (Fig. 4)	86	4.5 V		OUT				"				IN		4.5 V	4.5 V	A1	2	12	2	16	2	12	ns
		87	"			OUT			"			IN	"		"	"	A2	"	"	"	"	"	"	"
		88	"				OUT		"		IN	"			"	"	A3	"	"	"	"	"	"	"
		89	"					OUT	"	IN			OUT		"	"	A4	"	"	"	"	"	"	"
		90	GND		IN				"			OUT			GND	"	B1	"	"	"	"	"	"	"
		91	"			IN			"						"	"	B2	"	"	"	"	"	"	"
		92	"				IN		"		OUT				"	"	B3	"	"	"	"	"	"	"
		93	"					IN	"	OUT					"	"	B4	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 4)	94 to 101	Same tests, terminal conditions, and limits as specified above for t_{THL}.															2	"	2	"	2	"	"
t_{PHL2}	3003 (Fig. 4)	102	GND		IN				GND			OUT	OUT		GND	4.5 V	A1 to B1	4	17	4	23	4	17	"
		103	"			IN			"		OUT	"			"	"	A2 to B2	"	"	"	"	"	"	"
		104	"				IN		"		"	"			"	"	A3 to B3	"	"	"	"	"	"	"
		105	"					IN	"	OUT					"	"	A4 to B4	"	"	"	"	"	"	"
		106	4.5 V		OUT				"			OUT	IN		4.5 V	"	B1 to A1	"	"	"	"	"	"	"
		107	"			OUT			"		IN	"			"	"	B2 to A2	"	"	"	"	"	"	"
		108	"				OUT		"		"	"			"	"	B3 to A3	"	"	"	"	"	"	"
		109	"					OUT	"	IN					"	"	B4 to A4	"	"	"	"	"	"	"
t_{PLH2}	3003 (Fig. 4)	110 to 117	Same terminal conditions and measured terminals as specified above for t_{PHL2}.															4	"	4	"	4	"	"
t_{PZH1} 10/	3003 (Fig. 4)	118	IN		4.5 V				GND			OUT	OUT		GND	4.5 V	$\overline{GI}$ to B1	4	"	4	"	4	"	"
		119	"			4.5 V			"		OUT	"			"	"	$\overline{GI}$ to B2	"	"	"	"	"	"	"
		120	"				4.5 V		"		"	"			"	"	$\overline{GI}$ to B3	"	"	"	"	"	"	"
		121	"					4.5 V	"	OUT					"	"	$\overline{GI}$ to B4	"	"	"	"	"	"	"
		122	4.5 V		OUT				"			4.5 V	4.5 V		IN	"	G2 to A1	"	"	"	"	"	"	"
		123	"			OUT			"		OUT	"			"	"	G2 to A2	"	"	"	"	"	"	"
		124	"				OUT		"		"	"			"	"	G2 to A3	"	"	"	"	"	"	"
		125	"					OUT	"	4.5 V					"	"	G2 to A4	"	"	"	"	"	"	"
t_{PZL1} 9/	3003 (Fig. 4)	126	IN		GND				"			OUT	OUT		GND	"	$\overline{GI}$ to B1	"	27	"	36	"	27	"
		127	"			GND			"		OUT	"			"	"	$\overline{GI}$ to B2	"	"	"	"	"	"	"
		128	"				GND		"		"	"			"	"	$\overline{GI}$ to B3	"	"	"	"	"	"	"
		129	"					GND	"	OUT					"	"	$\overline{GI}$ to B4	"	"	"	"	"	"	"
		130	4.5 V		OUT				"			GND	GND		IN	"	G2 to A1	"	"	"	"	"	"	"
		131	"			OUT			"		OUT	"			"	"	G2 to A2	"	"	"	"	"	"	"
		132	"				OUT		"		"	"			"	"	G2 to A3	"	"	"	"	"	"	"
		133	"					OUT	"	GND					"	"	G2 to A4	"	"	"	"	"	"	"

See footnotes at end of table.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case 2 / Case C	Terminal conditions 1/														Measured terminal	Test limits						Unit
			2 / 1	3 / 2	4 / 3	6 / 4	8 / 5	9 / 6	10 / 7	12 / 8	13 / 9	14 / 10	16 / 11	18 / 12	19 / 13	20 / 14		Subgroup 9 T_C = +25°C 4/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 4/		
		Test no.	$\overline{GI}$	NC	A1	A2	A3	A4	GND	B4	B3	B2	B1	NC	G2	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHZ1} 10/	3003 (Fig. 4)	134	IN		4.5 V				GND				OUT		GND	4.5 V	$\overline{GI}$ to B1	4	32	4	42	4	32	ns
		135	"			4.5 V			"			OUT			"	"	$\overline{GI}$ to B2	"	"	"	"	"	"	"
		136	"				4.5 V		"		OUT				"	"	$\overline{GI}$ to B3	"	"	"	"	"	"	"
		137	"					4.5 V	"	OUT					"	"	$\overline{GI}$ to B4	"	"	"	"	"	"	"
		138	4.5 V		OUT				"				4.5 V		IN	"	G2 to A1	"	"	"	"	"	"	"
		139	"			OUT			"			4.5 V			"	"	G2 to A2	"	"	"	"	"	"	"
		140	"				OUT		"		4.5 V				"	"	G2 to A3	"	"	"	"	"	"	"
		141	"					OUT	"	4.5 V					"	"	G2 to A4	"	"	"	"	"	"	"
t_{PLZ1} 9/	3003 (Fig. 4)	142	IN		GND				"				OUT		GND	"	$\overline{GI}$ to B1	"	"	"	"	"	"	"
		143	"			GND			"			OUT			"	"	$\overline{GI}$ to B2	"	"	"	"	"	"	"
		144	"				GND		"		OUT				"	"	$\overline{GI}$ to B3	"	"	"	"	"	"	"
		145	"					GND	"	OUT					"	"	$\overline{GI}$ to B4	"	"	"	"	"	"	"
		146	4.5 V		OUT				"				GND		IN	"	G2 to A1	"	"	"	"	"	"	"
		147	"			OUT			"			GND			"	"	G2 to A2	"	"	"	"	"	"	"
		148	"				OUT		"		GND				"	"	G2 to A3	"	"	"	"	"	"	"
		149	"					OUT	"	GND					"	"	G2 to A4	"	"	"	"	"	"	"

See footnotes at end of table.

TABLE III. Group A inspection for device type 03.

Symbol	MIL-STD-883 method	Cases 2, R	Terminal conditions 1/																			Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{E}$	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1	1 mA									1/										GND	DIR	1/ 0.4	1.5					V
		2		1 mA								"										"	A1	"	"					"
		3			1 mA							"										"	A2	"	"					"
		4				1 mA						"										"	A3	"	"					"
		5					1 mA					"										"	A4	"	"					"
		6						1 mA				"										"	A5	"	"					"
		7							1 mA			"										"	A6	"	"					"
		8								1 mA		"										"	A7	"	"					"
		9									1 mA	"										"	A8	"	"					"
		10										"	1 mA									"	B8	"	"					"
		11										"		1 mA								"	B7	"	"					"
		12										"			1 mA							"	B6	"	"					"
		13										"				1 mA						"	B5	"	"					"
		14										"					1 mA					"	B4	"	"					"
		15										"						1 mA				"	B3	"	"					"
		16										"							1 mA			"	B2	"	"					"
		17										"								1 mA			"	B1	"	"					"
		18										"									1 mA	"	$\overline{E}$	"	"					"
V_{IC} (neg)		19	-1 mA									"										1/	DIR		-1.5					"
		20		-1 mA								"										"	A1	"	"					"
		21			-1 mA							"										"	A2	"	"					"
		22				-1 mA						"										"	A3	"	"					"
		23					-1 mA					"										"	A4	"	"					"
		24						-1 mA				"										"	A5	"	"					"
		25							-1 mA			"										"	A6	"	"					"
		26								-1 mA		"										"	A7	"	"					"
		27									-1 mA	"										"	A8	"	"					"
		28										"	-1 mA									"	B8	"	"					"
		29										"		-1 mA								"	B7	"	"					"
		30										"			-1 mA							"	B6	"	"					"
		31										"				-1 mA						"	B5	"	"					"
		32										"					-1 mA					"	B4	"	"					"
		33										"						-1 mA				"	B3	"	"					"
		34										"							-1 mA			"	B2	"	"					"
		35										"								-1 mA		"	B1	"	"					"
		36										"									-1 mA	"	$\overline{E}$	"	"					"
I_{CC}	3005	37	GND									GND	GND	6 V	GND	6 V	GND	6 V	GND	6 V	GND	6 V	V_{CC}		-2		20			mA
		38	GND									"	6 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	"	"	"		"		"			"
		39	6 V	GND	6 V	GND	6 V	GND	6 V	GND	6 V	"									"	"	"		"		"			"
		40	6 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	"									"	"	"		"		"			"
I_{CCZ}		41	GND									"									6 V	"	"		"		10			"

See footnotes at end of table.

TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Cases 2, R	Terminal conditions 1/																			Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{OH3}	3006	42	1.2 V	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA	GND	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	1.2 V	6 V	A1	5.95		5.95		5.95		V
		43	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2	"		"		"		"
		44	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A3	"		"		"		"
		45	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A4	"		"		"		"
		46	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A5	"		"		"		"
		47	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A6	"		"		"		"
		48	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A7	"		"		"		"
		49	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A8	"		"		"		"
		50	4.2 V	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B1	"		"		"		"
		51	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B2	"		"		"		"
		52	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B3	"		"		"		"
		53	"	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B4	"		"		"		"
		54	"	"	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B5	"		"		"		"
		55	"	"	"	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	B6	"		"		"		"
		56	"	"	"	"	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	B7	"		"		"		"
		57	"	"	"	"	"	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	B8	"		"		"		"
V_{OH5}	3006	58	1.2 V	-7.8 mA	-7.8 mA	-7.8 mA	-7.8 mA	-7.8 mA	-7.8 mA	-7.8 mA	-7.8 mA	"	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	"	"	A1	5.48		5.2		5.48		"
		59	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2	"		"		"		"
		60	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A3	"		"		"		"
		61	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A4	"		"		"		"
		62	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A5	"		"		"		"
		63	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A6	"		"		"		"
		64	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A7	"		"		"		"
		65	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A8	"		"		"		"
		66	4.2 V	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B1	"		"		"		"
		67	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B2	"		"		"		"
		68	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B3	"		"		"		"
		69	"	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B4	"		"		"		"
		70	"	"	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B5	"		"		"		"
		71	"	"	"	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	B6	"		"		"		"
		72	"	"	"	"	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	B7	"		"		"		"
		73	"	"	"	"	"	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	B8	"		"		"		"
V_{OL3}	3007	74	1.2 V	20 µA	20 µA	20 µA	20 µA	20 µA	20 µA	20 µA	20 µA	"	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	"	"	A1		.05		.05		.05	"
		75	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2		"		"		"	
		76	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A3		"		"		"	
		77	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A4		"		"		"	
		78	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A5		"		"		"	
		79	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A6		"		"		"	
		80	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A7		"		"		"	
		81	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A8		"		"		"	
		82	4.2 V	1.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B1		"		"		"	
		83	"	"	1.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B2		"		"		"	
		84	"	"	"	1.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B3		"		"		"	
		85	"	"	"	"	1.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B4		"		"		"	
		86	"	"	"	"	"	1.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B5		"		"		"	
		87	"	"	"	"	"	"	1.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	B6		"		"		"	
		88	"	"	"	"	"	"	"	1.2 V	"	"	"	"	"	"	"	"	"	"	"	"	B7		"		"		"	
		89	"	"	"	"	"	"	"	"	1.2 V	"	"	"	"	"	"	"	"	"	"	"	B8		"		"		"	

See footnotes at end of table.

TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Cases 2, R	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	OIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	E	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{OL5}	3007	90	1.2 V	7.8 mA	7.8 mA	7.8 mA	7.8 mA	7.8 mA	7.8 mA	7.8 mA	7.8 mA	GND	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	6 V	A1		0.26		0.4		0.26	V
		91	"									"											A2		"		"		"	"
		92	"									"											A3		"		"		"	"
		93	"									"											A4		"		"		"	"
		94	"									"											A5		"		"		"	"
		95	"									"											A6		"		"		"	"
		96	"									"											A7		"		"		"	"
		97	"									"											A8		"		"		"	"
		98	4.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	"	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	7.8 mA	"	"	B1		"		"		"	"
		99	"		1.2 V							"							7.8 mA		"	"	B2		"		"		"	"
		100	"			1.2 V						"						7.8 mA			"	"	B3		"		"		"	"
		101	"				1.2 V					"					7.8 mA				"	"	B4		"		"		"	"
		102	"					1.2 V				"				7.8 mA					"	"	B5		"		"		"	"
		103	"						1.2 V			"			7.8 mA						"	"	B6		"		"		"	"
		104	"							1.2 V		"		7.8 mA							"	"	B7		"		"		"	"
		105	"								1.2 V	"	7.8 mA								"	"	B8		"		"		"	"
I_{OSA}	3011	106	GND	GND	GND	GND	GND	GND	GND	GND	GND	"	4 V	4 V	4 V	4 V	4 V	4 V	4 V	4 V	GND	4 V	A1	-10	-135	-10	-135	-10	-135	mA
		107	"									"											A2		"		"		"	"
		108	"									"											A3		"		"		"	"
		109	"									"											A4		"		"		"	"
		110	"									"											A5		"		"		"	"
		111	"									"											A6		"		"		"	"
		112	"									"											A7		"		"		"	"
		113	"									"											A8		"		"		"	"
		114	4 V	4 V	4 V							"											B1		"		"		"	"
		115	"									"											B2		"		"		"	"
		116	"			4 V						"											B3		"		"		"	"
		117	"				4 V					"											B4		"		"		"	"
		118	"					4 V				"											B5		"		"		"	"
		119	"						4 V			"											B6		"		"		"	"
		120	"							4 V		"											B7		"		"		"	"
		121	"								4 V	"	GND	GND	GND	GND	GND	GND	GND	GND	"	"	B8		"		"		"	"
I_{IH}/ I_{OZH}		122	1.2 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	"	2/	2/	2/	2/	2/	2/	2/	2/	2/	6 V	A1		.1		2.0			µA
		123	"									"											A2		"		"			"
		124	"									"											A3		"		"			"
		125	"									"											A4		"		"			"
		126	"									"											A5		"		"			"
		127	"									"											A6		"		"			"
		128	"									"											A7		"		"			"
		129	"									"											A8		"		"			"
		130	4.2 V	2/	2/							"								6 V			B1		"		"			"
		131	"									"							6 V				B2		"		"			"
		132	"			2/						"						6 V					B3		"		"			"
		133	"				2/					"					6 V						B4		"		"			"
		134	"					2/				"				6 V							B5		"		"			"
		135	"						2/			"			6 V								B6		"		"			"
		136	"							2/		"		6 V									B7		"		"			"
		137	"								2/	"	6 V										B8		"		"			"

See footnotes at end of table.

TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Cases 2, R	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	E	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{IL}/ I_{OZL}		138	1.2 V	GND	GND	GND	GND	GND	GND	GND	GND	GND	2/	2/	2/	2/	2/	2/	2/	2/	2/	6 V	A1		-.1		-2.0			µA
		139	"									"									"	"	A2		"		"			
		140	"									"									"	"	A3		"		"			
		141	"									"									"	"	A4		"		"			
		142	"									"									"	"	A5		"		"			
		143	"									"									"	"	A6		"		"			
		144	"									"									"	"	A7		"		"			
		145	"									"									"	"	A8		"		"			
		146	4.2 V	2/								"									"	"	B1		"		"			
		147	"		2/							"									"	"	B2		"		"			
		148	"			2/						"									"	"	B3		"		"			
		149	"				2/					"									"	"	B4		"		"			
		150	"					2/				"									"	"	B5		"		"			
		151	"						2/			"									"	"	B6		"		"			
		152	"							2/		"									"	"	B7		"		"			
		153	"								2/	"									"	"	B8		"		"			
I_{IH}	3010	154	6 V									"									"	"	DIR		.05		.1			"
		155										"									6 V	"	E		.05		.1			"
I_{IL}	3009	156	GND									"									"	"	DIR		-.05		-.1			"
		157										"									GND	"	E		-.05		-.1			"
$C_{I/O}$		158	GND	3/	3/	3/	3/	3/	3/	3/	3/	"	3/	3/	3/	3/	3/	3/	3/	3/	6 V	"	A1		20					pF
		159	"									"									"	"	A2		"					
		160	"									"									"	"	A3		"					
		161	"									"									"	"	A4		"					
		162	"									"									"	"	A5		"					
		163	"									"									"	"	A6		"					
		164	"									"									"	"	A7		"					
		165	"									"									"	"	A8		"					
		166	6 V									"									"	"	B8		"					
		167	"									"									"	"	B7		"					
		168	"									"									"	"	B6		"					
		169	"									"									"	"	B5		"					
		170	"									"									"	"	B4		"					
		171	"									"									"	"	B3		"					
		172	"									"									"	"	B2		"					
		173	"									"									"	"	B1		"					
C_C		174	3/									"									3/	GND	DIR		10					"
		175										"										GND	E		10					"

			Test limits						
			Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		Unit
			Min	Max	Min	Max	Min	Max	
Truth table tests 5/		176	A	A	B	B	A	A	
		177	A	B	H	L	Z	Z	
		178	B	H	L	Z			
		179	B	L	Z				
		180	B	Z					

Truth table tests 5/		176	A	A	B	A	A	A	A	A	A	A	H	H	H	H	H	H	H	H	B	4.5 V	6/							
		177	A	B	H	B	B	B	B	B	B	B	L	L	L	L	L	L	L	L	B	"	"							
		178	B	H	L	H	H	H	H	H	H	H	A	A	A	A	A	A	A	A	B	"	"							
		179	B	L	Z	L	L	L	L	L	L	L	B	B	B	B	B	B	B	B	B	"	"							
		180	B	Z		Z	Z	Z	Z	Z	Z	Z	Z	Z	Z	Z	Z	Z	Z	Z	A	"	7/							

See footnotes at end of table.

TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Cases 2, R	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C 4/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 4/		
		Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{E}$	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{THL}	3004 (Fig. 4)	181	GND	OUT								GND									GND	4.5 V	A1	2	12	2	16	2	12	ns
		182	"		OUT							"						IN	IN		"	"	A2	"	"	"	"	"	"	"
		183	"			OUT						"								IN	"	"	A3	"	"	"	"	"	"	"
		184	"				OUT					"					IN				"	"	A4	"	"	"	"	"	"	"
		185	"					OUT				"				IN					"	"	A5	"	"	"	"	"	"	"
		186	"						OUT			"			IN						"	"	A6	"	"	"	"	"	"	"
		187	"							OUT		"		IN							"	"	A7	"	"	"	"	"	"	"
		188	"								OUT	"	IN								"	"	A8	"	"	"	"	"	"	"
		189	4.5 V	IN								"							OUT		"	"	B1	"	"	"	"	"	"	"
		190	"		IN							"								OUT	"	"	B2	"	"	"	"	"	"	"
		191	"			IN						"						OUT			"	"	B3	"	"	"	"	"	"	"
		192	"				IN					"					OUT				"	"	B4	"	"	"	"	"	"	"
		193	"					IN				"				OUT					"	"	B5	"	"	"	"	"	"	"
		194	"						IN			"			OUT						"	"	B6	"	"	"	"	"	"	"
		195	"							IN		"		OUT							"	"	B7	"	"	"	"	"	"	"
		196	"								IN	"	OUT								"	"	B8	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 4)	197 to 212	Same tests, terminal conditions, and limits as specified above for t_{THL}.																					2	12	2	16	2	12	ns
t_{PHL2}	3003 (Fig. 4)	213	4.5 V	IN								GND									GND	4.5 V	A1 to B1	4	25	4	33	4	25	ns
		214	"		IN							"							OUT		"	"	A2 to B2	"	"	"	"	"	"	"
		215	"			IN						"						OUT			"	"	A3 to B3	"	"	"	"	"	"	"
		216	"				IN					"					OUT				"	"	A4 to B4	"	"	"	"	"	"	"
		217	"					IN				"				OUT					"	"	A5 to B5	"	"	"	"	"	"	"
		218	"						IN			"			OUT						"	"	A6 to B6	"	"	"	"	"	"	"
		219	"							IN		"		OUT							"	"	A7 to B7	"	"	"	"	"	"	"
		220	"								IN	"	OUT								"	"	A8 to B8	"	"	"	"	"	"	"
		221	GND	OUT								"								OUT	"	"	B1 to A1	"	"	"	"	"	"	"
		222	"		OUT							"							IN		"	"	B2 to A2	"	"	"	"	"	"	"
		223	"			OUT						"						IN			"	"	B3 to A3	"	"	"	"	"	"	"
		224	"				OUT					"					IN				"	"	B4 to A4	"	"	"	"	"	"	"
		225	"					OUT				"				IN					"	"	B5 to A5	"	"	"	"	"	"	"
		226	"						OUT			"			IN						"	"	B6 to A6	"	"	"	"	"	"	"
		227	"							OUT		"		IN							"	"	B7 to A7	"	"	"	"	"	"	"
		228	"								OUT	"	IN								"	"	B8 to A8	"	"	"	"	"	"	"
t_{PLH2}	3003 (Fig. 4)	229 to 244	Same terminal conditions and measured terminals as specified above for t_{PHL2}.																					4	25	4	33	4	25	ns

See footnotes at end of table.

TABLE III. Group A inspection for device type 03 - Continued.

Symbol	Mil-STD-883 method	Cases 2, R	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test Limits: Subgroup 9 T_C = +25°C 4/ Min	Max	Subgroup 10 T_C = +125°C Min	Max	Subgroup 11 T_C = -55°C 4/ Min	Max	Unit
		Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}								
t_{PZH1} 10/	3003 (Fig. 4)	245	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	1N	4.5 V	G to A1	9	44	9	59	9	44	ns
		246																					G to A2							
		247																					G to A3							
		248																					G to A4							
		249																					G to A5							
		250																					G to A6							
		251																					G to A7							
		252																					G to A8							
		253	4.5 V	4.5 V	4.5 V	4.5 V																	G to B1							
		254																					G to B2							
		255																					G to B3							
		256					4.5 V																G to B4							
		257						4.5 V															G to B5							
		258							4.5 V														G to B6							
		259								4.5 V													G to B7							
		260									4.5 V												G to B8							
t_{PHZ1} 10/	3003 (Fig. 4)	261 to 276	Same terminal conditions and measured terminals as specified above for t_{PZH1}.																					8	41	8	54	8	41	ns
t_{PZL1} 9/	3003 (Fig. 4)	277	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	GND	GND	GND	GND	GND	GND	GND	GND	1N	4.5 V	G to A1	9	44	9	59	9	44	ns
		278																					G to A2							
		279																					G to A3							
		280																					G to A4							
		281																					G to A5							
		282																					G to A6							
		283																					G to A7							
		284																					G to A8							
		285	6 V	GND	GND	GND	GND	GND	GND	GND	GND												G to B1							
		286																					G to B2							
		287																					G to B3							
		288																					G to B4							
		289																					G to B5							
		290																					G to B6							
		291																					G to B7							
		292																					G to B8							
t_{PLZ1} 9/	3003 (Fig. 4)	293 to 308	Same terminal conditions and measured terminals as specified above for t_{PZL1}.																					8	41	8	54	8	41	ns

See footnotes at end of table.

TABLE III. Group A inspection for device type 06.

Symbol	Mil-STD-883 method	Cases 2, R	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	O1R	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B5	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1		1 mA	1 mA							1/																		
		2				1 mA						-																		
		3					1 mA					-																		
		4						1 mA				-																		
		5							1 mA			-																		
		6								1 mA		-																		
		7									1 mA	-																		
		8										-	1 mA																	
		9										-		1 mA																
		10										-			1 mA															
		11										-				1 mA														
		12										-					1 mA													
		13										-						1 mA												
		14										-							1 mA											
		15										-								1 mA										
		16										-																		
V_{IC} (neg)		17		-1 mA	-1 mA							GND										1/			-1.5					
		18				-1 mA						-										-								
		19					-1 mA					-										-								
		20						-1 mA				-										-								
		21							-1 mA			-										-								
		22								-1 mA		-										-								
		23									-1 mA	-										-								
		24										-	-1 mA									-								
		25										-		-1 mA								-								
		26										-			-1 mA							-								
		27										-				-1 mA						-								
		28										-					-1 mA					-								
		29										-						-1 mA				-								
		30										-							-1 mA			-								
		31										-								-1 mA		-								
		32										-								-1 mA		-								
I_{CC}	3005	33	6 V	6 V	GND	6 V	GND	6 V	GND	6 V	GND	-									GND	6 V	V_{CC}		-2		20			mA
		34	-	6 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	-	GND	6 V	GND	6 V	GND	6 V	GND	6 V	-	-			-		-				
		35	-									-	6 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	-	-			-		-				
		36	-									-									-	-			-		-				
I_{CCZ}	3005	37	-									-									6 V	-	-		-		10				-

See footnotes at end of table.

TABLE III. Group A inspection for device type 06 - Continued.

Symbol	MIL-STD-883 method	Cases 2, 4	Terminal conditions 17																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	G	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{OH3}	3006	38	1.2 V	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA	GND	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	6 V	A1	5.95		5.95		5.95		V
		39	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2	"		"		"		"
		40	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A3	"		"		"		"
		41	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A4	"		"		"		"
		42	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A5	"		"		"		"
		43	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A6	"		"		"		"
		44	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A7	"		"		"		"
		45	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A8	"		"		"		"
		46	4.2 V	1.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B1	"		"		"		"
		47	"	"	1.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B2	"		"		"		"
		48	"	"	"	1.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B3	"		"		"		"
		49	"	"	"	"	1.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B4	"		"		"		"
		50	"	"	"	"	"	1.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B5	"		"		"		"
		51	"	"	"	"	"	"	1.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	B6	"		"		"		"
		52	"	"	"	"	"	"	"	1.2 V	"	"	"	"	"	"	"	"	"	"	"	"	B7	"		"		"		"
		53	"	"	"	"	"	"	"	"	1.2 V	"	-20 µA	"	"	"	"	"	"	"	"	"	B8	"		"		"		"
V_{OH5}	3006	54 to 69	Same tests and terminal conditions as for V_{OH3} except I_{OH} = -7.8 mA.																					5.48		5.2		5.48		V
V_{OL3}	3007	70	1.2 V	20 µA	20 µA	20 µA	20 µA	20 µA	20 µA	20 µA	20 µA	"	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	"	"	A1		.05		.05		.05	V
		71	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2		"		"		"	
		72	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A3		"		"		"	
		73	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A4		"		"		"	
		74	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A5		"		"		"	
		75	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A6		"		"		"	
		76	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A7		"		"		"	
		77	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A8		"		"		"	
		78	4.2 V	4.2 V	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B1		"		"		"	
		79	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B2		"		"		"	
		80	"	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B3		"		"		"	
		81	"	"	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B4		"		"		"	
		82	"	"	"	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	"	B5		"		"		"	
		83	"	"	"	"	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	"	B6		"		"		"	
		84	"	"	"	"	"	"	"	"	4.2 V	"	"	"	"	"	"	"	"	"	"	"	B7		"		"		"	
		85	"	"	"	"	"	"	"	"	"	"	20 µA	"	"	"	"	"	"	"	"	"	B8		"		"		"	
V_{OL5}	3007	86 to 101	Same tests and terminal conditions as for V_{OL3} except I_{OH} = 7.8 mA.																						0.26		0.4		0.26	V
I_{OS4}	3011	102	GND	GND	GND	GND	GND	GND	GND	GND	GND	"	GND	GND	GND	GND	GND	GND	GND	GND	GND	4 V	A1	-10	-135	-10	-135	-10	-135	mA
		103	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2	"	"	"	"	"	"	"
		104	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A3	"	"	"	"	"	"	"
		105	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A4	"	"	"	"	"	"	"
		106	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A5	"	"	"	"	"	"	"
		107	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A6	"	"	"	"	"	"	"
		108	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A7	"	"	"	"	"	"	"
		109	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A8	"	"	"	"	"	"	"
		110	4 V	GND	GND	GND	GND	GND	GND	GND	GND	"	GND	GND	GND	GND	GND	GND	GND	GND	GND	"	B1	"	"	"	"	"	"	"
		111	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B2	"	"	"	"	"	"	"
		112	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B3	"	"	"	"	"	"	"
		113	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B4	"	"	"	"	"	"	"
		114	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B5	"	"	"	"	"	"	"
		115	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B6	"	"	"	"	"	"	"
		116	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B7	"	"	"	"	"	"	"
		117	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B8	"	"	"	"	"	"	"

See footnotes at end of table.

MIL-M-38510/655

TABLE III. Group A inspection for device type 06 - Continued.

Symbol	MIL-STD-883 method	Cases 2, 3, 4	Terminal conditions 17																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	G	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{IH}/ I_{OZH}	3010	118	1.2 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	GND	2/ 6 V	2/	2/	2/	2/	2/	2/	2/	2/	6 V	A1		.1		2.0			µA
		119	"									"		6 V	5 V	6 V							A2		"		"			
		120	"									"											A3		"		"			
		121	"									"											A4		"		"			
		122	"									"											A5		"		"			
		123	"									"											A6		"		"			
		124	"									"											A7		"		"			
		125	"									"											A8		"		"			
		126	4.2 V									"											B8		"		"			
		127	"									"											B7		"		"			
		128	"									"											B6		"		"			
		129	"									"											B5		"		"			
		130	"									"											B4		"		"			
		131	"									"											B3		"		"			
		132	"									"											B2		"		"			
		133	"									"											B1		"		"			
I_{IH}		134	6 V									"									"	"	DIR		.05		0.1			"
		135										"									6 V	"	G		.05		0.1			"
I_{IL}/ I_{OZL}	3010	136	6 V	GND	GND	GND	GND	GND	GND	GND	GND	"	2/ GND	2/ GND	2/	2/	2/	2/	2/	2/	GND	6 V	A1		-.1		-2.0			"
		137	"									"									"	"	A2		"		"			
		138	"									"											A3		"		"			
		139	"									"											A4		"		"			
		140	"									"											A5		"		"			
		141	"									"											A6		"		"			
		142	"									"											A7		"		"			
		143	"									"											A8		"		"			
		144	GND									"											B8		"		"			
		145	"									"											B7		"		"			
		146	"									"											B6		"		"			
		147	"									"											B5		"		"			
		148	"									"											B4		"		"			
		149	"									"											B3		"		"			
		150	"									"											B2		"		"			
		151	"									"											B1		"		"			
I_{IL}		152	GND									"									GND	"	DIR		-.05		-0.1			"
		153										"									"	"	G		-.05		-0.1			"

Symbol	MIL-STD-883 method	Cases 2, 3, 4	Terminal conditions 17																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 4 T_C = +25°C						
		Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	G	V_{CC}		Min	Max					
C_O		154	GND	3/	3/	3/	3/	3/	3/	3/	3/	GND	3/	3/	3/	3/	3/	3/	3/	3/	6 V	6 V	A1		20					pF
		155	"									"									"	"	A2		"					
		156	"									"											A3		"					
		157	"									"											A4		"					
		158	"									"											A5		"					
		159	"									"											A6		"					
		160	"									"											A7		"					
		161	"									"											A8		"					
		162	6 V									"											B8		"					
		163	"									"											B7		"					
		164	"									"											B6		"					
		165	"									"											B5		"					
		166	"									"											B4		"					
		167	"									"											B3		"					
		168	"									"											B2		"					
		169	"									"											B1		"					
C_C		170	3/									"									3/	GND	DIR		"					"
		171										"									"	GND	G		"					"

See footnotes at end of table.

TABLE III. Group A inspection for device type 06 - Continued.

Symbol	MIL-STD-883 method	Cases 2, R	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
		Test no.	D1R	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	G	V_{CC}		Min	Max	Min	Max	Min	Max	
Truth table tests 5/	3014	172	A	A	A	A	A	A	A	A	A	*	L	L	L	L	L	L	L	L	B	4.5 V	6/							
		173	A	B	B	B	B	B	B	B	B	*	H	H	H	H	H	H	H	H	*	*								
		174	B	L	L	L	L	L	L	L	L	*	A	A	A	A	A	A	A	A	*	*								
		175	B	H	H	H	H	H	H	H	H	*	B	B	B	B	B	B	B	B	*	*								
																								Subgroup 9 T_C = +25°C 4/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 4/		
																								Min	Max	Min	Max	Min	Max	
t_{THL}	3004 (Fig. 4)	176	GND	OUT								*											A1	2	12	2	16	2	12	ns
		177	*		OUT							*											A2	*	*	*	*	*	*	*
		178	*			OUT						*											A3	*	*	*	*	*	*	*
		179	*				OUT					*											A4	*	*	*	*	*	*	*
		180	*					OUT				*											A5	*	*	*	*	*	*	*
		181	*						OUT			*											A6	*	*	*	*	*	*	*
		182	*							OUT		*											A7	*	*	*	*	*	*	*
		183	*								OUT	*											A8	*	*	*	*	*	*	*
		184	6 V	IN								*	IN										B1	*	*	*	*	*	*	*
		185	*		IN							*		IN									B2	*	*	*	*	*	*	*
		186	*			IN						*			IN								B3	*	*	*	*	*	*	*
		187	*				IN					*				IN							B4	*	*	*	*	*	*	*
		188	*					IN				*					IN						B5	*	*	*	*	*	*	*
		189	*						IN			*						IN					B6	*	*	*	*	*	*	*
		190	*							IN		*							IN				B7	*	*	*	*	*	*	*
		191	*								IN	*	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT			B8	*	*	*	*	*	*	*
t_{TLH}	3004 (Fig. 4)	192 to 207	Same tests, terminal conditions, and limits as specified above t_{THL}.																					*	*	*	*	*	*	*
t_{PHL2}	3003 (Fig. 4)	208	GND	OUT								GND											B1 to A1	4	20	4	26	4	20	ns
		209	*		OUT							*											B2 to A2	*	*	*	*	*	*	*
		210	*			OUT						*											B3 to A3	*	*	*	*	*	*	*
		211	*				OUT					*											B4 to A4	*	*	*	*	*	*	*
		212	*					OUT				*											B5 to A5	*	*	*	*	*	*	*
		213	*						OUT			*											B6 to A6	*	*	*	*	*	*	*
		214	*							OUT		*											B7 to A7	*	*	*	*	*	*	*
		215	*								OUT	*											B8 to A8	*	*	*	*	*	*	*
		216	4.5 V	IN								*	IN										A1 to B1	*	*	*	*	*	*	*
		217	*		IN							*		IN									A2 to B2	*	*	*	*	*	*	*
		218	*			IN						*			IN								A3 to B3	*	*	*	*	*	*	*
		219	*				IN					*				IN							A4 to B4	*	*	*	*	*	*	*
		220	*					IN				*					IN						A5 to B5	*	*	*	*	*	*	*
		221	*						IN			*						IN					A6 to B6	*	*	*	*	*	*	*
		222	*							IN		*							IN				A7 to B7	*	*	*	*	*	*	*
		223	*								IN	*	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT			A8 to B8	*	*	*	*	*	*	*
t_{PLH2}	3003 (Fig. 4)	224 to 239	Same tests and terminal conditions as specified above t_{PHL1}.																					*	*	*	*	*	*	*

See footnotes at end of table.

TABLE III. Group A inspection for device type 06 - Continued.

Symbol	MIL-STD-883 method	Cases 2, 3	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C 4/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 4/		
		Test no.	$\overline{G}$ or DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHZ2} 10/	3003 (Fig. 4)	240	GND	OUT																	IN		to A1	8	40	8	53	8	40	ns
		241	"		OUT														GND		"		to A2	"	"	"	"	"	"	"
		242	"			OUT												GND			"		to A3	"	"	"	"	"	"	"
		243	"				OUT										GND				"		to A4	"	"	"	"	"	"	"
		244	"					OUT								GND					"		to A5	"	"	"	"	"	"	"
		245	"						OUT						GND						"		to A6	"	"	"	"	"	"	"
		246	"							OUT				GND							"		to A7	"	"	"	"	"	"	"
		247	"								OUT		GND								"		to A8	"	"	"	"	"	"	"
		248	4.5 V	GND															OUT		"		to B1	"	"	"	"	"	"	"
		249	"		GND																"		to B2	"	"	"	"	"	"	"
		250	"			GND												OUT			"		to B3	"	"	"	"	"	"	"
		251	"				GND										OUT				"		to B4	"	"	"	"	"	"	"
		252	"					GND								OUT					"		to B5	"	"	"	"	"	"	"
		253	"						GND						OUT						"		to B6	"	"	"	"	"	"	"
		254	"							GND				OUT							"		to B7	"	"	"	"	"	"	"
		255	"								GND		OUT								"		to B8	"	"	"	"	"	"	"
t_{PZH2} 10/	3003 (Fig. 4)	256 to 271	Same tests and terminal conditions as specified above t_{PHZ}.																					9	44	9	59	9	44	"
t_{PLZ2} 9/	3003 (Fig. 4)	272	GND	OUT																	IN		to A1	8	40	8	53	8	40	"
		273	"		OUT														4.5 V	4.5 V	"		to A2	"	"	"	"	"	"	"
		274	"			OUT												4.5 V			"		to A3	"	"	"	"	"	"	"
		275	"				OUT										4.5 V				"		to A4	"	"	"	"	"	"	"
		276	"					OUT								4.5 V					"		to A5	"	"	"	"	"	"	"
		277	"						OUT						1.5 V						"		to A6	"	"	"	"	"	"	"
		278	"							OUT				4.5 V							"		to A7	"	"	"	"	"	"	"
		279	"								OUT		4.5 V								"		to A8	"	"	"	"	"	"	"
		280	4.5 V	4.5 V															OUT		"		to B1	"	"	"	"	"	"	"
		281	"		4.5 V																"		to B2	"	"	"	"	"	"	"
		282	"			4.5 V											4.5 V				"		to B3	"	"	"	"	"	"	"
		283	"				4.5 V									4.5 V					"		to B4	"	"	"	"	"	"	"
		284	"					4.5 V													"		to B5	"	"	"	"	"	"	"
		285	"						4.5 V						OUT						"		to B6	"	"	"	"	"	"	"
		286	"							4.5 V											"		to B7	"	"	"	"	"	"	"
		287	"								4.5 V		OUT								"		to B8	"	"	"	"	"	"	"
t_{PZL2} 9/	3003 (Fig. 4)	288 to 303	Same tests and terminal conditions as specified above t_{PLZ}.																					9	44	9	59	9	44	"

See footnotes at end of table.

TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Cases 2, 3	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	D1A	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1		1 mA	1 mA	1 mA	1 mA	1 mA	1 mA	1 mA	1 mA	1/	1 mA	1 mA	1 mA	1 mA	1 mA	1 mA	1 mA	1 mA		GND	A1	0.4 1/	1.5					V
		2										"										"	A2	"	"					"
		3										"										"	A3	"	"					"
		4										"										"	A4	"	"					"
		5										"										"	A5	"	"					"
		6										"										"	A6	"	"					"
		7										"										"	A7	"	"					"
		8										"										"	A8	"	"					"
		9										"										"	B8	"	"					"
		10										"										"	B7	"	"					"
		11										"										"	B6	"	"					"
		12										"										"	B5	"	"					"
		13										"										"	B4	"	"					"
		14										"										"	B3	"	"					"
		15										"										"	B2	"	"					"
		16										"										"	B1	"	"					"
V_{IC} (neg)		17		-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	GND	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA		1/	A1		-1.5					"
		18										"										"	A2		"					"
		19										"										"	A3		"					"
		20										"										"	A4		"					"
		21										"										"	A5		"					"
		22										"										"	A6		"					"
		23										"										"	A7		"					"
		24										"										"	A8		"					"
		25										"										"	B8		"					"
		26										"										"	B7		"					"
		27										"										"	B6		"					"
		28										"										"	B5		"					"
		29										"										"	B4		"					"
		30										"										"	B3		"					"
		31										"										"	B2		"					"
		32										"										"	B1		"					"
I_{CC}	3005	33	6 V	6 V	GND	6 V	GND	6 V	GND	6 V	GND	"									GND	6 V	V_{CC}		.2		20			μA
		34	"	6 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	"	GND	6 V	GND	6 V	GND	6 V	GND	6 V	"	"	"		"		"				"
		35	"									"	6 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	"	"	"		"		"				"
		36	"									"									"	"	"		"		"				"
I_{CCZ}	3005	37	"									"									6 V	"	"		"		10				-

See footnotes at end of table.

TABLE III. Group A inspection for device type 07 - Continued.

Symbol	MIL-STD-883 method	Cases 2, R	Terminal conditions 17																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	J1x	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	G	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{OH3}	3006	38	1.2 V	-20 µA								GND									1.2 V	6 V	A1	5.95		5.95		5.95		V
		39	"		-20 µA							"							1.2 V		"	"	A2	"		"		"		"
		40	"			-20 µA						"						1.2 V		1.2 V	"	"	A3	"		"		"		"
		41	"				-20 µA					"					1.2 V				"	"	A4	"		"		"		"
		42	"					-20 µA				"				1.2 V					"	"	A5	"		"		"		"
		43	"						-20 µA			"			1.2 V						"	"	A6	"		"		"		"
		44	"							-20 µA		"		1.2 V							"	"	A7	"		"		"		"
		45	"								-20 µA	"	1.2 V								"	"	A8	"		"		"		"
		46	4.2 V	1.2 V								"							-20 µA	-20 µA	"	"	B1	"		"		"		"
		47	"		1.2 V							"						-20 µA			"	"	B2	"		"		"		"
		48	"			1.2 V						"									"	"	B3	"		"		"		"
		49	"				1.2 V					"					-20 µA				"	"	B4	"		"		"		"
		50	"					1.2 V				"				-20 µA					"	"	B5	"		"		"		"
		51	"						1.2 V			"			-20 µA						"	"	B6	"		"		"		"
		52	"							1.2 V		"		-20 µA							"	"	B7	"		"		"		"
		53	"								1.2 V	"	-20 µA								"	"	B8	"		"		"		"
V_{OH5}	3006	54 to 69	Same tests and terminal conditions as for V_{OH3} except I_{OH} = -7.8 mA.																					5.48		5.2		5.48		V
V_{OL3}	3007	70	1.2 V	20 µA								GND									4.2 V	6 V	A1		.05		.05		.05	V
		71	"		20 µA							"						4.2 V		4.2 V	"	"	A2		"		"		"	"
		72	"			20 µA						"							4.2 V		"	"	A3		"		"		"	"
		73	"				20 µA					"					4.2 V				"	"	A4		"		"		"	"
		74	"					20 µA				"				4.2 V					"	"	A5		"		"		"	"
		75	"						20 µA			"			4.2 V						"	"	A6		"		"		"	"
		76	"							20 µA		"		4.2 V							"	"	A7		"		"		"	"
		77	"								20 µA	"	4.2 V								"	"	A8		"		"		"	"
		78	4.2 V	4.2 V								"							20 µA	20 µA	"	"	B1		"		"		"	"
		79	"		4.2 V							"						20 µA			"	"	B2		"		"		"	"
		80	"			4.2 V						"									"	"	B3		"		"		"	"
		81	"				4.2 V					"					20 µA				"	"	B4		"		"		"	"
		82	"					4.2 V				"				20 µA					"	"	B5		"		"		"	"
		83	"						4.2 V			"			20 µA						"	"	B6		"		"		"	"
		84	"							4.2 V		"		20 µA							"	"	B7		"		"		"	"
		85	"								4.2 V	"	20 µA								"	"	B8		"		"		"	"
V_{OL5}	3007	86 to 101	Same tests and terminal conditions as for V_{OL3} except I_{OH} = 7.8 mA.																						.26		0.4		.26	V
I_{OS4}	3011	102	GND	GND																	GND	4 V	A1	-10	-135	-10	-135	-10	-135	mA
		103	"		GND																"	"	A2	"	"	"	"	"	"	"
		104	"			GND															"	"	A3	"	"	"	"	"	"	"
		105	"				GND														"	"	A4	"	"	"	"	"	"	"
		106	"					GND													"	"	A5	"	"	"	"	"	"	"
		107	"						GND												"	"	A6	"	"	"	"	"	"	"
		108	"							GND											"	"	A7	"	"	"	"	"	"	"
		109	"								GND										"	"	A8	"	"	"	"	"	"	"
		110	4 V	GND									4 V								"	"	B1	"	"	"	"	"	"	"
		111	"		GND									4 V							"	"	B2	"	"	"	"	"	"	"
		112	"			GND															"	"	B3	"	"	"	"	"	"	"
		113	"				GND														"	"	B4	"	"	"	"	"	"	"
		114	"					GND													"	"	B5	"	"	"	"	"	"	"
		115	"						GND												"	"	B6	"	"	"	"	"	"	"
		116	"							GND											"	"	B7	"	"	"	"	"	"	"
		117	"								GND										"	"	B8	"	"	"	"	"	"	"

See footnotes at end of table.

TABLE III. Group A inspection for device type 07 - Continued.

Symbol	MIL-STD-883 method	Cases 2, R	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	Min	Max	Min	Max	
[illegible]	3013	118	4.2 V	0 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	GND	2/ 6 V	2/	2/	2/	2/	2/	2/	2/	2/	6 V	A1		.1		2.0			
		119																					A2							
		120																					A3							
		121																					A4							
		122																					A5							
		123																					A6							
		124																					A7							
		125																					A8							
		126	4.2 V																				B8							
		127																					B7							
		128																					B6							
		129																					B5							
		130																					B4							
		131																					B3							
		132																					B2							
		133																					B1							
[illegible]		134	0 V																				DIR		.05		.1			
		135																					$\overline{G}$		.05		.1			
[illegible]	3013	136	4.2 V	GND	GND	GND	GND	GND	GND	GND	GND	GND	2/ GND	2/ GND	2/ GND	2/ GND	2/ GND	2/ GND	2/ GND	2/ GND	2/	6 V	A1		-.1		-2.0			
		137																					A2							
		138																					A3							
		139																					A4							
		140																					A5							
		141																					A6							
		142																					A7							
		143																					A8							
		144	4.2 V																				B8							
		145																					B7							
		146																					B6							
		146																					B5							
		148																					B4							
		149																					B3							
		150																					B2							
		151																					B1							
[illegible]		152	GND																		GND		DIR		-.05		-.1			
		153																					$\overline{G}$		-.05		-.1			
																								Subgroup 4 T_C = +25°C						
																								Min	Max					
[illegible]		154	GND	3/	3/	3/	3/	3/	3/	3/	3/	GND	3/	3/	3/	3/	3/	3/	3/	3/	6 V	6 V	A1		20					
		155																					A2							
		156																					A3							
		157																					A4							
		158																					A5							
		159																					A6							
		160																					A7							
		161																					A8							
		162	6 V																				B8							
		163																					B7							
		164																					B6							
		165																					B5							
		166																					B4							
		167																					B3							
		168																					B2							
		169																					B1							
I_{CC}		170	3/																		3/	GND	DIR		-					
		171																				GND	$\overline{G}$		-					

See footnotes at end of table.

TABLE III. Group A inspection for device type 07 - Continued.

Symbol	MIL-STD-883 method	Cases 2, 3	Terminal conditions 1/																					Measured terminal	Test limits						Unit
		R	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -5°C		
		Test no.	D14	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	G	V_{CC}		Min	Max	Min	Max	Min	Max	
Truth table tests 5/	3014	172	A	A	A	A	A	A	A	A	A	-	L	L	L	L	L	L	L	L	B	4.5 V	6/							
		173	A	B	B	B	B	B	B	B	B	-	H	H	H	H	H	H	H	H	-	-								
		174	B	L	L	L	L	L	L	L	L	-	B	B	B	B	B	B	B	B	-	-								
		175	B	H	H	H	H	H	H	H	H	-	A	A	A	A	A	A	A	A	-	-								
																								Subgroup 9 T_C = +25°C 4/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 4/		
																								Min	Max	Min	Max	Min	Max	
t_{THL}	3004 (Fig. 4)	176	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	-	IN	IN	IN	IN	IN	IN	IN	IN	GND	4.5 V	A1	2	12	2	16	2	12	ns
		177	-									-									-	-	A2	-	-	-	-	-	-	-
		178	-									-									-	-	A3	-	-	-	-	-	-	-
		179	-									-									-	-	A4	-	-	-	-	-	-	-
		180	-									-									-	-	A5	-	-	-	-	-	-	-
		181	-									-									-	-	A6	-	-	-	-	-	-	-
		182	-									-									-	-	A7	-	-	-	-	-	-	-
		183	-									-									-	-	A8	-	-	-	-	-	-	-
		184	4.5 V	IN	IN	IN						-								OUT	-	-	B1	-	-	-	-	-	-	-
		185	-									-							OUT		-	-	B2	-	-	-	-	-	-	-
		186	-				IN					-						OUT			-	-	B3	-	-	-	-	-	-	-
		187	-									-					OUT				-	-	B4	-	-	-	-	-	-	-
		188	-					IN				-				OUT					-	-	B5	-	-	-	-	-	-	-
		189	-						IN			-			OUT						-	-	B6	-	-	-	-	-	-	-
		190	-							IN		-		OUT							-	-	B7	-	-	-	-	-	-	-
		191	-								IN	-	OUT								-	-	B8	-	-	-	-	-	-	-
t_{TLH}	3004 (Fig. 4)	192 to 207	Same tests, terminal conditions, and limits as specified above t_{THL}.																					-	-	-	-	-	-	-
t_{PHL2}	3003 (Fig. 4)	208	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	IN	IN	IN	IN	IN	IN	IN	IN	GND	4.5 V	B1 to A1	4	20	4	26	4	20	ns
		209	-									-									-	-	B2 to A2	-	-	-	-	-	-	-
		210	-									-									-	-	B3 to A3	-	-	-	-	-	-	-
		211	-									-									-	-	B4 to A4	-	-	-	-	-	-	-
		212	-									-									-	-	B5 to A5	-	-	-	-	-	-	-
		213	-									-									-	-	B6 to A6	-	-	-	-	-	-	-
		214	-									-									-	-	B7 to A7	-	-	-	-	-	-	-
		215	-									-									-	-	B8 to A8	-	-	-	-	-	-	-
		216	6 V	IN	IN	IN						-							OUT		-	-	A1 to B1	-	-	-	-	-	-	-
		217	-									-						OUT			-	-	A2 to B2	-	-	-	-	-	-	-
		218	-				IN					-					OUT				-	-	A3 to B3	-	-	-	-	-	-	-
		219	-									-				OUT					-	-	A4 to B4	-	-	-	-	-	-	-
		220	-					IN				-			OUT						-	-	A5 to B5	-	-	-	-	-	-	-
		221	-						IN			-		OUT							-	-	A6 to B6	-	-	-	-	-	-	-
		222	-							IN		-	OUT								-	-	A7 to B7	-	-	-	-	-	-	-
		223	-								IN	-									-	-	A8 to B8	-	-	-	-	-	-	-
t_{PLH2}	3003 (Fig. 4)	224 to 239	Same tests and terminal conditions as specified above t_{PHL1}.																					-	-	-	-	-	-	-

See footnotes at end of table.

MIL-M-38510/655

TABLE III. Group A inspection for device type 07 - Continued.

Symbol	MIL-STD-883 method	Cases 2, R	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C 4/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 4/		
		Test no.	D13	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	G	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHZ2} 10/	3003 (Fig. 4)	240	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	IN	4.5 V	G to A1	8	40	8	53	8	40	ns
		241	"									"									"	"	G to A2	"	"	"	"	"	"	"
		242	"									"									"	"	G to A3	"	"	"	"	"	"	"
		243	"									"									"	"	G to A4	"	"	"	"	"	"	"
		244	"									"									"	"	G to A5	"	"	"	"	"	"	"
		245	"									"									"	"	G to A6	"	"	"	"	"	"	"
		246	"									"									"	"	G to A7	"	"	"	"	"	"	"
		247	"									"									"	"	G to A8	"	"	"	"	"	"	"
		248	4.5 V	GND	GND	GND	GND	GND	GND	GND	GND	"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	OUT	OUT	"	"	G to B1	"	"	"	"	"	"	"
		249	"									"									"	"	G to B2	"	"	"	"	"	"	"
		250	"									"									"	"	G to B3	"	"	"	"	"	"	"
		251	"									"									"	"	G to B4	"	"	"	"	"	"	"
		252	"									"									"	"	G to B5	"	"	"	"	"	"	"
		253	"									"									"	"	G to B6	"	"	"	"	"	"	"
		254	"									"									"	"	G to B7	"	"	"	"	"	"	"
		255	"									"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	G to B8	"	"	"	"	"	"	"
t_{PZH2} 10/	3003 (Fig. 4)	256 to 271	Same tests and terminal conditions as specified above t_{PHZ}.																					9	44	9	59	9	44	"
t_{PLZ2} 9/	3003 (Fig. 4)	272	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	GND	GND	GND	GND	GND	GND	GND	GND	IN	4.5 V	G to A1	8	40	8	53	8	40	ns
		273	"									"									"	"	G to A2	"	"	"	"	"	"	"
		274	"									"									"	"	G to A3	"	"	"	"	"	"	"
		275	"									"									"	"	G to A4	"	"	"	"	"	"	"
		276	"									"									"	"	G to A5	"	"	"	"	"	"	"
		277	"									"									"	"	G to A6	"	"	"	"	"	"	"
		278	"									"									"	"	G to A7	"	"	"	"	"	"	"
		279	"									"									"	"	G to A8	"	"	"	"	"	"	"
		280	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	GND	GND	GND	GND	GND	GND	OUT	OUT	"	"	G to B1	"	"	"	"	"	"	"
		281	"									"									"	"	G to B2	"	"	"	"	"	"	"
		282	"									"									"	"	G to B3	"	"	"	"	"	"	"
		283	"									"									"	"	G to B4	"	"	"	"	"	"	"
		284	"									"									"	"	G to B5	"	"	"	"	"	"	"
		285	"									"									"	"	G to B6	"	"	"	"	"	"	"
		286	"									"									"	"	G to B7	"	"	"	"	"	"	"
		287	"									"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	G to B8	"	"	"	"	"	"	"
t_{PZL2} 9/	3003 (Fig. 4)	288 to 303	Same tests and terminal conditions as specified above t_{PLZ}.																					9	44	9	59	9	44	"

See footnotes at end of table.

TABLE III. Group A inspection for device type 08.

Symbol	MIL-STD-883 method	Case X	Terminal conditions 1/																								Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24				Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Case 3	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24										
		Test no.	CAB	SAB	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	C	SBA	CBA	V_{CC}			Min	Max	Min	Max	Min	Max	
V_{OH} (1Ω)		1	1 mA											1/													GND	CA0	0.6	1/	1.5					V
		2		1 mA										"													"	SA0	"		"					"
		3			1 mA									"													"	DI0	"		"					"
		4				1 mA								"													"	A1	"		"					"
		5					1 mA							"													"	A2	"		"					"
		6						1 mA						"													"	A3	"		"					"
		7							1 mA					"													"	A4	"		"					"
		8								1 mA				"													"	A5	"		"					"
		9									1 mA			"													"	A6	"		"					"
		10										1 mA		"													"	A7	"		"					"
		11											1 mA	"													"	A8	"		"					"
		12												"	1 mA												"	B8	"		"					"
		13												"		1 mA											"	B7	"		"					"
		14												"			1 mA										"	B6	"		"					"
		15												"				1 mA									"	B5	"		"					"
		16												"					1 mA								"	B4	"		"					"
		17												"						1 mA							"	B3	"		"					"
		18												"							1 mA						"	B2	"		"					"
		19												"								1 mA					"	B1	"		"					"
		20												"									1 mA				"	C	"		"					"
		21												"										1 mA			"	SBA	"		"					"
		22												"											1 mA		"	CBA	"		"					"
I_{IC} (neg)		23	-1 mA											GND													1/	CA0			-1.5					
		24		-1 mA										"													"	SA0			"					
		25			-1 mA									"													"	DI0			"					
		26				-1 mA								"													"	A1			"					
		27					-1 mA							"													"	A2			"					
		28						-1 mA						"													"	A3			"					
		29							-1 mA					"													"	A4			"					
		30								-1 mA				"													"	A5			"					
		31									-1 mA			"													"	A6			"					
		32										-1 mA		"													"	A7			"					
		33											-1 mA	"													"	A8			"					
		34												"	-1 mA												"	B8			"					
		35												"		-1 mA											"	B7			"					
		36												"			-1 mA										"	B6			"					
		37												"				-1 mA									"	B5			"					
		38												"					-1 mA								"	B4			"					
		39												"						-1 mA							"	B3			"					
		40												"							-1 mA						"	B2			"					
		41												"								-1 mA					"	B1			"					
		42												"									-1 mA				"	C			"					
		43												"										-1 mA			"	SBA			"					
		44												"											-1 mA		"	CBA			"					

See footnotes at end of table.

TABLE III. Group A inspection for device type 06 - Continued.

Symbol	MIL-STD-883 method	Case C / Case 3	Terminal conditions 1/																								Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
			-	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24								
		Test no.	CAd	AB	DIR	A1	A2	A3	A4	A5	Ao	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	C	SBA	CBA	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{CC}	3005	43 46 47 48	GND " " "	GND " " "	0 V 6 V GND GND	6 V 6 V	GND 6 V	6 V 6 V	GND 6 V	6 V 6 V	GND 6 V	0 V 6 V	GND 6 V	GND " "		GND 6 V	6 V 0 V	GND 0 V	6 V 6 V	GND 6 V	6 V 6 V	GND 6 V	6 V 0 V	GND " "	GND " "	GND " "	6 V " "	V_{CC} " "		.2 " "		20 " "			mA " "
I_{CCZ}	3005	49 50	GND GND	GND GND	GND GND	6 V GND	6 V GND	6 V GND	6 V GND	6 V GND	6 V GND	6 V GND	6 V GND	6 V GND	" "	GND 6 V	GND 6 V	GND 6 V	GND 6 V	GND 6 V	GND 6 V	GND 6 V	GND 6 V	6 V 6 V	GND GND	GND GND	6 V 6 V	" "		0.4 0.2		10 10			" "
V_{OH3}	3006	51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66	1.2 V " " " " " " " " " " " " " " "	1.2 V " " " " " " " " " " " " " " "	1.2 V " " " " " " 4.2 V " " " " " " " "	-20 µA " " " " " " 4.2 V " " " " " " " "	-20 µA " " " " " " 4.2 V " " " " " " " "	-20 µA " " " " " " 4.2 V " " " " " " " "	-20 µA " " " " " " 4.2 V " " " " " " " "	-20 µA " " " " " " 4.2 V " " " " " " " "	-20 µA " " " " " " 4.2 V " " " " " " " "	-20 µA " " " " " " 4.2 V " " " " " " " "	-20 µA " " " " " " 4.2 V " " " " " " " "	-20 µA " " " " " " 4.2 V " " " " " " " "	" " " " " " " " " " " " " " " "	4.2 V " " " " " " " " " " " " " " -20 µA	4.2 V " " " " " " " " " " " " " -20 µA "	4.2 V " " " " " " " " " " " " -20 µA " "	4.2 V " " " " " " " " " " " -20 µA " " "	4.2 V " " " " " " " " " " -20 µA " " " "	4.2 V " " " " " " " " " -20 µA " " " " "	4.2 V " " " " " " " " -20 µA " " " " " "	4.2 V " " " " " " " -20 µA " " " " " " "	1.2 V " " " " " " " " " " " " " " "	1.2 V " " " " " " " " " " " " " " "	1.2 V " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	A1 A2 A3 A4 A5 A6 A7 A8 B1 B2 B3 B4 B5 B6 B7 B8	5.95 " " " " " " " " " " " " " " "		5.95 " " " " " " " " " " " " " " "		5.95 " " " " " " " " " " " " " " "		V " " " " " " " " " " " " " " "
V_{OH5}	3006	67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82	1.2 V " " " " " " " " " " " " " " "	1.2 V " " " " " " " " " " " " " " "	1.2 V " " " " " " 4.2 V " " " " " " " "	-7.8 mA " " " " " " 4.2 V " " " " " " " "	-7.8 mA " " " " " " 4.2 V " " " " " " " "	-7.8 mA " " " " " " 4.2 V " " " " " " " "	-7.8 mA " " " " " " 4.2 V " " " " " " " "	-7.8 mA " " " " " " 4.2 V " " " " " " " "	-7.8 mA " " " " " " 4.2 V " " " " " " " "	-7.8 mA " " " " " " 4.2 V " " " " " " " "	-7.8 mA " " " " " " 4.2 V " " " " " " " "	" " " " " " " " " " " " " " " "	4.2 V " " " " " " " " " " " " " " -7.8 mA	4.2 V " " " " " " " " " " " " " -7.8 mA "	4.2 V " " " " " " " " " " " " -7.8 mA " "	4.2 V " " " " " " " " " " " -7.8 mA " " "	4.2 V " " " " " " " " " " -7.8 mA " " " "	4.2 V " " " " " " " " " -7.8 mA " " " " "	4.2 V " " " " " " " " -7.8 mA " " " " " "	4.2 V " " " " " " " -7.8 mA " " " " " " "	1.2 V " " " " " " " " " " " " " " "	1.2 V " " " " " " " " " " " " " " "	1.2 V " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	A1 A2 A3 A4 A5 A6 A7 A8 B1 B2 B3 B4 B5 B6 B7 B8	5.48 " " " " " " " " " " " " " " "		5.4 " " " " " " " " " " " " " " "		5.48 " " " " " " " " " " " " " " "		" " " " " " " " " " " " " " " "
V_{OL3}	3007	33 34 35 36 37 38 89 90 91 92 93 94 95 96 97 98	1.2 V " " " " " " " " " " " " " " "	1.2 V " " " " " " " " " " " " " " "	1.2 V " " " " " " 4.2 V " " " " " " " "	20 µA " " " " " " 1.2 V " " " " " " " "	20 µA " " " " " " 1.2 V " " " " " " " "	20 µA " " " " " " 1.2 V " " " " " " " "	20 µA " " " " " " 1.2 V " " " " " " " "	20 µA " " " " " " 1.2 V " " " " " " " "	20 µA " " " " " " 1.2 V " " " " " " " "	20 µA " " " " " " 1.2 V " " " " " " " "	20 µA " " " " " " 1.2 V " " " " " " " "	" " " " " " " " " " " " " " " "	1.2 V " " " " " " " " " " " " " " 20 µA	1.2 V " " " " " " " " " " " " " 20 µA "	1.2 V " " " " " " " " " " " " 20 µA " "	1.2 V " " " " " " " " " " " 20 µA " " "	1.2 V " " " " " " " " " " 20 µA " " " "	1.2 V " " " " " " " " 20 µA " " " " " "	1.2 V " " " " " " " 20 µA " " " " " " "	1.2 V " " " " " " 20 µA " " " " " " " "	1.2 V " " " " " 20 µA " " " " " " " " "	1.2 V " " " " 20 µA " " " " " " " " " "	1.2 V " " " 20 µA " " " " " " " " " " "	1.2 V " " 20 µA " " " " " " " " " " " "	1.2 V " 20 µA " " " " " " " " " " " " "	1.2 V 20 µA " " " " " " " " " " " " " "	1.2 V " " " " " " " " " " " " " " "	1.2 V " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	A1 A2 A3 A4 A5 A6 A7 A8 B1 B2 B3 B4 B5 B6 B7 B8		.05 " " " " " " " " " " " " " " "		.05 " " " " " " " " " " " " " " "		.05 " " " " " " " " " " " " " " "

See footnotes at end of table.

TABLE III. Group A inspection for device type 08 - Continued.

Symbol	MIL-STD-883 method	Terminal conditions 1/																								Measured terminal	Test limits						Unit
		Case 3	2	3	4	5	6	7	8	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Case 2	1	2	3	4	5	6	7	8	9	10	11	12	GND	13	14	15	16	17	18	19	20	21	22	23	24		Min	Max	Min	Max	Min	Max	
		Test no.	$\overline{CAS}$	$\overline{SAS}$	$\overline{SAR}$	A1	A2	A3	A4	A5	A6	A7	A8	$\overline{CAS}$	GND	B8	B7	B6	B5	B4	B3	B2	B1	C	$\overline{SBA}$	$\overline{CBA}$	V_{CC}							
V_{OL5}	3007	99	1.2 V	1.2 V	1.2 V	7.8 mA	7.8 mA	7.8 mA	7.8 mA	7.8 mA	7.8 mA	7.8 mA	7.8 mA	7.8 mA	GND	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	6 V	All		.26		0.4		.26	V
		100																																	
		101																																	
		102																																	
		103																																	
		104																																	
		105																																	
		106																																	
		107				4.2 V	1.2 V																												
		108						1.2 V	1.2 V		1.2 V								7.8 mA		7.8 mA	7.8 mA	7.8 mA												
		109								1.2 V		1.2 V						7.8 mA		7.8 mA															
		110											1.2 V				7.8 mA																		
		111												1.2 V		7.8 mA																			
		112																																	
		113																																	
		114																																	
I_{OS4}	3011	115	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND		6 V	6 V	6 V	4 V	4 V	4 V	4 V	4 V	GND	GND	GND	4 V	All	-10	-135	-10	-135	-10	-135	mA
		116																																	
		117																																	
		118																																	
		119																																	
		120																																	
		121																																	
		122																																	
		123				4 V	4 V																												
		124						4 V																											
		125							6 V		4 V																								
		126								6 V		4 V																							
		127																																	
		128																																	
		129																																	
		130																																	
		131				1.2 V	6 V																												
		132						6 V																											
		133							6 V																										
		134								6 V																									
		135									6 V																								
		136										6 V																							
		137											6 V																						
		138												6 V																					
		139	2/	GND	GND	4.2 V	2/									2/	2/	2/	2/	2/	2/	2/	2/	2/	GND	2/	6 V								
		140						2/																											
		141							2/																										
		142								2/																									
		143									2/																								
		144										2/																							
		145											2/																						
		146												2/																					

See footnotes at end of table.

TABLE III. Group A inspection for device type 06 - Continued.

Symbol	MIL-STD-883 method	Terminal condition (pins) 1/ Case 3	1	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28	Measured terminal	Test limits Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C	
		Case C																											Min	Max	Min	Max	Min	Max
		Test no.	SAd	SAd	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	E	SBA	CBA	V_{CC}								
I_{IL}/ I_{IH}		147			1.2 V	GND	GND	GND	GND	GND	GND	GND	GND	GND	2/	2/	2/	2/	2/	2/	2/	2/	2/	GND	2/	6 V	A1		-.1		-2.0			
		148																									A2							
		149																									A3							
		150																									A4							
		151																									A5							
		152																									A6							
		153																									A7							
		154																									A8							
		155	2/	GND	4.2 V	2/	2/	2/	2/	2/	2/	2/	2/	2/	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	B1							
		156																									B2							
		157																									B3							
		158																									B4							
		159																									B5							
		160																									B6							
		161																									B7							
		162																									B8							
		163	0 V		6 V	6 V																		6 V	6 V	6 V		CAd		.05		3.1		
		164																										SAB							
		165																										DIR							
		166																										E							
		167																										SBA							
		168																										CBA							
		169	GND	GND	GND	GND																		GND	GND	GND		CAB		-0.5		-.1		
		170																										SAB							
		171																										DIR							
		172																										E							
		173																										SBA							
		174																										CBA							
		175	GND	GND	GND	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	6 V	GND	GND		A1		20					
		176																										A2							
		177																										A3							
		178																										A4							
		179																										A5							
		180																										A6							
		181																										A7							
		182																										A8							
		183				6 V																						B8							
		184																										B7							
		185																										B6							
		186																										B5							
		187																										B4							
		188																										B3							
		189																										B2							
		190																										B1							
		191	3/		3/	3/																		3/	3/	3/	GND	CAB		10					
		192																										SAB							
		193																										DIR							
		194																										E							
		195																										SBA							
		196																										CBA							

See footnotes at end of table.

TABLE III. Group A inspection for device type 08 - Continued.

Symbol	MIL-STD-883 method	Case T	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28	Measured terminal	Test limits: Subgroup 7 T_C = +25°C Min	Subgroup 7 T_C = +25°C Max	Subgroup 8 T_C = +125°C Min	Subgroup 8 T_C = +125°C Max	Subgroup 9 T_C = -55°C Min	Subgroup 9 T_C = -55°C Max	Units
		Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24								
		Test no.	CA3	CA8	CA4	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	E	SBA	CBA	V_{CC}								
Truth table tests 2	3014	197	3	3	A	A	A	A	A	A	A	A	A	GND	H	H	H	H	H	H	H	H	0	0	0	4.5 V	6/						
		198	-	-	A	A	H	H	H	H	H	H	H	-	L	L	L	L	L	L	L	L	-	-	-	-								
		199	-	-	A	A	L	L	L	L	L	L	L	-	A	A	A	A	A	A	A	A	-	-	-	-								
		230	-	A	-	A	H	H	H	H	H	H	H	-	H	H	H	H	H	H	H	H	-	-	-	-								
		231	-	A	-	A	L	L	L	L	L	L	L	-	A	A	A	A	A	A	A	A	-	-	-	-								
		232	A	-	-	A	H	H	H	H	H	H	H	-	H	H	H	H	H	H	H	H	-	-	-	-								
		233	A	-	-	A	L	L	L	L	L	L	L	-	A	A	A	A	A	A	A	A	-	-	-	-								
		234	A	-	-	A	H	H	H	H	H	H	H	-	H	H	H	H	H	H	H	H	-	-	-	-								
		235	A	A	-	A	L	L	L	L	L	L	L	-	A	A	A	A	A	A	A	A	-	-	-	-								
		236	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-								
		237	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-								
		238	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-								
		239	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-								
t_{MAX} 6/	3333	213	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	IN	IN	Id	IN	Id	IN	Id	IN	GND	GND	GND	4.5 V	A1	30		23		30		MHz
		214	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		A2	-		-		-		-
		215	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		A3	-		-		-		-
		216	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		A4	-		-		-		-
		217	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		A5	-		-		-		-
		218	-	-	-	4.5 V	IN	Id	IN	Id	IN	Id	IN	-	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	-	-	-	-		A6	-		-		-		-
		219	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		A7	-		-		-		-
		220	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		A8	-		-		-		-
		221	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		B8	-		-		-		-
		222	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		B7	-		-		-		-
		223	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		B6	-		-		-		-
		224	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		B5	-		-		-		-
		225	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		B4	-		-		-		-
																												B3	-		-		-		-
																												B2	-		-		-		-
																												B1	-		-		-		-
t_{PHL}	3004 (Fig. 6)	226	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	IN	IN	IN	IN	IN	IN	IN	IN	GND	GND	GND	4.5 V	A1	2	12	2	16	2	12	ns
		227	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		A2	-	-	-	-	-	-	-
		228	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		A3	-	-	-	-	-	-	-
		229	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		A8	-	-	-	-	-	-	-
		230	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		A5	-	-	-	-	-	-	-
		231	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		A6	-	-	-	-	-	-	-
		232	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		A7	-	-	-	-	-	-	-
		233	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		A4	-	-	-	-	-	-	-
		234	-	-	-	4.5 V	IN	Id	IN	Id	IN	Id	IN	-	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	-	-	-	-		B8	-	-	-	-	-	-	-
		235	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		B7	-	-	-	-	-	-	-
		236	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		B6	-	-	-	-	-	-	-
		237	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		B5	-	-	-	-	-	-	-
		238	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		B4	-	-	-	-	-	-	-
		239	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		B3	-	-	-	-	-	-	-
		240	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		B2	-	-	-	-	-	-	-
		241	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		B1	-	-	-	-	-	-	-
t_{PLH}	3004 (Fig. 6)	242 to 257	Same tests, terminal conditions, and limits as specified above for t_{PHL}.																										-	-	-	-	-	-	-

See footnotes at end of table.

TABLE III. Group A inspection for device type 08 - Continued.

Symbol	MIL-STD-883 method	Terminal conditions 1/																									Measured terminal	Test limits						Unit
		Case 3	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 9 T_C = +25°C 4/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 4/		
		Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24								
		Test no.	1A3	1A8	2A4	3A1	2A2	3A3	4A4	5A5	6A6	7A7	8A8	GND	8B8	7B7	6B6	5B5	4B4	3B3	2B2	1B1	G	SBA	CBA	V_{CC}		t_{PLH}	t_{PHL}	t_{PLH}	t_{PHL}	t_{PLH}	t_{PHL}	
t_{PHL1}	3003 (Fig. 4)	256					OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	IN	IN	IN	IN	IN	IN	IN	IN	GND	4.5 V	IN	4.5 V	ICBA to A1	8	38	8	51	8	38	ns
		257																									ICBA to A2							
		258																									ICBA to A3							
		260																									ICBA to A4							
		261																									ICBA to A5							
		262																									ICBA to A6							
		264																									ICBA to A7							
		265																									ICBA to A8							
		266																									ICAB to B1							
		267																									ICAB to B2							
		268																									ICAB to B3							
		269																									ICAB to B4							
		270																									ICAB to B5							
		271																									ICAB to B6							
		272																									ICAB to B7							
		273																									ICAB to B8							
t_{PLH1}	3003 (Fig. 4)	274 to 289	Same terminal conditions and measured terminals as specified above for t_{PHL1}.																										-	-	-	-	-	-	
t_{PHL2}	3003 (Fig. 4)	290					OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	IN	IN	IN	IN	IN	IN	IN	IN	GND	GND	GND	4.5 V	IB1 to A1	5	30	5	40	5	30	ns
		291																									IB2 to A2							
		292																									IB3 to A3							
		293																									IB4 to A4							
		294																									IB5 to A5							
		295																									IB6 to A6							
		296																									IB7 to A7							
		298																									IB8 to A8							
		299				4.5 V	IN																					IA1 to B1							
		300																									IA2 to B2							
		301																									IA3 to B3							
		302																									IA4 to B4							
		303																									IA5 to B5							
		304																									IA6 to B6							
		305																									IA7 to B7							
																											IA8 to B8							
t_{PLH2}	3003 (Fig. 4)	305 to 321	Same terminal conditions and measured terminals as specified above for t_{PHL2}.																										-	-	-	-	-	-	
t_{PHL3}	3003 (Fig. 4)	322	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT															ISBA to A1	10	38	10	51	10	38	ns
		323																									ISBA to A2							
		324																									ISBA to A3							
		325																									ISBA to A4							
		326																									ISBA to A5							
		327																									ISBA to A6							
		328																									ISBA to A7							
		329																									ISBA to A8							
		330	5/	IN	4.5 V	5/	5/	5/	5/	5/	5/	5/	5/		5/	5/	5/	5/	5/	5/	5/	5/	5/	GND	IN	5/	4.5 V	ISAB to B1							
		331																									ISAB to B2							
		332																									ISAB to B3							
		333																									ISAB to B4							
		334																									ISAB to B5							
		335																									ISAB to B6							
		336																									ISAB to B7							
		337																									ISAB to B8							
t_{PLH3}	3003 (Fig. 4)	338 to 353	Same terminal conditions and measured terminals as specified above for t_{PHL3}.																										-	-	-	-	-	-	

See footnotes at end of table.

TABLE III. Group A inspection for device type 08 - Continued.

Symbol	MIL-STD-883 method	Terminal condition (pin no.)																									Measured terminal	Test limits						Unit
		Case 1	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Case 2	1	2	3	4	5	6	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	G	SBA	CBA	V_{CC}		Min	Max	Min	Max	Min	Max	
		Test no.	A3	A4	A5	A1	A2	A3																										
t_{PLH8}	3303 (Fig. 41)	354	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	-	6/	6/	6/	6/	6/	6/	6/	6/	GND	IN	5/	4.5 V	SBA to A1	-	-	-	-	-	-	-
		355	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SAB to A1	-	-	-	-	-	-	-
		356	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SBA to A3	-	-	-	-	-	-	-
		357	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SAB to A4	-	-	-	-	-	-	-
		358	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SBA to A5	-	-	-	-	-	-	-
		359	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SAB to A6	-	-	-	-	-	-	-
		360	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SBA to A7	-	-	-	-	-	-	-
		361	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SAB to A8	-	-	-	-	-	-	-
		362	5/	5/	4.5 V	6/	6/	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	GND	GND	GND	SAB to B1	-	-	-	-	-	-	-
		363	-	-	-	-	-	6/	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SAB to B2	-	-	-	-	-	-	-
		364	-	-	-	-	-	-	6/	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SAB to B3	-	-	-	-	-	-	-
		365	-	-	-	-	-	-	-	6/	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SAB to B4	-	-	-	-	-	-	-
		366	-	-	-	-	-	-	-	-	6/	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SAB to B5	-	-	-	-	-	-	-
		367	-	-	-	-	-	-	-	-	-	6/	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SAB to B6	-	-	-	-	-	-	-
		368	-	-	-	-	-	-	-	-	-	-	6/	-	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	-	-	-	-	SAB to B7	-	-	-	-	-	-	-
		369	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SAB to B8	-	-	-	-	-	-	-
																												10	30	10	51	10	30	ns
t_{PLH8}	3303 (Fig. 41)	370 to 385	Same terminal conditions and measured terminals as specified above for t_{PHL8}.																															
t_{PZH2} 12/	3303 (Fig. 41)	386	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	-	4.5 V	4.5 V	4.5 V	4.5 V	14.5 V	4.5 V	4.5 V	4.5 V	4.5 V	IN	GND		4.5 V	IA to A1	6	43	6	57	6	43	ns
		387	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	IA to A2	-	-	-	-	-	-	-
		388	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	IA to A3	-	-	-	-	-	-	-
		389	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	IA to A4	-	-	-	-	-	-	-
		390	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	IA to A5	-	-	-	-	-	-	-
		391	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	IA to A6	-	-	-	-	-	-	-
		392	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	IA to A7	-	-	-	-	-	-	-
		393	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	IA to A8	-	-	-	-	-	-	-
		394	-	-	-	4.5 V	4.5 V	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	IA to B1	-	-	-	-	-	-	-
		395	-	-	-	-	-	4.5 V	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	IA to B2	-	-	-	-	-	-	-
		396	-	-	-	-	-	-	14.5 V	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	IA to B3	-	-	-	-	-	-	-
		397	-	-	-	-	-	-	-	4.5 V	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	IA to B4	-	-	-	-	-	-	-
		398	-	-	-	-	-	-	-	-	4.5 V	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	IA to B5	-	-	-	-	-	-	-
		399	-	-	-	-	-	-	-	-	-	4.5 V	-	-	-	-	-	-	-	-	-	-	-	-	-	-	IA to B6	-	-	-	-	-	-	-
		400	-	-	-	-	-	-	-	-	-	-	4.5 V	-	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	-	-	-	-	IA to B7	-	-	-	-	-	-	-
		401	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	IA to B8	-	-	-	-	-	-	-
t_{PZL2} 9/	3303 (Fig. 41)	402 to 417	Same terminal conditions (except GND replaces 4.5 V) and measured terminals as specified above for t_{PZH2}.																															
t_{PHZ2} 12/	3303 (Fig. 41)	418 to 433	Same terminal conditions and measured terminals as specified above for t_{PLH2}.																															
t_{PLZ3} 9/	3303 (Fig. 41)	434 to 449	Same terminal conditions and measured terminals as specified above for t_{PZL2}.																															

See footnotes at end of table.

MIL-M-38510/355

| | | |

TABLE III. Group A inspection for device type 09.

table failure modes in the new design are considered

predict

TABLE III. Group A inspection for device type 09 - Continued.

Symbol	MIL-STD-883 method	Case X	Terminal conditions 17																								Measured terminal	Test limits						Unit
		Case C	1	2	3	4	5	6	7	8	9	10	11	12			14	15	16	17	18	19	20	21	22	23	24			Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}_A$	$\overline{OE}_B$	SEL_{AB}	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	$\overline{OE}_{BA}$	I		B_7	B_6	B_5	B_4	B_3	B_2	B_1	$\overline{OE}_{AB}$	SEL_{BA}	V_{CC}			Min	Max	Min	Max	Min	Max	
I_{CC}	3005	45	4.5 V	4.5 V	0 V	0 V	GND	0 V	GND	6 V	GND	0 V	GND	GND	0		6 V	GND	6 V	GND	6 V	GND	6 V	GND	GND	GND	5.5 V	V_{CC}				20			μA
		46	"	"	6 V	5 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	5.5 V		6 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	"								
		47	"	"	GND																														
		48	"	"	GND																														
I_{OZZ}	3035	49	GND	GND	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	0		GND	GND	GND	GND	GND	GND	GND	5.5 V	5.5 V	5.5 V	"				0.4		10			
		50	"	"	"	GND	GND	GND	GND	GND	GND	GND	GND	GND	5.5 V		6 V	6 V	6 V	6 V	6 V	6 V	6 V	GND	GND	GND	"				0.2		10			
V_{IK3}	3006	51	1.2 V	1.2 V	1.2 V	-20 μA	-20 μA	-20 μA	-20 μA	-20 μA	-20 μA	-20 μA	-20 μA	-20 μA	1	V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	6 V	A1	5.95		5.95		5.95		V
		52	"	"	"																									A2							
		53	"	"	"																									A3							
		54	"	"	"																									A4							
		55	"	"	"																									A5							
		56	"	"	"																									A6							
		57	"	"	"																									A7							
		58	"	"	"																									A8							
		59	"	"	4.2 V	1.2 V																								B1							
		60	"	"	"		1.2 V		1.2 V																					B2							
		61	"	"	"																									B3							
		62	"	"	"					1.2 V																				B4							
		63	"	"	"							1.2 V																		B5							
		64	"	"	"								1.2 V																	B6							
		65	"	"	"									1.2 V																B7							
		66	"	"	"																									B8							
V_{OL3}	3007	67	1.2 V	1.2 V	1.2 V	7.8 mA	7.8 mA	7.8 mA	7.8 mA	7.8 mA	7.8 mA	7.8 mA	7.8 mA	7.8 mA	1	V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	1.2 V	6 V	A1	5.48		5.4		5.48		V
		68	"	"	"																									A2							
		69	"	"	"																									A3							
		70	"	"	"																									A4							
		71	"	"	"																									A5							
		72	"	"	"																									A6							
		73	"	"	"																									A7							
		74	"	"	"																									A8							
		75	"	"	4.2 V	1.2 V																								B1							
		76	"	"	"		1.2 V		1.2 V																					B2							
		77	"	"	"																									B3							
		78	"	"	"					1.2 V																				B4							
		79	"	"	"							1.2 V																		B5							
		80	"	"	"								1.2 V																	B6							
		81	"	"	"									1.2 V																B7							
		82	"	"	"										.7	mA														B8							
V_{OL3}	3007	83	1.2 V	1.2 V	1.2 V	20 μA	20 μA	20 μA	20 μA	20 μA	20 μA	20 μA	20 μA	20 μA	6	V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.2 V	4.5 V	A1		.05		.05		.05	V
		84	"	"	"																									A2							
		85	"	"	"																									A3							
		86	"	"	"																									A4							
		87	"	"	"																									A5							
		88	"	"	"																									A6							
		89	"	"	"																									A7							
		90	"	"	"																									A8							
		91	"	"	4.2 V	4.2 V																								B1							
		92	"	"	"		4.2 V		4.2 V																					B2							
		93	"	"	"																									B3							
		94	"	"	"					4.2 V																				B4							
		95	"	"	"							4.2 V																		B5							
		96	"	"	"								4.2 V																	B6							
		97	"	"	"									4.2 V																B7							
		98	"	"	"										2	mA														B8							

See footnotes at end of table.

TABLE III. Group A inspection for device type 09 - Continued.

Symbol	Mil-STD-883 method	Terminal conditions 1/																										Measured terminal	Test limits						Unit
		Case 3	2	1	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Case 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24									
		Test no.	CAd	ΔAd	ΔCd	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	G	SBA	CBA	V_{CC}			Min	Max	Min	Max	Min	Max	
V_{OLS}	3007	33 130 131 132 133 134 135 136 137 138 139 110 111 112 113 114	1.2 V "	1.2 V "	1.2 V "	7.0 mA " " " " " " " " " " " " " " "	7.0 mA " " " " " " " " " " " " " " "	7.0 mA " " " " " " " " " " " " " " "	7.0 mA " " " " " " " " " " " " " " "	7.0 mA " " " " " " " " " " " " " " "	7.0 mA " " " " " " " " " " " " " " "	7.0 mA " " " " " " " " " " " " " " "	7.0 mA " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	4.2 V " " " " " " " " " " " " " " 7.0 mA	4.2 V " " " " " " " " " " " " " " 7.0 mA	4.2 V " " " " " " " " " " " " " " 7.0 mA	4.2 V " " " " " " " " " " " " " " 7.0 mA	4.2 V " " " " " " " " " " " " " " 7.0 mA	4.2 V " " " " " " " " " " " " " " 7.0 mA	4.2 V " " " " " " " " " " " " " " 7.0 mA	4.2 V " " " " " " " " " " " " " " 7.0 mA	4.2 V " " " " " " " " " " " " " " 7.0 mA	1.2 V " " " " " " " " " " " " " " "	1.2 V " " " " " " " " " " " " " " "	1.2 V " " " " " " " " " " " " " " "	1.2 V " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	A1 A2 A3 A4 A5 A6 A7 A8 B1 B2 B3 B4 B5 B6 B7 B8		.26 " " " " " " " " " " " " " " "		0.4 " " " " " " " " " " " " " " "		.26 " " " " " " " " " " " " " " "	V " " " " " " " " " " " " " " "
I_{DLK}	3021	115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	A1 A2 A3 A4 A5 A6 A7 A8 B1 B2 B3 B4 B5 B6 B7 B8	-10 " " " " " " " " " " " " " " "	-135 " " " " " " " " " " " " " " "	-10 " " " " " " " " " " " " " " "	-135 " " " " " " " " " " " " " " "	-10 " " " " " " " " " " " " " " "	-135 " " " " " " " " " " " " " " "	nA " " " " " " " " " " " " " " "
I_{1H} I_{2H}		131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146	2/ " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	4.2 V " " " " " " " " " " " " " " "	2/ " " " " " " " " " " " " " " "	2/ " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	2/ " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	2/ " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	2/ " " " " " " " " " " " " " " "	2/ " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "	2/ " " " " " " " " " " " " " " "	6 V " " " " " " " " " " " " " " "	A1 A2 A3 A4 A5 A6 A7 A8 B1 B2 B3 B4 B5 B6 B7 B8		.1 " " " " " " " " " " " " " " "		2.0 " " " " " " " " " " " " " " "			µA " " " " " " " " " " " " " " "

See footnotes at end of table.

TABLE III. Group A inspection for device type 09 - Continued.

Symbol	MIL-STD-883 method	Case 3	Terminal conditions 1/																							Measured terminal	Test limits						Unit
		Case 5	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24			Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	CAB	SAB	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	G	SBA	CBA	V_{CC}			Min	Max	Min	Max	Min	Max	
I_{IL} I_{IL2}		147			1.2 V	GND	GND	GND	GND	GND	GND	GND	GND	GND														A1		-.1		-2.0			μA
		148																										A2							
		149																										A3							
		150																										A4							
		151																										A5							
		152																										A6							
		153																										A7							
		154																										A8							
		155	2/	GND	4.2 V	2/	2/	2/	2/	2/	2/	2/	2/		2/		2/	2/	2/	2/	2/	2/	2/	GND	2/	6 V		B1							
		156																										B2							
		157																										B3							
		158																										B4							
		159																										B5							
		160																										B6							
		161																										B7							
		162																GND	GND	GND	GND	GND	GND					B8							
I_{IH}	3010	163	6 V																									CAB		.05		1			μA
		164		6 V																								SAB							
		165			6 V																							DIR							
		166																						6 V				C							
		167																							6 V			SBA							
		168																								6 V		CBA							
I_{IL}	3009	169	GND																									CAB		-0.5		-.1			μA
		170		GND																								SAB							
		171			GND																							DIR							
		172				GND																		GND				C							
		173																							GND			SBA							
		174																								GND		CBA							
I_{CEX}		175	GND	GND	GND	3/	3/	3/	3/	3/	3/	3/	3/		3/		3/	3/	3/	3/	3/	3/	3/	6 V	GND	GND		A1		20					μA
		176																										A2							
		177																										A3							
		178																										A4							
		179																										A5							
		180																										A6							
		181																										A7							
		182																										A8							
		183																										B8							
		184			0 V																							B7							
		185																										B6							
		186																										B5							
		187																										B4							
		188																										B3							
		189																										B2							
		190																										B1							
I_{CC}		191	3/		3/																			3/	3/	3/	GND	CAB		10					μA
		192																										SAB							
		193				3/																						DIR							
		194																										C							
		195																										SBA							
		196																										CBA							

See footnotes at end of table.

MIL-M-38510/65

TABLE III. Group A inspection for device type 09 - Continued.

Symbol	MIL-STD-883 method	Terminal condition 1/																										Measured terminal	Test limits						Unit
		Case 3	2	3	4	5	6	7	8	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 9 T_C = -55°C 4/		
		Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24			Min	Max	Min	Max	Min	Max	
		Test no.	CA0	SA0	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	G	SBA	CBA	V_{CC}									
Truth table tests 5/		227	B	B	A	B	B	B	B	B	B	B	B	4.5 V	H	H	H	H	H	H	H	H	B	B	B	4.5 V	6/								
		228	-	-	-	A	A	A	A	A	A	A	A	-	L	L	L	L	L	L	L	L	-	-	-	-									
		229	-	-	-	-	-	-	-	-	-	-	-	-	B	B	B	B	B	B	B	B	-	-	-	-									
		230	-	A	-	-	-	-	-	-	-	-	-	-	A	A	A	A	A	A	A	A	-	-	-	-									
		231	-	-	-	B	B	B	B	B	B	B	B	-	H	H	H	H	H	H	H	H	-	-	-	-									
		232	A	-	-	-	-	-	-	-	-	-	-	-	L	L	L	L	L	L	L	L	-	-	-	-									
		233	B	B	-	-	-	-	-	-	-	-	-	-	B	B	B	B	B	B	B	B	-	-	-	-									
		234	-	-	A	-	-	-	-	-	-	-	-	-	A	A	A	A	A	A	A	A	-	-	-	-									
		235	B	B	B	-	-	-	-	-	-	-	-	-	H	H	H	H	H	H	H	H	-	-	-	-									
		236	-	-	-	A	A	A	A	A	A	A	A	-	L	L	L	L	L	L	L	L	-	-	-	-									
		237	-	-	-	-	-	-	-	-	-	-	-	-	B	B	B	B	B	B	B	B	-	-	-	-									
		238	-	-	-	B	B	B	B	B	B	B	B	-	A	A	A	A	A	A	A	A	-	-	-	-									
		239	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-									
t_{PHL}	3333	240	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	IN	IN	IN	IN	IN	IN	IN	IN	GND	GND	GND	4.5 V	A1	30		23		30		MHz
		241	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A2	-		-		-		-
		242	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A3	-		-		-		-
		243	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A4	-		-		-		-
		244	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A5	-		-		-		-
		245	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A6	-		-		-		-
		246	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A7	-		-		-		-
		247	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A8	-		-		-		-
		248	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B8	-		-		-		-
		249	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B7	-		-		-		-
		250	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B6	-		-		-		-
		251	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B5	-		-		-		-
		252	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B4	-		-		-		-
		253	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B3	-		-		-		-
		254	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B2	-		-		-		-
		255	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B1	-		-		-		-
t_{PLH}	3334 (Fig. 4)	256	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	IN	IN	IN	IN	IN	IN	IN	IN	GND	GND	GND	4.5 V	A1	2	12	2	16	2	12	ns
		257	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A2	-	-	-	-	-	-	-
		258	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A3	-	-	-	-	-	-	-
		259	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A4	-	-	-	-	-	-	-
		260	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A5	-	-	-	-	-	-	-
		261	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A6	-	-	-	-	-	-	-
		262	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A7	-	-	-	-	-	-	-
		263	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A8	-	-	-	-	-	-	-
		264	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B8	-	-	-	-	-	-	-
		265	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B7	-	-	-	-	-	-	-
		266	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B6	-	-	-	-	-	-	-
		267	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B5	-	-	-	-	-	-	-
		268	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B4	-	-	-	-	-	-	-
		269	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B3	-	-	-	-	-	-	-
		270	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B2	-	-	-	-	-	-	-
		271	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B1	-	-	-	-	-	-	-
t_{THL}	3334 (Fig. 4)	272 to 287	Same tests, terminal conditions, and limits as specified above for t_{PHL}.																																

See footnotes at end of table.

5962-8853/855

TABLE III. Group A inspection for device type 09 - Continued.

Symbol	MIL-STD-883 method	Case C	Terminal conditions 1/																								Measured terminal	Test limits						Unit
			2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 9 T_C = +25°C 6/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 6/		
		Test no.	$\overline{OE}_A$	$\overline{OE}_B$	$SAB/\overline{SBA}$	$CAB/\overline{CBA}$	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{OE}$	SBA	CBA	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 (Fig. 4)	254	-	-	-	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	IN	IN	IN	IN	IN	IN	IN	IN	GND	4.5 V	IN	4.5 V	ICBA to A1	8	42	8	56	8	42	ns
		255	-	-	-																						ICBA to A2	-	-	-	-	-	-	
		256	-	-	-																						ICBA to A3	-	-	-	-	-	-	
		263	-	-	-																						ICBA to A4	-	-	-	-	-	-	
		264	-	-	-																						ICBA to A5	-	-	-	-	-	-	
		265	-	-	-																						ICBA to A6	-	-	-	-	-	-	
		266	-	4.5 V	4.5 V	IN																					ICBA to A7	-	-	-	-	-	-	
		267	-	-	-		IN	IN	IN	IN	IN	IN	IN		IN	IN	IN	IN	IN	IN	IN	OUT					ICBA to A8	-	-	-	-	-	-	
		268	-	-	-																						ICAB to B1	-	-	-	-	-	-	
		269	-	-	-																						ICAB to B2	-	-	-	-	-	-	
		270	-	-	-																						ICAB to B3	-	-	-	-	-	-	
		271	-	-	-																						ICAB to B4	-	-	-	-	-	-	
		272	-	-	-																						ICAB to B5	-	-	-	-	-	-	
		273	-	-	-																						ICAB to B6	-	-	-	-	-	-	
																											ICAB to B7	-	-	-	-	-	-	
																											ICAB to B8	-	-	-	-	-	-	
t_{PHL1}	3003 (Fig. 4)	273 to 289	Same terminal conditions and measured terminals as specified above for t_{PHL1}.																									-	-	-	-	-	-	
t_{PHL2}	3003 (Fig. 4)	290	-	-	-	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	IN	IN	IN	IN	IN	IN	IN	IN	GND	GND	GND	4.5 V	B1 to A1	5	30	5	40	5	30	ns
		291	-	-	-																						B2 to A2	-	-	-	-	-	-	
		292	-	-	-																						B3 to A3	-	-	-	-	-	-	
		293	-	-	-																						B4 to A4	-	-	-	-	-	-	
		294	-	-	-																						B5 to A5	-	-	-	-	-	-	
		295	-	-	-																						B6 to A6	-	-	-	-	-	-	
		296	-	-	-																						B7 to A7	-	-	-	-	-	-	
		297	-	-	-																						B8 to A8	-	-	-	-	-	-	
		298	-	4.5 V	-	IN	IN	IN	IN	IN	IN	IN	IN		IN	IN	IN	IN	IN	IN	IN	OUT					A1 to B1	-	-	-	-	-	-	
		299	-	-	-																						A2 to B2	-	-	-	-	-	-	
		300	-	-	-																						A3 to B3	-	-	-	-	-	-	
		301	-	-	-																						A4 to B4	-	-	-	-	-	-	
		302	-	-	-																						A5 to B5	-	-	-	-	-	-	
		303	-	-	-																						A6 to B6	-	-	-	-	-	-	
		304	-	-	-																						A7 to B7	-	-	-	-	-	-	
		305	-	-	-																						A8 to B8	-	-	-	-	-	-	
t_{PHL2}	3003 (Fig. 4)	306 to 321	Same terminal conditions and measured terminals as specified above for t_{PHL2}.																									-	-	-	-	-	-	
t_{PHL3}	3003 (Fig. 4)	322	-	-	-	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	5/	5/	5/	5/	5/	5/	5/	5/	GND	IN	5/	4.5 V	SBA to A1	10	38	10	51	10	38	ns
		323	-	-	-																						SAA to A2	-	-	-	-	-	-	
		324	-	-	-																						SBA to A3	-	-	-	-	-	-	
		325	-	-	-																						SAA to A4	-	-	-	-	-	-	
		326	-	-	-																						SBA to A5	-	-	-	-	-	-	
		327	-	-	-																						SAA to A6	-	-	-	-	-	-	
		328	-	-	-																						SBA to A7	-	-	-	-	-	-	
		329	-	-	-																						SAA to A8	-	-	-	-	-	-	
		330	5/	IN	4.5 V	5/	5/								5/								GND		GND		SAB to B1	-	-	-	-	-	-	
		331	-	-	-			5/								5/											SAB to B2	-	-	-	-	-	-	
		332	-	-	-				5/								5/										SAB to B3	-	-	-	-	-	-	
		333	-	-	-					5/								5/									SAB to B4	-	-	-	-	-	-	
		334	-	-	-						5/								5/								SAB to B5	-	-	-	-	-	-	
		335	-	-	-							5/								5/							SAB to B6	-	-	-	-	-	-	
		336	-	-	-								5/								5/						SAB to B7	-	-	-	-	-	-	
		337	-	-	-																						SAB to B8	-	-	-	-	-	-	

See footnotes at end of table.

MIL-M-38510/655

TABLE III. Group A inspection for device type 09 - Continued.

Symbol	MIL-STD-883 method	Terminal conditions and 17																											Measured terminal	Test limits						Unit
		Case 3	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28			Subgroup 9 T_C = +25°C 4/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 4/		
		Case 4	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24										
		Test No.	CAB	SAB	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	G	SRA	CBA	V_{CC}			Min	Max	Min	Max	Min	Max	
t_{PLH3}	3003 (Fig. 4)	338 to 353	Same terminal conditions and measured terminals as specified above for t_{PHL3}.																										10	30	10	51	10	30	ns
t_{PHL4}	3003 (Fig. 4)	354	-	GND	GND	OUT	OUT	-	-	-	-	-	-	-	6/	-	-	-	-	6/	6/	6/	6/	GND	IN	6/	4.5 V	ISBA to A1	-	-	-	-	-	-	-
		355	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	6/	-	-	-	-	-	-	-	-		ISJA to A2	-	-	-	-	-	-	-
		356	-	-	-	-	-	OUT	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		ISBA to A3	-	-	-	-	-	-	-
		357	-	-	-	-	-	-	OUT	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		ISJA to A4	-	-	-	-	-	-	-
		358	-	-	-	-	-	-	-	OUT	-	-	-	-	-	-	-	6/	-	-	-	-	-	-	-	-	-		ISBA to A5	-	-	-	-	-	-	-
		359	-	-	-	-	-	-	-	-	OUT	-	-	-	-	-	6/	-	-	-	-	-	-	-	-	-	-		ISJA to A6	-	-	-	-	-	-	-
		360	-	-	-	-	-	-	-	-	-	OUT	-	-	-	6/	-	-	-	-	-	-	-	-	-	-	-		ISBA to A7	-	-	-	-	-	-	-
		361	-	-	-	-	-	-	-	-	-	-	OUT	-	6/	-	-	-	-	-	-	-	-	-	-	-	-		ISJA to A8	-	-	-	-	-	-	-
		362	2	5V	1 or 3	6/	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	OUT	-	GND	GND	-		ISAB to B1	-	-	-	-	-	-	-
		363	-	-	-	-	6/	-	-	-	-	-	-	-	-	-	-	-	-	-	OUT	-	-	-	-	-	-		ISAB to B2	-	-	-	-	-	-	-
		364	-	-	-	-	-	6/	-	-	-	-	-	-	-	-	-	-	-	OUT	-	-	-	-	-	-	-		ISAB to B3	-	-	-	-	-	-	-
		365	-	-	-	-	-	-	6/	-	-	-	-	-	-	-	-	-	OUT	-	-	-	-	-	-	-	-		ISAB to B4	-	-	-	-	-	-	-
		366	-	-	-	-	-	-	-	6/	-	-	-	-	-	-	-	OUT	-	-	-	-	-	-	-	-	-		ISAB to B5	-	-	-	-	-	-	-
		367	-	-	-	-	-	-	-	-	6/	-	-	-	-	-	OUT	-	-	-	-	-	-	-	-	-	-		ISAB to B6	-	-	-	-	-	-	-
		368	-	-	-	-	-	-	-	-	-	6/	-	-	-	OUT	-	-	-	-	-	-	-	-	-	-	-		ISAB to B7	-	-	-	-	-	-	-
		369	-	-	-	-	-	-	-	-	-	-	6/	-	OUT	-	-	-	-	-	-	-	-	-	-	-	-		ISAB to B8	-	-	-	-	-	-	-
t_{PLH4}	3003 (Fig. 4)	370 to 385	Same terminal conditions and measured terminals as specified above for t_{PHL4}.																										10	30	10	51	10	30	ns
t_{PZH2} 13/	3003 (Fig. 4)	386	-	GND	GND	OUT	OUT	-	-	-	-	-	-	GND	-	-	-	-	-	-	GND	GND	GND	IN	GND	4.5 V	IG to A1	8	43	6	57	6	43	ns
		387	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		IG to A2	-	-	-	-	-	-	-
		388	-	-	-	-	-	OUT	-	-	-	-	-	-	-	-	-	-	-	GND	-	-	-	-	-	-	-		IG to A3	-	-	-	-	-	-	-
		389	-	-	-	-	-	-	OUT	-	-	-	-	-	-	-	-	-	GND	-	-	-	-	-	-	-	-		IG to A4	-	-	-	-	-	-	-
		390	-	-	-	-	-	-	-	OUT	-	-	-	-	-	-	-	GND	-	-	-	-	-	-	-	-	-		IG to A5	-	-	-	-	-	-	-
		391	-	-	-	-	-	-	-	-	OUT	-	-	-	-	-	GND	-	-	-	-	-	-	-	-	-	-		IG to A6	-	-	-	-	-	-	-
		392	-	-	-	-	-	-	-	-	-	OUT	-	-	-	GND	-	-	-	-	-	-	-	-	-	-	-		IG to A7	-	-	-	-	-	-	-
		393	-	-	-	-	-	-	-	-	-	-	OUT	-	GND	-	-	-	-	-	-	-	-	-	-	-	-		IG to A8	-	-	-	-	-	-	-
		394	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		IG to B1	-	-	-	-	-	-	-
		395	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		IG to B2	-	-	-	-	-	-	-
		396	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		IG to B3	-	-	-	-	-	-	-
		397	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		IG to B4	-	-	-	-	-	-	-
		398	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		IG to B5	-	-	-	-	-	-	-
		399	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		IG to B6	-	-	-	-	-	-	-
		400	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		IG to B7	-	-	-	-	-	-	-
		401	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		IG to B8	-	-	-	-	-	-	-
t_{PLZ2} 9/	3003 (Fig. 4)	402 to 417	Same terminal conditions (except GND replaces 4.5 V) and measured terminals as specified above for t_{PZH2}.																										-	-	-	-	-	-	-
t_{PHZ2} 13/	3003 (Fig. 4)	418 to 433	Same terminal conditions and measured terminals as specified above for t_{PZH2}.																										-	-	-	-	-	-	-
t_{PLZ3} 9/	3003 (Fig. 4)	434 to 449	Same terminal conditions and measured terminals as specified above for t_{PZL2}.																										-	-	-	-	-	-	-

See footnotes at end of table.

TABLE III. Group A inspection for device type 09 - Continued.

Symbol	MIL-STD-883 method	Case T	Terminal conditions 1/																								Measured terminal	Test limits						Unit
		Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24			Subgroup 9 T_C = +25°C 4/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 4/		
		Test no.	A0	A3	A4	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12	B5	B7	B6	B5	B4	B3	B2	B1	G	S8A	C8A	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{pZH} 3/	3003 (Fig. 4)	430				GND	OUT									GND								GND	GND	GND		4.5 V	IOIR to A1	6	63	6	57	6	63	ns
		431				-	-									-								-	-	-		-	IOIR to A2	-	-	-	-	-	-	-
		432				-	-	OUT								-							GND	-	-	-		-	IOIR to A3	-	-	-	-	-	-	-
		433				-	-		OUT							-								-	-	-		-	IOIR to A4	-	-	-	-	-	-	-
		434				-	-			OUT						-						GND		-	-	-		-	IOIR to A5	-	-	-	-	-	-	-
		435				-	-				OUT					-				GND				-	-	-		-	IOIR to A6	-	-	-	-	-	-	-
		436				-	-									-		GND						-	-	-		-	IOIR to A7	-	-	-	-	-	-	-
		437				-	-								OUT	-			GND					-	-	-		-	IOIR to A8	-	-	-	-	-	-	-
		438			GND	-	-									-								-	-	-		-	IOIR to B1	-	-	-	-	-	-	-
		439				-	-	GND								-								-	-	-		-	IOIR to B2	-	-	-	-	-	-	-
		440				-	-		GND							-							OUT	-	-	-		-	IOIR to B3	-	-	-	-	-	-	-
		441				-	-			GND						-								-	-	-		-	IOIR to B4	-	-	-	-	-	-	-
		442				-	-				GND					-								-	-	-		-	IOIR to B5	-	-	-	-	-	-	-
		443				-	-					GND				-								-	-	-		-	IOIR to B6	-	-	-	-	-	-	-
		444				-	-						GND			-								-	-	-		-	IOIR to B7	-	-	-	-	-	-	-
		445				-	-							GND		-								-	-	-		-	IOIR to dB	-	-	-	-	-	-	-
t_{pZL} 3/	3003 (Fig. 4)	446 to 491	Same terminal conditions (except GND replaces 4.5 V) and measured terminals as specified above for t_{pZH}.																										-	-	-	-	-	-	-
t_{PHZ} 3/	3003 (Fig. 4)	492 to 537	Same terminal conditions and measured terminals as specified above for t_{pZL}.																										-	-	-	-	-	-	ns
t_{PLZ} 3/	3003 (Fig. 4)	498 to 513	Same terminal conditions and measured terminals as specified above for t_{pZL}.																										-	-	-	-	-	-	ns

See footnotes at end of table.

TABLE III. Group A inspection for device type 53.

Symbol	MIL-STD-883 method	Cases 2, 3	Terminal conditions 1/																				Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1	1 mA									1/										GND	DIR	1/						
		2		1 mA								"										"	A1	"	1.5					V
		3			1 mA							"										"	A2	"	"					"
		4				1 mA						"										"	A3	"	"					"
		5					1 mA					"										"	A4	"	"					"
		6						1 mA				"										"	A5	"	"					"
		7							1 mA			"										"	A6	"	"					"
		8								1 mA		"										"	A7	"	"					"
		9									1 mA	"										"	A8	"	"					"
		10										"	1 mA									"	B8	"	"					"
		11										"		1 mA								"	B7	"	"					"
		12										"			1 mA							"	B6	"	"					"
		13										"				1 mA						"	B5	"	"					"
		14										"					1 mA					"	B4	"	"					"
		15										"						1 mA				"	B3	"	"					"
		16										"							1 mA			"	B2	"	"					"
		17										"								1 mA			"	B1	"	"					"
		18										"									1 mA	"	$\overline{G}$	"	"					"
V_{IC} (neg)		19	-1 mA																			1/	DIR		-1.5					
		20		-1 mA																		"	A1		"					V
		21			-1 mA																	"	A2		"					"
		22				-1 mA																"	A3		"					"
		23					-1 mA															"	A4		"					"
		24						-1 mA														"	A5		"					"
		25							-1 mA													"	A6		"					"
		26								-1 mA												"	A7		"					"
		27									-1 mA											"	A8		"					"
		28											-1 mA									"	B8		"					"
		29												-1 mA								"	B7		"					"
		30													-1 mA							"	B6		"					"
		31														-1 mA						"	B5		"					"
		32															-1 mA					"	B4		"					"
		33																-1 mA				"	B3		"					"
		34																	-1 mA			"	B2		"					"
		35																		-1 mA		"	B1		"					"
		36																			-1 mA	"	$\overline{G}$		"					"
I_{CO}	3005	37	GND									GND	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V	GND	5.5 V	V_{CC}		.2		20			µA
		38	GND									"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	"		"		"			"
		39	6 V	GND	6 V	GND	6 V	GND	6 V	GND	6 V	"	GND	GND	GND	GND	GND	GND	GND	GND	"	"	"		"		"			"
		40	6 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	6 V	"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	"		"		"			"
I_{CCZ}		41	GND									"									5.5 V	"	"		-		10			-
I_{CCa}		42	2.4 V									"									5.5 V	"	"		.430		1.5			mA
		43	0.8 V									"									0.8 V	"	"		.195		"			"
		44	0.8 V	0.8 V	0.8 V	0.1 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	"	2.4 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	2.4 V	"	"		.735		"			"
		45	2.4 V	2.4 V	0.8 V	0.1 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	"									5.5 V	"	"		.195		"			"

See footnotes at end of table.

TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 method	Cases 2, R	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	D14	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{OH6}	3006	46	0.8 V	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA	GND	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	0.8 V	4.5 V	A1	4.4		4.4		4.4		V
		47	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2	"		"		"		"
		48	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A3	"		"		"		"
		49	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A4	"		"		"		"
		50	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A5	"		"		"		"
		51	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A6	"		"		"		"
		52	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A7	"		"		"		"
		53	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A8	"		"		"		"
		54	2.0 V	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B1	"		"		"		"
		55	"	"	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B2	"		"		"		"
		56	"	"	"	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B3	"		"		"		"
		57	"	"	"	"	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B4	"		"		"		"
		58	"	"	"	"	"	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B5	"		"		"		"
		59	"	"	"	"	"	"	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	B6	"		"		"		"
		60	"	"	"	"	"	"	"	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	B7	"		"		"		"
		61	"	"	"	"	"	"	"	"	2.0 V	"	-20 µA	-20 µA	"	"	"	"	"	"	"	"	B8	"		"		"		"
V_{OH7}	3006	62	0.8 V	-6.0 mA	-6.0 mA	-6.0 mA	-6.0 mA	-6.0 mA	-6.0 mA	-6.0 mA	-6.0 mA	"	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	5.5 V	A1	3.98		3.7		3.98		"
		63	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2	"		"		"		"
		64	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A3	"		"		"		"
		65	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A4	"		"		"		"
		66	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A5	"		"		"		"
		67	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A6	"		"		"		"
		68	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A7	"		"		"		"
		69	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A8	"		"		"		"
		70	2.0 V	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B1	"		"		"		"
		71	"	"	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B2	"		"		"		"
		72	"	"	"	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B3	"		"		"		"
		73	"	"	"	"	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B4	"		"		"		"
		74	"	"	"	"	"	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B5	"		"		"		"
		75	"	"	"	"	"	"	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	B6	"		"		"		"
		76	"	"	"	"	"	"	"	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	B7	"		"		"		"
		77	"	"	"	"	"	"	"	"	2.0 V	"	-6.0 mA	-6.0 mA	"	"	"	"	"	"	"	"	B8	"		"		"		"
V_{OL6}	3007	78	0.8 V	20 µA	20 µA	20 µA	20 µA	20 µA	20 µA	20 µA	20 µA	"	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	"	4.5 V	A1		0.1		0.1		0.1	"
		79	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2		"		"		"	"
		80	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A3		"		"		"	"
		81	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A4		"		"		"	"
		82	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A5		"		"		"	"
		83	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A6		"		"		"	"
		84	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A7		"		"		"	"
		85	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A8		"		"		"	"
		86	2.0 V	0.8 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B1		"		"		"	"
		87	"	"	0.8 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B2		"		"		"	"
		88	"	"	"	0.8 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B3		"		"		"	"
		89	"	"	"	"	0.8 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B4		"		"		"	"
		90	"	"	"	"	"	0.8 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B5		"		"		"	"
		91	"	"	"	"	"	"	0.8 V	"	"	"	"	"	"	"	"	"	"	"	"	"	B6		"		"		"	"
		92	"	"	"	"	"	"	"	0.8 V	"	"	"	"	"	"	"	"	"	"	"	"	B7		"		"		"	"
		93	"	"	"	"	"	"	"	"	0.8 V	"	20 µA	20 µA	"	"	"	"	"	"	"	"	B8		"		"		"	"

See footnotes at end of table.

TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 method	Cases 2, 3 R	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	G	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{OL7}	3007	94	0.8 V	6.0 mA	6.0 mA	6.0 mA	6.0 mA	6.0 mA	6.0 mA	6.0 mA	6.0 mA	GND	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	5.5 V	A1		.26		0.4		.26	V
		95	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2		"		"		"	
		96	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A3		"		"		"	
		97	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A4		"		"		"	
		98	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A5		"		"		"	
		99	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A6		"		"		"	
		100	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A7		"		"		"	
		101	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A8		"		"		"	
		102	2.0 V	0.8 V	6.0 mA	"	"	"	"	"	"	"	"	"	"	"	"	"	6.0 mA	6.0 mA	"	"	B1		"		"		"	
		103	"	"	0.8 V	0.8 V	"	"	"	"	"	"	"	"	"	"	"	6.0 mA	"	"	"	"	B2		"		"		"	
		104	"	"	"	"	0.8 V	"	"	"	"	"	"	"	"	"	6.0 mA	"	"	"	"	"	B3		"		"		"	
		105	"	"	"	"	"	0.8 V	"	"	"	"	"	"	"	6.0 mA	"	"	"	"	"	"	B4		"		"		"	
		106	"	"	"	"	"	"	0.8 V	"	"	"	"	"	6.0 mA	"	"	"	"	"	"	"	B5		"		"		"	
		107	"	"	"	"	"	"	"	0.8 V	"	"	"	6.0 mA	"	"	"	"	"	"	"	"	B6		"		"		"	
		108	"	"	"	"	"	"	"	"	0.8 V	"	6.0 mA	"	"	"	"	"	"	"	"	"	B7		"		"		"	
		109	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B8		"		"		"	
I_{OS4}	3011	110	GND	GND	GND	GND	GND	GND	GND	GND	GND	"	4 V	4 V	4 V	4 V	4 V	4 V	4 V	4 V	GND	4 V	A1	-10	-135	-10	-135	-10	-135	mA
		111	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2	"	"	"	"	"	"	
		112	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A3	"	"	"	"	"	"	
		113	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A4	"	"	"	"	"	"	
		114	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A5	"	"	"	"	"	"	
		115	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A6	"	"	"	"	"	"	
		116	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A7	"	"	"	"	"	"	
		117	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A8	"	"	"	"	"	"	
		118	4 V	4 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	GND	"	"	B1	"	"	"	"	"	"	
		119	"	"	4 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B2	"	"	"	"	"	"	
		120	"	"	"	4 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B3	"	"	"	"	"	"	
		121	"	"	"	"	4 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B4	"	"	"	"	"	"	
		122	"	"	"	"	"	4 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B5	"	"	"	"	"	"	
		123	"	"	"	"	"	"	4 V	"	"	"	"	"	"	"	"	"	"	"	"	"	B6	"	"	"	"	"	"	
		124	"	"	"	"	"	"	"	4 V	"	"	"	"	"	"	"	"	"	"	"	"	B7	"	"	"	"	"	"	
		125	"	"	"	"	"	"	"	"	4 V	"	GND	GND	GND	GND	GND	GND	"	"	"	"	B8	"	"	"	"	"	"	
[illegible]		126	.8 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	2/	2/	2/	2/	2/	2/	2/	2/	2/	5.5 V	A1		.1		2.0			μA
[illegible]		127	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2		"		"			
		128	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A3		"		"			
		129	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A4		"		"			
		130	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A5		"		"			
		131	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A6		"		"			
		132	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A7		"		"			
		133	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A8		"		"			
		134	2.0 V	2/	2/	"	"	"	"	"	"	"	"	"	"	"	"	"	5.5 V	5.5 V	"	"	B1		"		"			
		135	"	"	"	2/	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B2		"		"			
		136	"	"	"	"	2/	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B3		"		"			
		137	"	"	"	"	"	2/	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B4		"		"			
		138	"	"	"	"	"	"	2/	"	"	"	"	"	"	"	"	"	"	"	"	"	B5		"		"			
		139	"	"	"	"	"	"	"	2/	"	"	"	"	"	"	"	"	"	"	"	"	B6		"		"			
		140	"	"	"	"	"	"	"	"	2/	"	"	"	"	"	"	"	"	"	"	"	B7		"		"			
		141	"	"	"	"	"	"	"	"	"	"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	"	"	B8		"		"			

See footnotes at end of table.

TABLE III. Group A inspection for device type 53 - Continued.

Symbol	Mil-STD-883 method	Cases 2, R	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{G}$	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{IL}/ I_{OZL}		142	0.4 V	GND	GND	GND	GND	GND	GND	GND	GND	GND	2/	2/	2/	2/	2/	2/	2/	2/	2/	5.5 V	A1		-.1		-2.0			µA
		143	"																				A2		"		"			"
		144	"																				A3		"		"			"
		145	"																				A4		"		"			"
		146	"																				A5		"		"			"
		147	"																				A6		"		"			"
		148	"																				A7		"		"			"
		149	"																				A8		"		"			"
		150	2.0 V	2/																			B1		"		"			"
		151	"		2/																		B2		"		"			"
		152	"			2/																	B3		"		"			"
		153	"				2/																B4		"		"			"
		154	"					2/															B5		"		"			"
		155	"						2/														B6		"		"			"
		156	"							2/													B7		"		"			"
		157	"								2/												B8		"		"			"
I_{IH}	3010	158	5.5 V																				DIR		.05		.1			"
		159	"																		5.5 V		$\overline{G}$		.05		.1			"
I_{IL}	3009	160	GND																				DIR		-.05		-.1			"
		161	"																		GND		$\overline{G}$		-.05		-.1			"
$C_{I/O}$		162	GND	3/	3/	3/	3/	3/	3/	3/	3/		3/	3/	3/	3/	3/	3/	3/	3/	5.5 V		A1		20					pF
		163	"																		"		A2		"					"
		164	"																		"		A3		"					"
		165	"																		"		A4		"					"
		166	"																		"		A5		"					"
		167	"																		"		A6		"					"
		168	"																		"		A7		"					"
		169	"																		"		A8		"					"
		170	5.5 V																		"		B8		"					"
		171	"																		"		B7		"					"
		172	"																		"		B6		"					"
		173	"																		"		B5		"					"
		174	"																		"		B4		"					"
		175	"																		"		B3		"					"
		176	"																		"		B2		"					"
		177	"																		"		B1		"					"
C_C		178	3/																		3/	GND	DIR		10					"
		179	"																		"	GND	$\overline{G}$		10					"

																								Test limits						
																								Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 9 T_C = -55°C		
																								Min	Max	Min	Max	Min	Max	
Truth table tests 5/		180	A	A	A	A	A	A	A	A	A		H	H	H	H	H	H	H	H	B	4.5 V	6/							
		181	B	B	B	B	B	B	B	B	B		L	L	L	L	L	L	L	L	B	"	"							
		182	B	H	H	H	H	H	H	H	H		A	A	A	A	A	A	A	A	B	"	"							
		183	L	L	L	L	H	H	H	H	H		B	B	B	B	B	B	B	B	B	"	"							
		184	H	H	H	H	L	H	H	H	H		H	H	H	H	H	H	H	H	A	"	"							

See footnotes at end of table.

TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 method	Cases 2, R	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C 4/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 4/		
		Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	G	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{THL}	3004 (Fig. 4)	185	GND	OUT	OUT							GND									GND	4.5 V	A1	2	12	2	16	2	12	ns
		186	"			OUT						"						IN	IN		"	"	A2	"	"	"	"	"	"	"
		187	"				OUT					"					IN				"	"	A3	"	"	"	"	"	"	"
		188	"					OUT				"				IN					"	"	A4	"	"	"	"	"	"	"
		189	"						OUT			"			IN						"	"	A5	"	"	"	"	"	"	"
		190	"							OUT		"		IN							"	"	A6	"	"	"	"	"	"	"
		191	"								OUT	"	IN								"	"	A7	"	"	"	"	"	"	"
		192	"									"								OUT	"	"	A8	"	"	"	"	"	"	"
		193	4.5 V	IN	IN							"							OUT		"	"	B1	"	"	"	"	"	"	"
		194	"			IN						"						OUT			"	"	B2	"	"	"	"	"	"	"
		195	"				IN					"					OUT				"	"	B3	"	"	"	"	"	"	"
		196	"					IN				"				OUT					"	"	B4	"	"	"	"	"	"	"
		197	"						IN			"			OUT						"	"	B5	"	"	"	"	"	"	"
		198	"							IN		"		OUT							"	"	B6	"	"	"	"	"	"	"
		199	"								IN	"	OUT								"	"	B7	"	"	"	"	"	"	"
		200	"									"									"	"	B8	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 4)	201 to 216	Same tests, terminal conditions, and limits as specified above for t_{THL}.																					-	-	-	-	-	-	-
t_{PHL2}	3003 (Fig. 4)	217	4.5 V	IN	IN							GND								OUT	GND	4.5 V	A1 to B1	4	23	4	33	4	23	ns
		218	"			IN						"							OUT		"	"	A2 to B2	"	"	"	"	"	"	"
		219	"				IN					"						OUT			"	"	A3 to B3	"	"	"	"	"	"	"
		220	"					IN				"					OUT				"	"	A4 to B4	"	"	"	"	"	"	"
		221	"						IN			"				OUT					"	"	A5 to B5	"	"	"	"	"	"	"
		222	"							IN		"			OUT						"	"	A6 to B6	"	"	"	"	"	"	"
		223	"								IN	"	OUT								"	"	A7 to B7	"	"	"	"	"	"	"
		224	"									"								OUT	"	"	A8 to B8	"	"	"	"	"	"	"
		225	GND	OUT	OUT							"							IN		"	"	B1 to A1	"	"	"	"	"	"	"
		226	"			OUT						"						IN			"	"	B2 to A2	"	"	"	"	"	"	"
		227	"				OUT					"					IN				"	"	B3 to A3	"	"	"	"	"	"	"
		228	"					OUT				"				IN					"	"	B4 to A4	"	"	"	"	"	"	"
		229	"						OUT			"			IN						"	"	B5 to A5	"	"	"	"	"	"	"
		230	"							OUT		"		IN							"	"	B6 to A6	"	"	"	"	"	"	"
		231	"								OUT	"	IN								"	"	B7 to A7	"	"	"	"	"	"	"
		232	"									"									"	"	B8 to A8	"	"	"	"	"	"	"
t_{PLH2}	3003 (Fig. 4)	223 to 248	Same terminal conditions and measured terminals as specified above for t_{PHL2}.																					-	-	-	-	-	-	-

See footnotes at end of table.

TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 method	Cases 2, R	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C 4/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 4/		
		Test no.	D1R	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PZH1} 10/	3003 (Fig. 4)	249	GND	OUT								GND									IN	4.5 V	to A1	6	41	6	54	6	61	ns
		250	"		OUT							"							4.5 V		"	"	to A2	"	"	"	"	"	"	"
		251	"			OUT						"						4.5 V			"	"	to A3	"	"	"	"	"	"	"
		252	"				OUT					"					4.5 V				"	"	to A4	"	"	"	"	"	"	"
		253	"					OUT				"				4.5 V					"	"	to A5	"	"	"	"	"	"	"
		254	"						OUT			"			4.5 V						"	"	to A6	"	"	"	"	"	"	"
		255	"							OUT		"		4.5 V							"	"	to A7	"	"	"	"	"	"	"
		256	"								OUT	"	4.5 V								"	"	to A8	"	"	"	"	"	"	"
		257	4.5 V	4.5 V								"								OUT	"	"	to B1	"	"	"	"	"	"	"
		258	"		4.5 V							"							OUT		"	"	to B2	"	"	"	"	"	"	"
		259	"			4.5 V						"						OUT			"	"	to B3	"	"	"	"	"	"	"
		260	"				4.5 V					"					OUT				"	"	to B4	"	"	"	"	"	"	"
		261	"					4.5 V				"				OUT					"	"	to B5	"	"	"	"	"	"	"
		262	"						4.5 V			"			OUT						"	"	to B6	"	"	"	"	"	"	"
		263	"							4.5 V		"		OUT							"	"	to B7	"	"	"	"	"	"	"
		264	"								4.5 V	"	OUT								"	"	to B8	"	"	"	"	"	"	"
t_{PHZ1} 10/	3003 (Fig. 4)	265 to 280	Same terminal conditions and measured terminals as specified above for t_{PZH1}.																					-	-	-	-	-	-	-
t_{PZL1} 9/	3003 (Fig. 4)	281	GND	OUT								GND								GND	IN	4.5 V	to A1	-	-	-	-	-	-	-
		282	"		OUT							"							GND		"	"	to A2	"	"	"	"	"	"	"
		283	"			OUT						"						GND			"	"	to A3	"	"	"	"	"	"	"
		284	"				OUT					"					GND				"	"	to A4	"	"	"	"	"	"	"
		285	"					OUT				"				GND					"	"	to A5	"	"	"	"	"	"	"
		286	"						OUT			"			GND						"	"	to A6	"	"	"	"	"	"	"
		287	"							OUT		"		GND							"	"	to A7	"	"	"	"	"	"	"
		288	"								OUT	"	GND								"	"	to A8	"	"	"	"	"	"	"
		289	6 V	GND								"								OUT	"	"	to B1	"	"	"	"	"	"	"
		290	"		GND							"							OUT		"	"	to B2	"	"	"	"	"	"	"
		291	"			GND						"						OUT			"	"	to B3	"	"	"	"	"	"	"
		292	"				GND					"					OUT				"	"	to B4	"	"	"	"	"	"	"
		293	"					GND				"				OUT					"	"	to B5	"	"	"	"	"	"	"
		294	"						GND			"			OUT						"	"	to B6	"	"	"	"	"	"	"
		295	"							GND		"		OUT							"	"	to B7	"	"	"	"	"	"	"
		296	"								GND	"	OUT								"	"	to B8	"	"	"	"	"	"	"
t_{PLZ1} 9/	3003 (Fig. 4)	297 to 312	Same terminal conditions and measured terminals as specified above for t_{PZL1}.																					-	-	-	-	-	-	-

1/ Pins not designated may be "High" level logic, "Low" level logic, or open. Exceptions are as follows:

a. $V_{IC(pos)}$ tests, the GND terminal shall be open. A minimum limit of 0.4 V applies to tests being performed on equipment not capable of opening the GND pin during testing.

b. $V_{IC(neg)}$ tests, the V_{CC} terminal shall be open.

c. I_{CC} tests, the output terminal shall be open.

2/ Three-state output conditions are required. For I_{OZL}, set output to high state. For I_{OZH}, set output to low state. Set input pins to $V_{IL} = V_{IL(max)}$ or to $V_{IH} = V_{IH(min)}$, as required.

3/ See 4.4.1c.

4/ See 4.4.1d.

5/ Tests shall be performed in sequence, attributes data only.

6/ H > 1.5 V; L < 1.5 V; A = 3.7 V; B = 0.4 V; for the 53 device type A = 2.4 V; B = 0.4 V.

7/ Three-state output conditions are required.

8/ f_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

9/ Add 1 kΩ minimum resistor between V_{CC} and output to pull output to high state.

10/ Add 1 kΩ minimum resistor between GND and output to pull output to low state.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.1). When group E testing is performed it shall be in accordance with table V of method 5005 of MIL-STD-883 and 4.5.4 herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. Unless otherwise specified, all voltages given are referenced to the microcircuit GND terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When the burn-in and life tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to within 10°C of their power stable condition at room temperature; then, electrical parameter end-point measurements shall be performed.

TABLE IV. Delta limits at 25°C.

Parameter 1/	Device types	
	All	
I_{CC}		±30 nA

1/ The above parameters shall be recorded before and after the required burn-in and life tests to determine deltas (Δ).

4.5.3 Quiescent supply current (I_{CC} test). When performing quiescent supply current measurements (I_{CC}), the meter shall be placed so that all currents flow through the meter.

4.5.4 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in table V of method 5005 of MIL-STD-883 and herein:

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (group A subgroup 1 at 25°C) and also be subjected to the threshold-voltage test in table V in order to calculate the delta threshold (ΔV_T) after irradiation.

b. The devices shall be subjected to a total radiation dose as specified in MIL-M-38510 for the radiation hardness assurance (RHA) level being tested, and meet the end point electrical parameters as defined in table VI at 25°C, after exposure. The start and completion of the end point electrical parameter measurements shall not exceed 2 hours following irradiation.

c. Threshold-voltage test circuit conditions shall be as specified in table V and figure 5. In situ and remote testing, the tests shall be performed with the devices biased in accordance with table VII and bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.

d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required, then an equivalent truth table test shall be performed.

TABLE V. Threshold-voltage test circuit conditions.

Device type	GND	5 V	V_{TN} measured at -20 µA supply	GND	-5.0 V	V_{TP} measured at 20 µA supply
01	1	14	3,4,5,6,7,13	1	7,8,9,10,11	13, 14
02	1	14	3,4,5,6,7,13	1	7,8,9,10,11	13, 14
03, 53	1	20	10-19	1	2-10,19	20
04						
05						
06	1	20	10-19	1	2-10,19	20
07	1	20	10-19	1	2-10,19	20
08	3	24	1,2, 12-23	3	1,2 4-12, 21-23	24
09	3	24	1,2, 12-23	3	1,2 4-12, 21-23	24

TABLE VI. Radiation hardening end-point electrical parameters at +25°C.

Parameter	All device types	V_{CC} Device types 01-09	V_{CC} Device types 53
V_{TN}	-0.3 V min	5 V	5 V
V_{TP}	2.8 V max	5 V	5 V
ΔV_T	1.0 V max	5 V	5 V
I_{CC}	100 x max limit	6 V	5.5 V
t_{PLH}	1.35 x max limit	4.5 V	4.5 V
t_{PHL}	1.35 x max limit	4.5 V	4.5 V

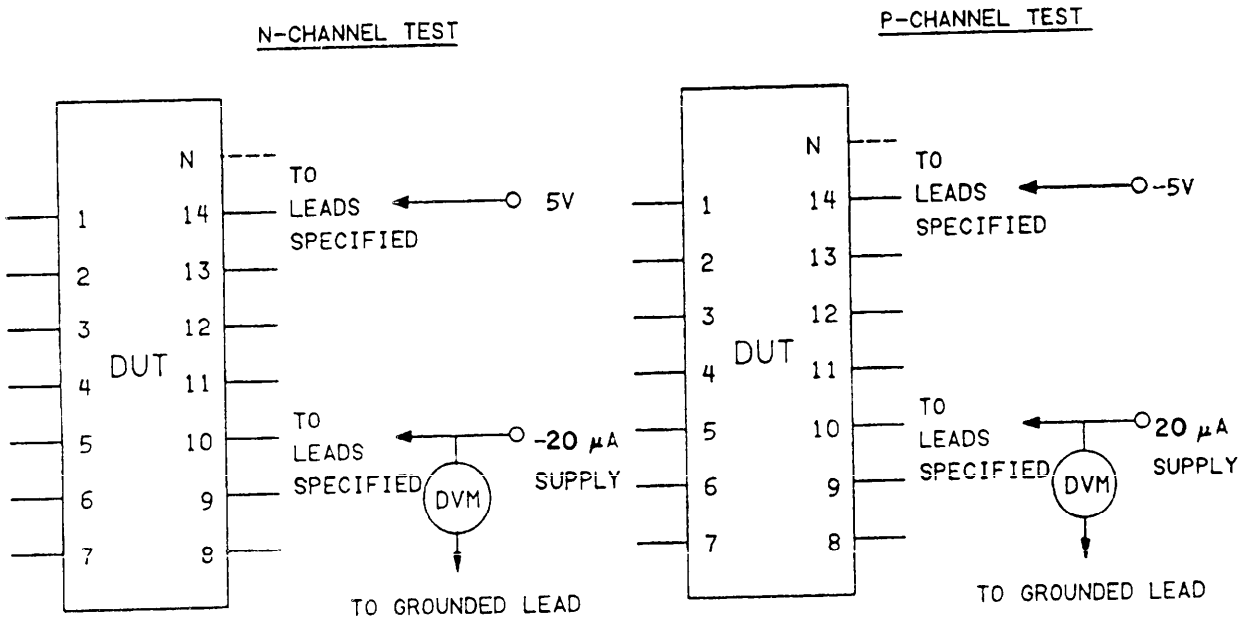


FIGURE 5. Threshold - voltage test circuit.

TABLE VII. Bias during exposure to radiation.

Device type	Pin connections		
	V_{CC} = 4.5 V dc (through a 30- to 60-kilohm resistor)	GND	V_{CC} = 4.5 V dc
01	1, 8-11, 13	7	14
02	1, 8-11, 13	7	14
03, 53	11-18	1,10,19	20
04			
05			
06	11-18	1,10,19	20
07	11-18	1,10,19	20
08	1, 13-20, 23	2,3,12, 22,23	24
09	1, 13-20, 23	2,3,12, 22,23	24

Pins not designated are open or connected to 4.5 V dc through a 30- to 60-kilohm resistor.

4.6 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, shall be supplied.

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady-state life tests (see 3.5).

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parameters data (see 4.2c).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by, or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
V_{IN}	Voltage level at an input terminal.
I_{IN}	Current flowing into an input terminal.
t_{PHZ}	Output disable time (of a three-state output) from high-level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined high-level to a high-impedance (off) state.
t_{PLZ}	Output disable time (of a three-state output) from low-level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined low-level to a high-impedance (off) state.
t_{PZH}	Output enable time (of a three-state output) to high-level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined high-level.
t_{PZL}	Output enable time (of a three-state output) to low-level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined low-level.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54HC242
02	54HC243
03	54HC245
04 1/	54HC620
05 1/	54HC623
06	54HC640
07	54HC643
08	54HC646
09	54HC648
53	54HCT245

1/ These device types to be included in a later revision of this specification.

6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static build up. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, OC, MC, SH

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-0850)

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE AIR FORCE

Rome Air Development Center
Griffiss AFB, NY 13441

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

Commander
Rome Air Development Center
Attn: RBE-2
Griffiss AFB, NY 13441

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/655	MICROCIRCUITS, DIGITAL, HIGH SPEED, CMOS, BUS TRANSCEIVERS WITH THREE-STATE OUTPUTS, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording: